

AIRCRAFT FINAL ASSEMBLY LINE TRANSFORMATION THROUGH ARTIFICIAL INTELLIGENCE IN AN INDUSTRY 5.0 ENVIRONMENT v1

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OBJECTIVE

This project aims to study and develop the different transformations required in an aircraft final assembly line in order to implement an Industry 5.0 model.

- As the core element of the transformation we will use the potential of generative artificial intelligence and intelligent agents, touching to a lesser extent on classic types of AI (Industry 4.0) such as expert systems, artificial neural networks, deep learning, .. historically used in applications such as computer vision, data analytics,…

- The starting point will be a military aircraft final assembly line framed within the Industry 4.0 model.

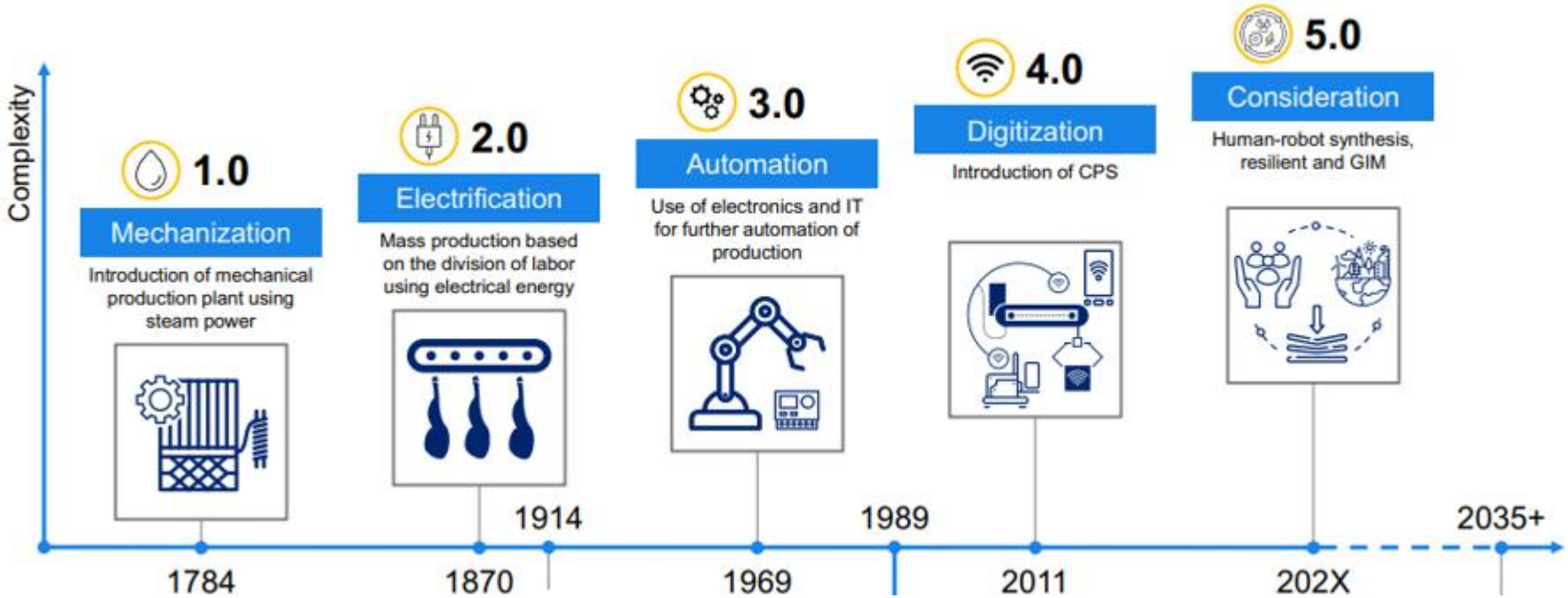
- It will be assumed to have MES (Manufacturing Execution System) and ERP (Enterprise Resource Planning) production control systems, as well as deployed operational excellence (Lean Manufacturing) and asset management systems.

- In terms of technologies, the classic I4.0 technologies are assumed to be established, such as: IoT (RFID, LORA, GPS, BLE, Environmental Sensing,..), Augmented Reality (Hololens,..), Cobots, AGV, 3D Printing, Classic AI, Digital Twins,..

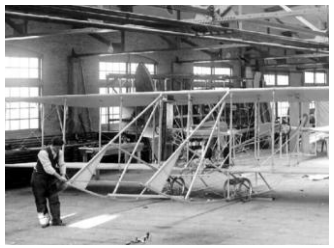
- The normal flow of activities for the complete assembly of the aircraft will be studied, and those that must be transformed to be in line with the Industry 5.0 model will be selected.

- The difficulties, costs and benefits will be evaluated.

INDUSTRIAL MODEL EVOLUTION



SECTOR AEROSPACE





AEROSPACE INDUSTRIAL MODEL

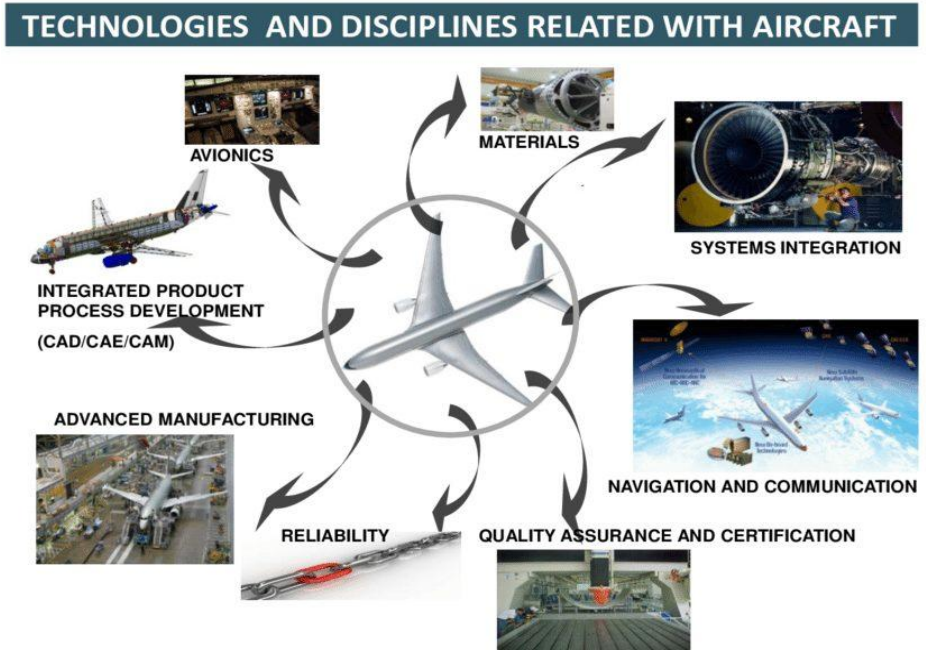
AEROSPACE INDUSTRIAL MODEL

Aircraft manufacturing is a complex process involving design, material selection, component manufacturing, assembly and rigorous testing.

This process requires the collaboration of engineers, designers and technicians, as well as the use of advanced technologies that ensure efficiency and safety.

In addition:

- **The manufacturing of components**, systems and subsystems is usually carried out independently, on demand, by numerous suppliers around the world before being integrated into the final assembly line (FAL).
- **Production and assembly planning**, as well as external and internal production supply logistics, are therefore a very complex but integral part of aircraft production, which nevertheless depends on a complex global supply chain.
- **The wide variety of systems**, such as the aircraft cabin, where each customer has different requirements, and the low order volume further increase the complexity of planning and production tasks.
- **Manual work predominates** in aircraft production, and final assembly takes place in a fixed position or in pulses rather than on a continuous moving line like car assembly lines.
- **All design, production and testing processes must be qualified,** documented and supervised by the competent authorities, original equipment manufacturers and customers to guarantee aircraft safety.





AIRCRAFT FINAL ASSEMBLY LINE UNDER INDUSTRY 4.0



INDUSTRY 4.0 - MODEL

I4.0 CONCEPT

Industry 4.0 is based on the idea of smart factories, where machines, systems and people are interconnected, optimizing production and decision-making.

The key technologies are:

- **Internet of Things (IoT):** enables devices and machines to connect over the Internet, allowing real-time data collection and exchange. It is used to analyze large amounts of data, predict patterns, optimize processes and automate tasks.
- **Robotics and Automation:** the integration of robots and automated systems increases efficiency and precision.
- **Augmented Reality and Virtual Reality:** Used to improve training and real-time data visualization.
- **Cloud Computing:** provides the infrastructure for storing and processing large amounts of data.
- **Big Data:** enables the analysis of large datasets to obtain valuable insights and make better-informed decisions.
- **Cyber-Physical Systems:** systems that integrate computing, communication and physical control capabilities to interact with the physical world.

IMPACT ON INDUSTRY 3.0

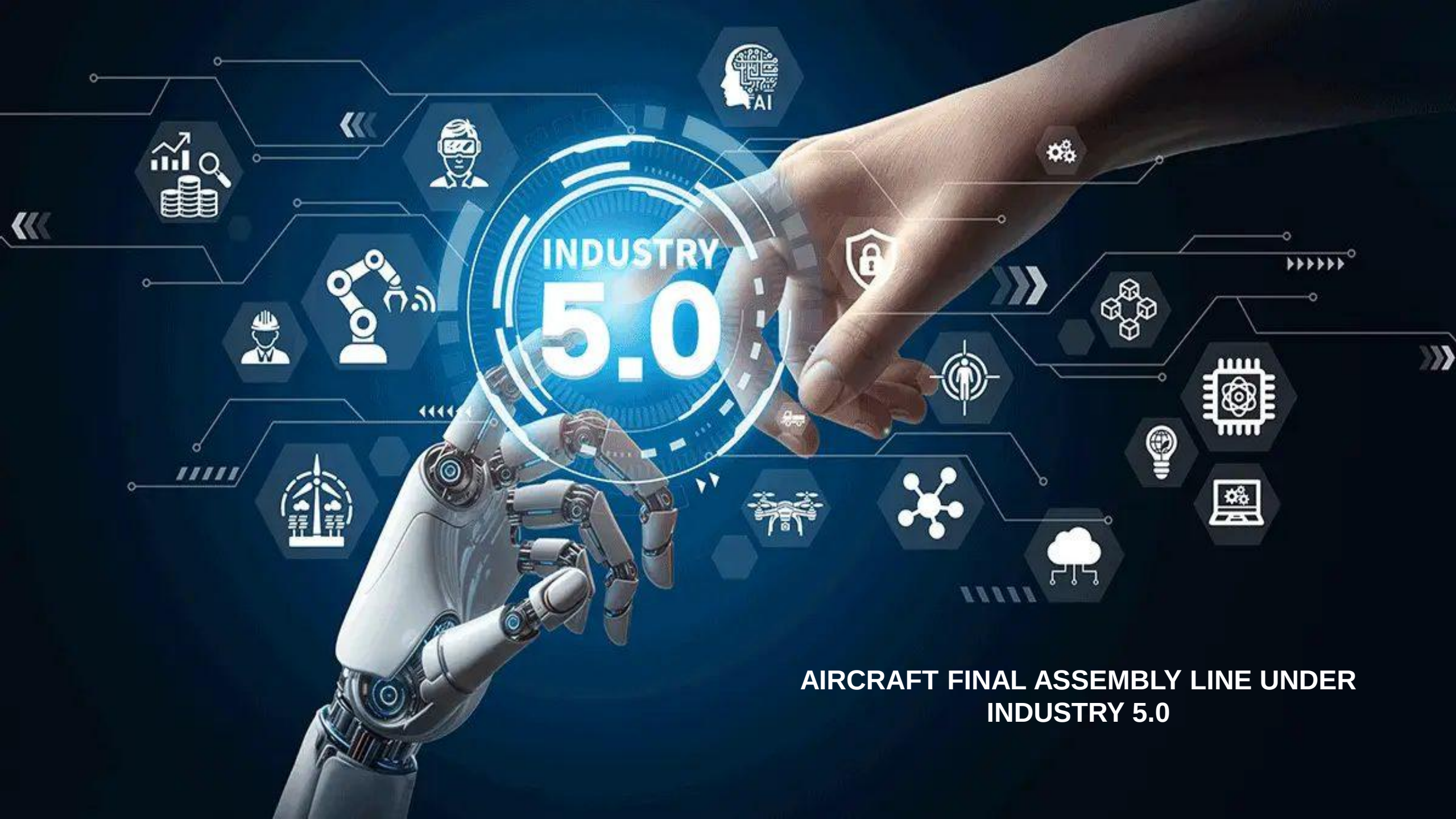
- Greater efficiency, productivity, flexibility and customization
- Analysis and decision-making become faster and more efficient.
- New business models
- Change in the job profile





PROBLEMS OF THE I4.0 MODEL

- **High costs:** Industrial automation requires substantial investment in hardware, software and specialized engineering.
- **Slow implementations:** Long PLC programming cycles delay adoption, deployment and scaling. In addition, traditional industrial controllers are obsolete and cannot harness the transformative power of industrial AI.
- **Rigid systems:** Traditional solutions struggle to adapt to changing production demands.
- **Labor shortage:** Skilled labor for advanced automation is scarce.
- **Sustainability concerns:** Inefficient systems and high energy consumption conflict with modern sustainability goals.



AIRCRAFT FINAL ASSEMBLY LINE UNDER INDUSTRY 5.0

INDUSTRY 5.0 - MODEL

HUMAN-CENTRIC

I5.0 promotes a socio-technical system that humanizes technology and interactions with the environment by reorienting the values that underpin production systems.

It seeks to emphasize collaboration between humans and machines, leveraging human skills and creativity together with the precision and efficiency of robots and AI.

It requires not only a change in the work environment and philosophy but also in the role of the operator in this production model.

This role has evolved in line with the industrial model:

- **Op 1.0 (I1.0-1770):** the operator who manually used tools based on experience and skill, but aided by steam-powered machines
- **Op 2.0 (I2.0-1870):** the operator whose work was determined by their place on the line (monotonous and non-creative), aided by electric machines.
- **Op. 3.0 (I3.0-1970):** the operator whose work was assisted by information systems (computers) and numerically controlled machines
- **Op. 4.0 (I4.0-2010):** the operator who works collaboratively with classic robots, automated machines and computers within an interconnected cyber-physical system.
- **Op. 5.0 (I5.0-2025):** the operator who, building on 4.0 cyber-physical systems, is oriented towards the symbiosis of the human-cyber-physical system aided by artificial intelligence (AI), natural language processing (NLP), intelligent robotics..





SUSTAINABLE

It addresses the need for environmental sustainability through resource optimization processes and a circular economy that preserves the habitat. A key aspect is the focus on reducing waste, energy consumption and the overall environmental impact of aerospace manufacturing and operations.

RESILIENT

The term resilience refers not only to a system's ability to recover from disruptions (supply chain failures, geopolitical crises, natural disasters..), but also to its ability to adapt and continue operating effectively.

Implementing resilience in manufacturing systems begins with designing robust systems that consider strategic redundancy and flexibility, together with strategic management to identify possible scenarios and establish contingency plans.

INDUSTRY 5.0 – ROLE OF ARTIFICIAL INTELLIGENCE

AI EVOLUTION

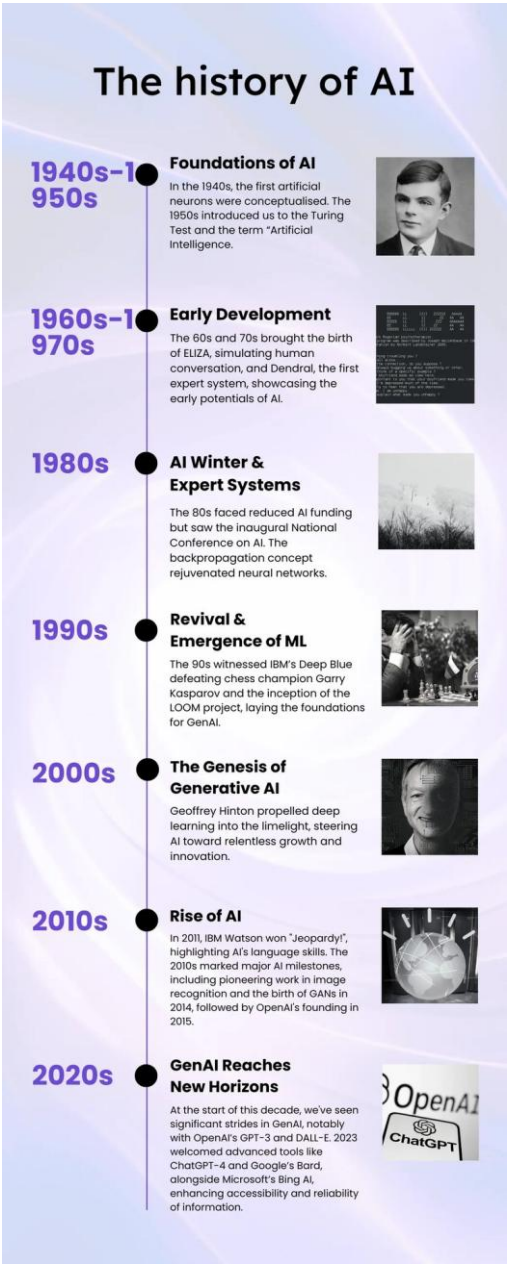
- Artificial intelligence (AI) has evolved from the theoretical concepts of the early 20th century and its formal beginnings in the 1950s to modern, complex systems, marked by periods of great optimism and "AI winters" in which progress stalled.
- Key milestones include Alan Turing's Turing Test, the 1956 Dartmouth Conference where the term AI was coined, the rise of expert systems in the 1980s, and advances in machine learning and deep learning driven by big data and greater processing power.
- Today's AI, including generative models such as transformers, is widely used in everyday applications and continues to advance rapidly, raising incredible opportunities and ethical considerations.

ROLE OF AI IN I5.0

- Artificial intelligence (AI) is fundamental to Industry 5.0, which evolves from Industry 4.0's focus on automation to a human-centric approach that prioritizes collaboration between humans and advanced machines (such as AI-powered robots) to achieve more sustainable, resilient and customized production systems.
- AI enhances human capabilities by handling repetitive tasks, while humans contribute creativity and critical thinking to solving complex problems and to innovation. This integration results in intelligent, adaptive production lines, predictive maintenance, greater efficiency and sustainable practices such as waste reduction and better energy management.

KEY ASPECTS OF AI IN I5.0

- **Human-machine collaboration**: AI and robots work alongside humans. Machines, increasingly humanized, perform routine tasks and free workers to focus on supervisory activities and more creative, strategic and innovative tasks.
- **Customization:** AI enables hyper-personalization, allowing products and solutions tailored to customers' individual needs and preferences.
- **Intelligent and adaptive production:** AI makes production systems smarter and more adaptable, going beyond simple automation towards real-time decision-making and self-regulation.
- **Predictive maintenance:** AI analyzes data to predict equipment failures, enabling proactive maintenance that minimizes downtime and improves overall efficiency.
- **Resilience:** AI-driven adaptive production lines can respond in real time to market changes, resource availability and unexpected disruptions, improving supply chain resilience.
- **Sustainability:** AI optimizes resource consumption, reduces waste and facilitates the integration of renewable energy sources and circular economy principles into production systems.



INDUSTRY 5.0 – IMPACT ON INDUSTRY 4.0

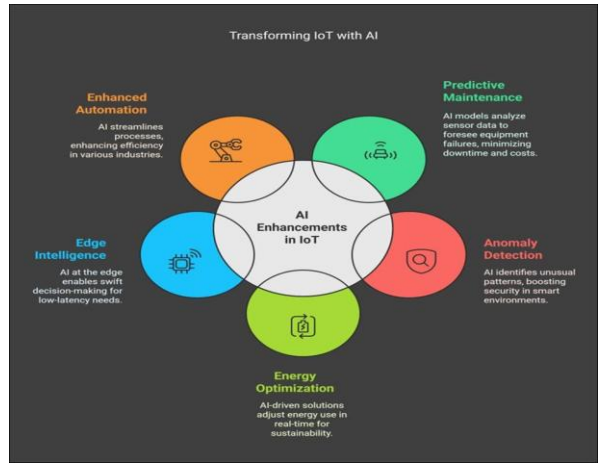
Cognitive Control

- **Let the machines program:** Automates code generation for applications, robot control, CNC and hybrid systems, with no need for manual coding.
- **One engine for all machines:** generates control intelligence on commercial hardware, from robots to CNC and everything in between.
- **Built for change:** adapts instantly to production shifts, product variation, minimizing downtime and manual intervention (supervision).
- **Always smarter:** continuously learns and optimizes for maximum machine performance.



Rapid Deployment

- **Faster than ever:** reduces deployment time with end-to-end, fully automated programming.
- **No more manual coding for robots/PLCs:** automates programming generation for industrial robotics**.**
- **Integrates with any line, old or new:** Integrates with legacy equipment and cutting-edge systems.
- **Accelerates commissioning and simplifies changes:** reconfigure and scale production in hours, not weeks



Smarter Manufacturing Control

- **Designed to scale across all your operations:** from a single cell to global factories.
- **Optimized for any production style:** supports both high-volume, low-variation runs and high-mix, high-variation environments.
- **Fewer errors, less rework:** consistent and precise logic execution on every machine.
- **Reusable logic, transferable intelligence:** move automation logic between machines and lines with minimal effort.
- **Connected from the shop floor to the supply chain.**

INDUSTRY 5.0 – AI INTEGRATION STEPS

AI INTEGRATION INTO PROCESSES

- 1. **Assess line needs:** identify the manufacturing processes that would benefit most from AI. Analyze current weak points (bottlenecks in the assembly process, recurring defects, downtime, etc.) to focus on priority areas, taking strategic objectives into account (lead time reduction, efficiency increase, cost reduction…).

- **2. Select suitable AI technologies:** look for solutions that can be easily integrated into existing systems and upgraded as they are rolled out. Ensure that solutions have good technical support and an active community to share best practices.

- **3. Plan and implement AI solutions:** rigorous planning, clear steps and instructions to test and deploy the AI solutions. Pilot projects in a specific production unit (assembly or test station).

- **4. Train and support teams:** AI adoption requires upskilling staff to ensure a smooth and efficient transition. Specific training is essential to ensure work teams understand and make the most of the potential of the new technologies. Engaging partners specialized in AI and manufacturing transformation can greatly facilitate adoption.

- **5. Measure results:** use key performance indicators (KPIs) to evaluate the AI implementation and adjust strategies accordingly. For example, the reduction in downtime and the improvement in product quality can be measured. Periodic analysis.





INDUSTRY 5.0 - AIRCRAFT FINAL ASSEMBLY LINE

I4.0

ASSISTED LOGISTIC



ASSISTED ASSEMBLY



ASSISTED TESTS



ASSISTED QUALITY



AI AUGMENTED HUMANS



AI AGENTS



AI COBOTS & ROBOTS



HYBRID WORKFORCE

ASSISTED MANUFACTURING

EMPOWERED AND CREATIVE WORKERS

I5.0



COLLABORATIVE MANUFACTURING
HUMANS - MACHINES- AI

COGNITIVE FACTORY

OPERATIONAL EXCELLENCE

AI PLANNING



AI LEAN



TBD



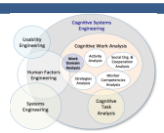
INTELLIGENT ENVIRONMENT



COGNITIVE MACHINES



COGNITIVE SYSTEMS



INTELLIGENT PRODUCTS



HYBRID WORKFORCE



HYBRID WORKFORCE

AI IN THE COMPANY

The arrival of generative AI in the company represents a paradigm shift in which humans can offload many of the tasks that add no value to them as people and focus on more human values such as creativity, innovation, equality,.. (Industry 5.0).

For example:

- **Efficiency:** AI enables people and teams to be more efficient and have more impact. It performs the most monotonous jobs and those requiring the most mental load, although work will remain people-centered so that they can contribute their creativity, which is one of the best things humans have.
- **Specialization:** generative AI opens new possibilities to transform the workforce and prepare people so that they move from being specialized in one or two fields (with the corresponding skills) to mastering multiple interconnected capabilities. This is always rewarding for the person.
- **Democratization of Knowledge:** removes hierarchies between blue and white collars, making knowledge accessible to everyone regardless of education or work experience. This facilitates human relationships, professional progress,…

This situation will lead to a hybrid workforce of humans, software and hardware, all under the umbrella of artificial intelligence.



HYBRID WORKFORCE

In Industry 5.0 the workforce is made up not only of people but also of AI-controlled software and hardware devices that make it possible to create pleasant, safe and highly efficient production environments at a lower cost. A typical workforce would consist of:

- **Augmented Humans:** through AI-controlled hardware and software, they will take on a more supervisory, creative and innovative role with less physical or mental effort.
- **AI Cobots:** robotic arms and robotic machines which, being more sensorized and AI-controlled, allow more comfortable and safer work alongside humans.
- **AI Humanoid Robots:** will replace humans in the hardest, most monotonous physical jobs,… also companionship.
- **AI Agents**: software entities that will perform many human cognitive tasks, reducing mental effort and fostering creativity and innovation in a pleasant environment.



AUGMENTED WORKER: STRENGTH, INTELLIGENCE AND SAFETY

AUGMENTED WORKER: STRENGTH, INTELLIGENCE, SAFETY AND HEALTH

DEFINITION

- **An augmented worker** is an employee whose physical and cognitive capabilities are enhanced by technology, such as augmented reality (AR) devices, artificial intelligence (AI) tools or strength-augmenting devices (exoskeletons), to perform tasks with greater precision, efficiency and safety.
- **This technological integration** improves productivity by automating repetitive or dangerous tasks and providing real-time guidance, allowing workers to focus on creativity, critical thinking and solving complex problems.

METHODS

To achieve this growth in the augmented worker's capabilities we can use:

- Digital technologies for physical, cognitive, mental and perceptual augmentation
- Wearable and ubiquitous computing to enhance human capabilities
- Exoskeletons and assistive devices for amplifying physical strength and physiological perception
- Bionic systems for amplifying and augmenting human capabilities
- Brain-machine interfaces
- Implantable technologies and interfaces
- Sensors and actuators for smart artifacts and textiles
- Sensing for context awareness
- Augmented, virtual and mixed reality
- Telemanipulation and telexistence through robotic technology
- Training and rehabilitation technology
- Technologies to improve the human experience and reduce stress



INDUSTRY 4.0 **INDUSTRY 5.0**

BENEFITS

- **Higher productivity:** technology helps reduce errors, streamline processes and optimize complex tasks.
- **Improved skills:** workers acquire new skills and knowledge, enabling them to perform more specialized and complex tasks.
- **Improved safety:** technologies such as AR and exoskeletons can guide workers and provide physical assistance, creating safer work environments.
- **Empowered workforce:** by taking over repetitive tasks, technology allows humans to concentrate on higher-value activities that require judgment, creativity and teamwork.

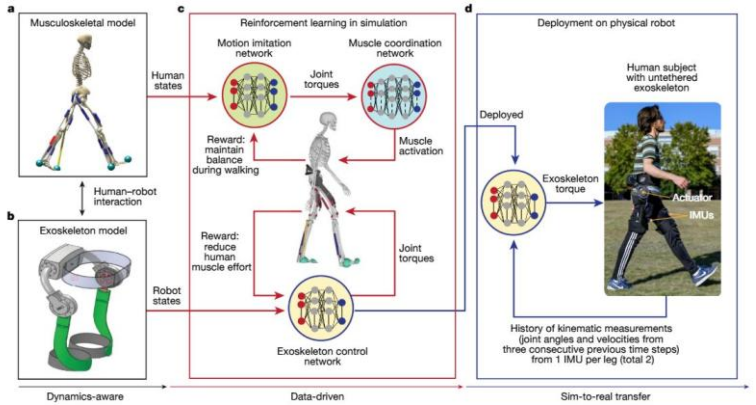
AUGMENTED WORKER – PHYSICAL STRENGTH

AI EXOSKELETONS

In general, exoskeletons, already deployed in Industry 4.0, reduce the risk of injury, increase productivity and extend working life in physically demanding jobs.

However, introducing artificial intelligence into the control system significantly improves their behavior and makes them more adaptable in an Industry 5.0 environment, as follows:

- Using AI and a large amount of data collected in a variety of real work environments, it understands how people lift, carry, bend and move in the context of their actual jobs, enabling smoother and more natural movements.
- In addition, embedding AI in the system allows it to keep learning and, as it is used, adjust its behavior to each user and their specific tasks, becoming even more a part of the body.
- It is like reproducing the way the human brain works. Thus, anything done with the body, the machine can also do. For example, AI exoskeletons feature new algorithms capable of assisting on slopes and stairs, and they also incorporate vision sensors.
- Incorporating vision sensors will further improve the device's ability to interact with its environment, providing an additional layer of safety and functionality.





WANDERCRAFT EXOSKELETON



GERMAN BIONIC: AI EXOSKELETON EXIA

AUGMENTED WORKER - INTELLIGENCE

INTELLIGENCES

Augmented intelligence (AuI): is an approach that combines human intelligence (HI) with artificial intelligence (AI) to improve and enhance human cognitive capabilities and decision-making, rather than replace them.

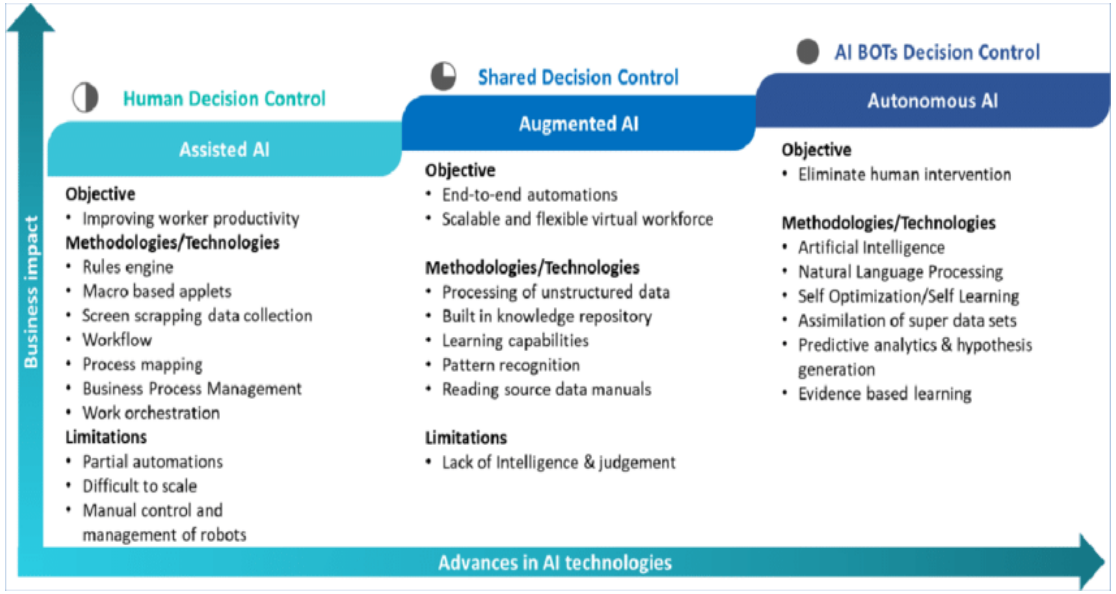
It differs from traditional AI in that it does not seek full automation of processes but a collaboration in which the machine assists humans by providing data analysis, information and tools, allowing people to focus on creativity, intuition and ethical judgment.

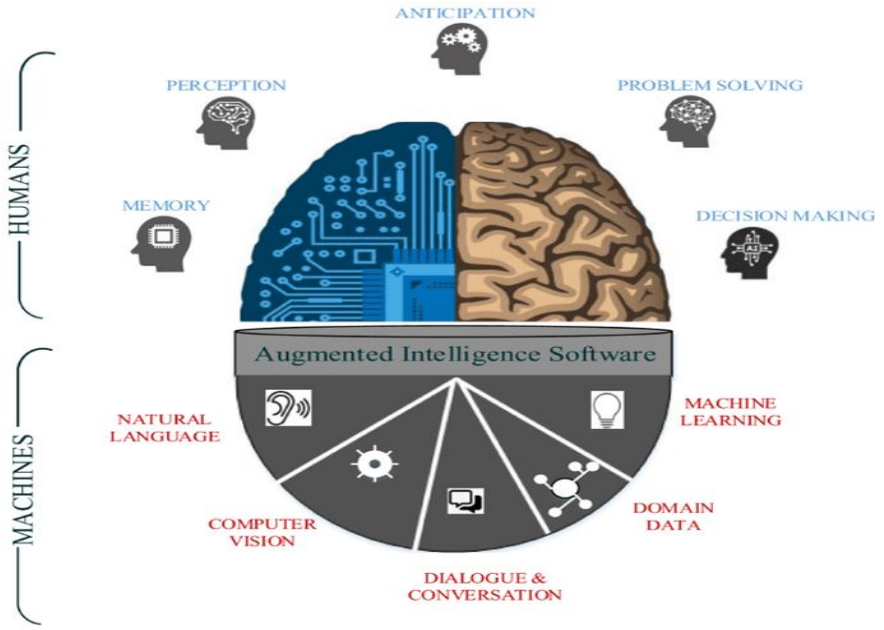
This synergy results in greater productivity, efficiency and capacity for innovation.

Depending on how artificial intelligence interacts with the operator, we can define three types:

- **Assisted Intelligence:** technology that helps operators perform tasks more efficiently and accurately (e.g. language translator, data search engine).
- **Augmented Intelligence:** more active collaboration between the operator and AI, with the latter helping in decision-making (e.g. chatbots)
- **Autonomous Intelligence:** involves autonomous control over the process (e.g. self-driving car).

In an Industry 5.0 model, **augmented intelligence** is most relevant because it leverages the best of the human and artificial worlds.

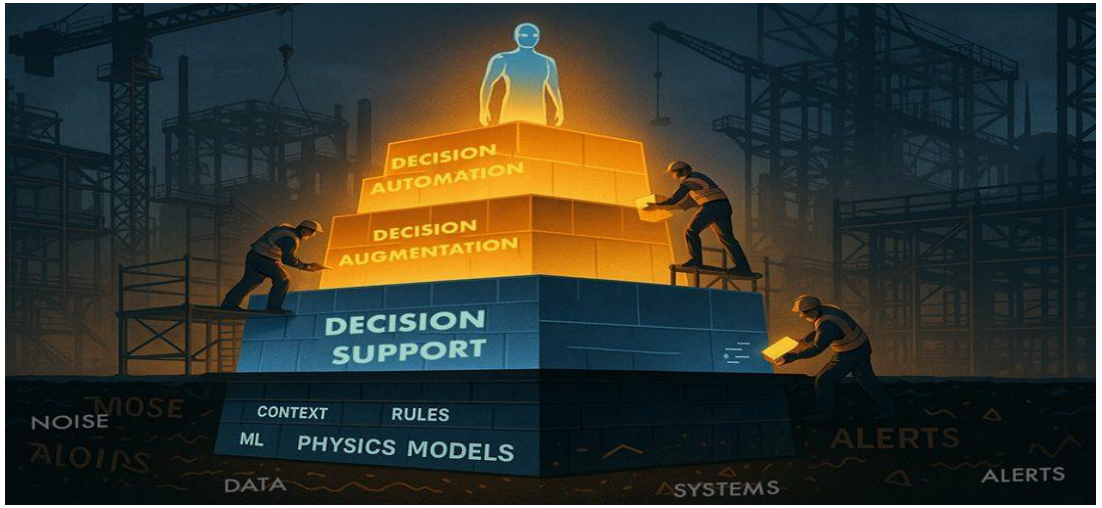




AUGMENTED WORKER - INTELLIGENCE

OPERATOR AUGMENTED INTELLIGENCE APPLICATIONS

- **Work Organization:** assists team leaders in selecting the work to be done (feasible and optimal operations according to the theoretical sequence and current production circumstances (precedences, shortages, incompatibilities,..)). It also helps with the optimal distribution of work according to the operators' situation (availability, skill, physical, mental,..).
- **Operator-Environment Assistance**: assists the operator by indicating the production status of the line (simultaneous operations in progress, incompatibility of work zones on the aircraft, specific environmental safety situations,....)
- **Operator Assistance - Operation Execution**: helps the operator by indicating the materials and tools needed to carry out the work order. It also helps if any training is needed to perform the job and provides information on other executions of the operation on other aircraft, flagging issues such as non-conformities, defects, faulty documentation,.. It also assists with safety matters related to each operation. It assists with languages, hands-free voice control,…. It also assists with closing the operation in corporate systems…
- **Operator Assistance - Operation Feedback:** AI will record anomalies during execution and, together with historical data, propose changes to the content of the order (materials, tools, steps,..) or even actions on the materials or the associated tools and industrial resources. These changes can reach, in real time, the production engineer responsible for the activity or the departments related to the problem (logistics, industrial services,..). At the operator's request it also records expertise information that may need to be added to the operation. In turn, once the operation is completed, it will show the associated performance compared to previous executions and, based on the observed data, will propose whether additional or specific training is needed or recommend actions regarding the operator and operation (e.g. depending on age, certain operations may not be suitable).



AUGMENTED INTELLIGENCE BENEFITS

- **Higher Productivity:** by providing workers with the tools and information they need, Industry 5.0 technologies can significantly improve efficiency and productivity.
- **Better Decision-Making:** AI-based tools can analyze large amounts of data and provide insights that allow workers to make more informed decisions.
- **Greater Safety:** augmented reality and other technologies can help reduce the risk of accidents by providing workers with real-time safety information and guidance.
- **Greater Job Satisfaction:** by empowering workers and making their jobs more engaging, Industry 5.0 can generate greater job satisfaction and retention.
- **More Resilient Operations:** predictive maintenance capabilities help anticipate and respond to disruptions, increasing the resilience of operations.

AUGMENTED WORKER - TRAINING

OBJECTIVE

AI-based training leverages artificial intelligence to create personalized, adaptive and efficient learning experiences.

This approach analyzes employee data to identify individual needs and preferences, adapting training content and delivery accordingly.

AI can also predict future skill needs, personalize learning paths, automate administrative tasks and provide real-time feedback and support.



FEATURES

- **Virtual Teacher:** AI acts as content generator, instructor and assessor, reducing costs and making training resources available 24 hours a day.
- **Continuous and Instant Training:** instant answers to queries, explanations and guidance for operators through training materials. A smooth, uninterrupted learning experience.
- **Improved Content Quality:** training materials updated in real time both with new information (new aircraft versions or new systems) and with information obtained from previous executions of operations (provided by other operators). Translation and processing of information into the learner's language. Tailored multimodal content (video, audio,..)
- **Personalized Methodology and Content:** Training is adjusted in both content and format to present and future needs, personal situation, prior knowledge, execution history, previous performance,.. This level of personalization not only improves the learning experience but also improves knowledge retention and the application of new skills in the workplace while optimizing costs.
- **Adaptive assessments:** personalized assessments adapted to the answers provide a more dynamic way of evaluating learner progress and ensure the right challenge is set to test their competencies.
- **Personalized metrics:** With data generated from operator interactions, AI can provide valuable metrics and analysis on their performance, engagement level and knowledge gaps.

AUGMENTED WORKER – OCCUPATIONAL HEALTH AND SAFETY

OBJECTIVES

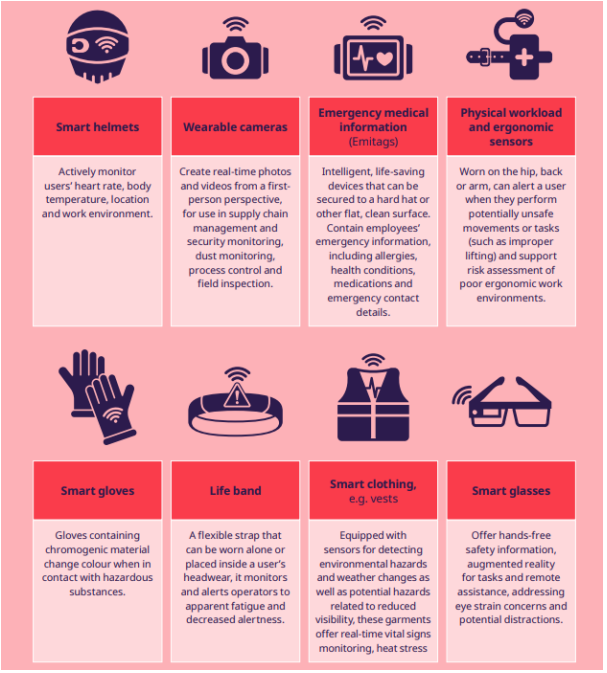
From a safety point of view, it involves making efforts and creating organized procedures to identify occupational risks and reduce accidents and exposure to harmful situations and substances. It also includes training staff in accident prevention, accident response, emergency preparedness and the use of protective clothing and equipment.

From a health point of view, it involves developing safe, high-quality, environmentally friendly processes, work practices and systemic activities that prevent or reduce the risk of harm to people in general, operators or patients.

From an environmental point of view, it involves creating a systematic approach to comply with environmental regulations, such as waste management or atmospheric emissions.

The benefits of AI in workplace safety are significant: by reducing human error and providing actionable information, AI helps the plant improve regulatory safety compliance, minimize downtime caused by incidents and reduce associated costs.

As these technologies become more accessible, AI is expected to play an increasingly vital role in creating safer workplaces across various sectors.



Industry 4.0 Wearable Technologies

AI TECHNIQUES

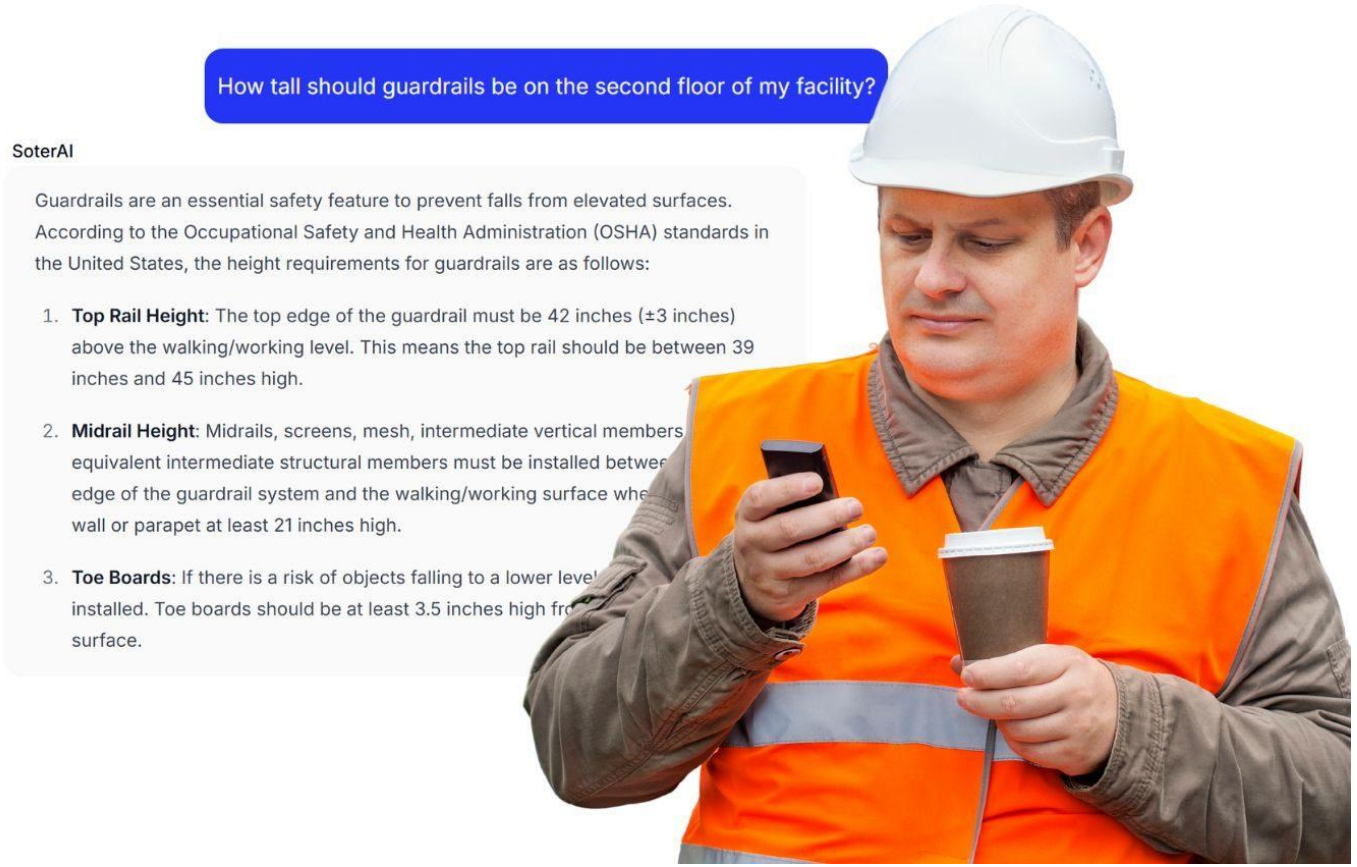
- Artificial intelligence (AI) is becoming an indispensable tool for improving workplace safety through:
- **Machine learning algorithms and data analysis:** AI can identify hazards, predict incidents and optimize overall safety management systems.
- **Predictive analytics:** AI systems can analyze historical safety data along with real-time information to anticipate risks. For example, sensors located in work areas can detect hazardous conditions and generate alerts before an accident occurs.
- **Safety training:** through virtual simulations, workers can experience realistic but risk-free work scenarios, allowing them to develop better hazard awareness and skills.
- **Compliance Checking:** AI uses cameras and software to monitor operator behavior and identify unsafe practices, such as not wearing personal protective equipment (PPE).
- **Safety Recommendations:** AI can suggest actions to correct compliance gaps, both in documentation and in actual practices on the shop floor.



AUGMENTED WORKER – OCCUPATIONAL HEALTH AND SAFETY – OHS AGENT

OHS AGENT

- Based on internal safety regulations, procedures and policies, AI provides contextual and reliable answers to safety and compliance questions in seconds.



AUGMENTED WORKER – OCCUPATIONAL HEALTH AND SAFETY - HAZARDS

HAZARD IDENTIFICATION

- From an image (photo or video) of the site and the applicable safety regulations, elements with potential risk can be found.
- Corrective actions can also be determined

SoterAI

Executive Summary

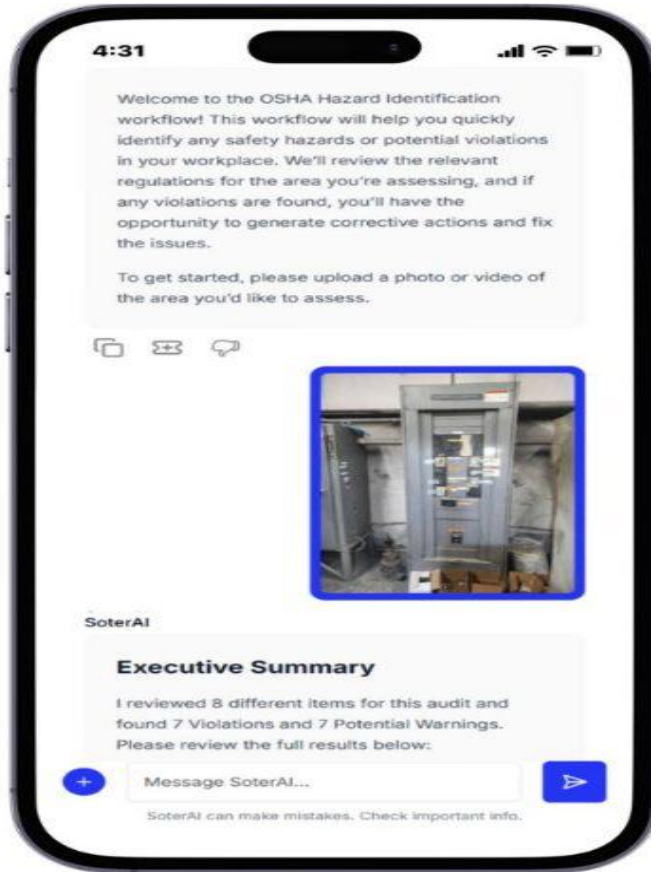
I reviewed 10 different items for this audit and found 2 Violations and 5 Potential Warnings. Please review the full results below:

Hazards

- Violations
- **Hazard Name:** Falling loads and materials
- **Hazard Name:** Excavation and trench safety
- Warnings

SoterAI

View document

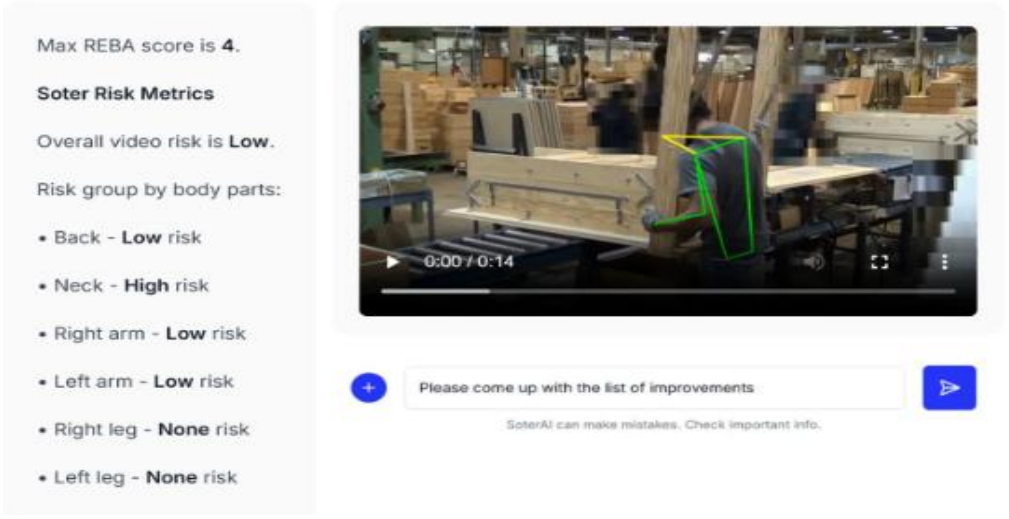


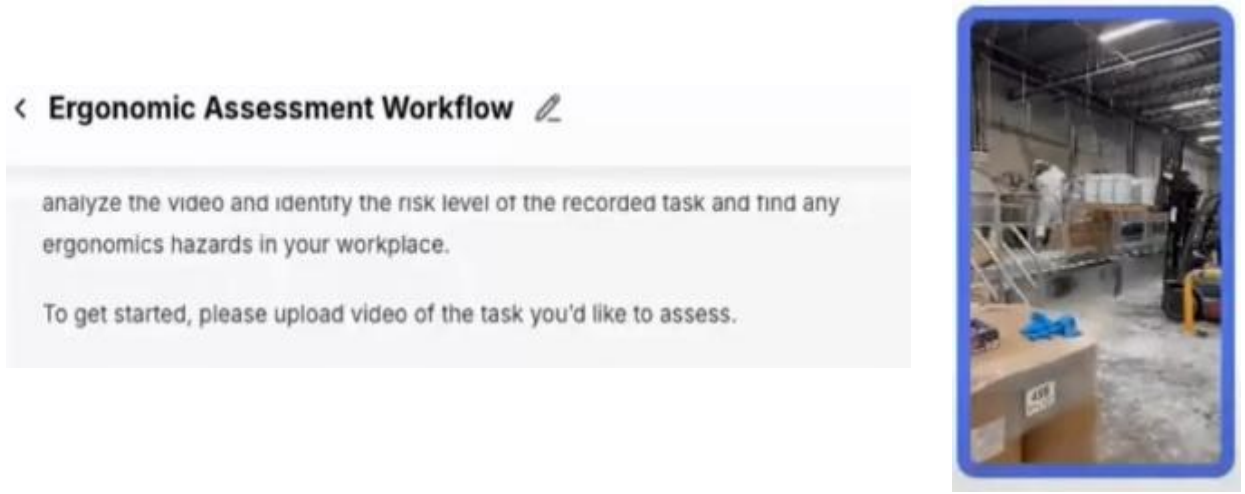
AUGMENTED WORKER – OCCUPATIONAL HEALTH AND SAFETY - ERGONOMICS

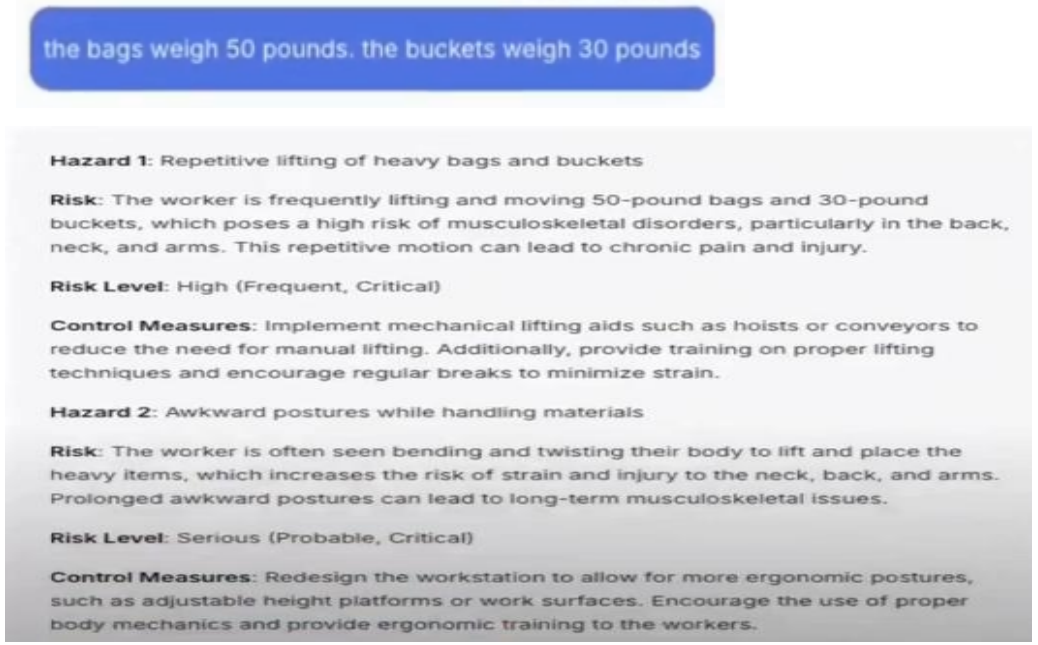
ERGONOMIC ASSESSMENT

- From videos of tasks in the workplace, AI instantly generates reports with high-risk postures visually highlighted in the recording.
- It also indicates corrective actions and improvements without the need for specialists or long lead times, doing the work efficiently and accurately in just minutes.

Ergonomic Assessment through Computer Vision



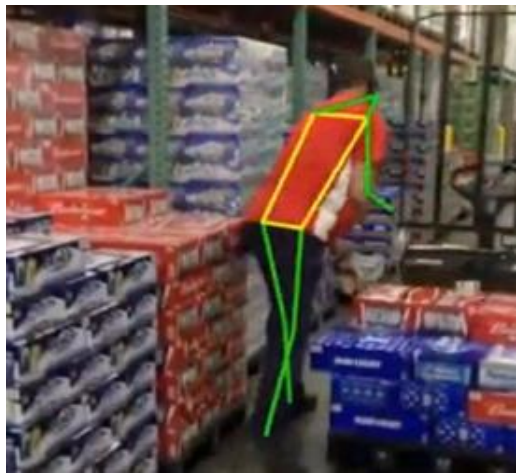


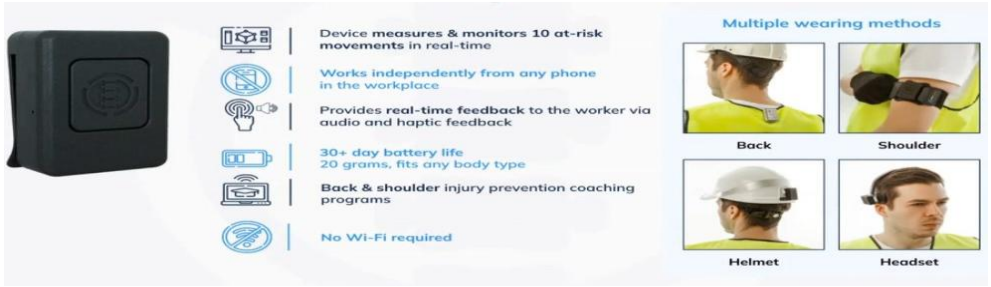


AUGMENTED WORKER – OCCUPATIONAL HEALTH AND SAFETY - ERGONOMICS

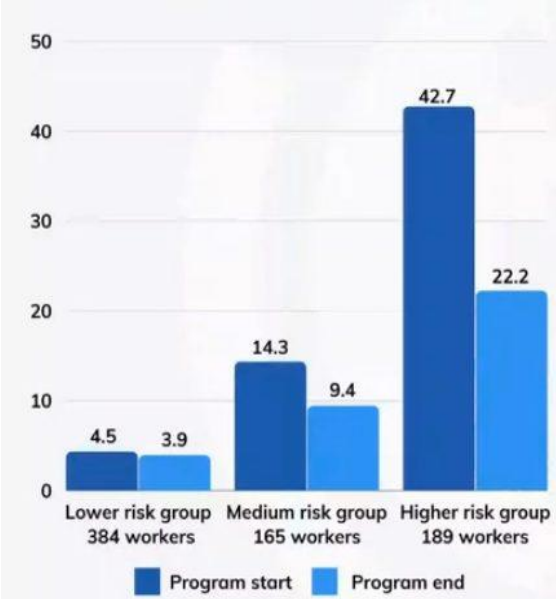
ERGONOMICS ON THE ASSEMBLY LINE (SAFETY OPTIMIZATION)

- Identification of Ergonomic Risks:
- Operator education:
- Changes in operators
- Evaluation of Interventions
- Barriers to ergonomic improvements









Risk group	Hazards per hour - start	Hazards per hour - end	Hazards per hour improvement, %
Lower risk group (384 workers)	4.5	3.9	13.3
Medium risk group (165 workers)	14.3	9.4	34.2
Higher risk group (189 workers)	42.7	22.2	48.0

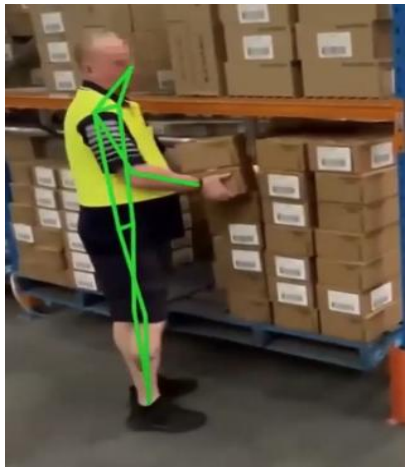
Higher risk group users - users who have at least 20 H/H during baseline period.

Medium risk group users - users who have between 10 H/H and 20 H/H during baseline period.

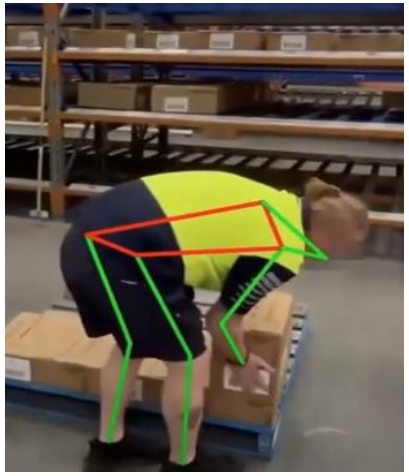
Lower risk group users - users who have less than 10 H/H during baseline period.

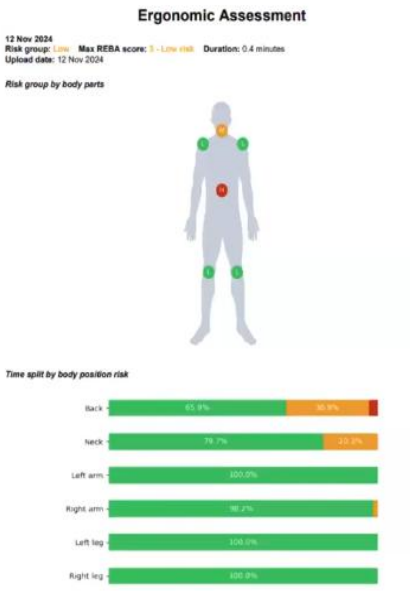
AUGMENTED WORKER – OCCUPATIONAL HEALTH AND SAFETY - ERGONOMICS

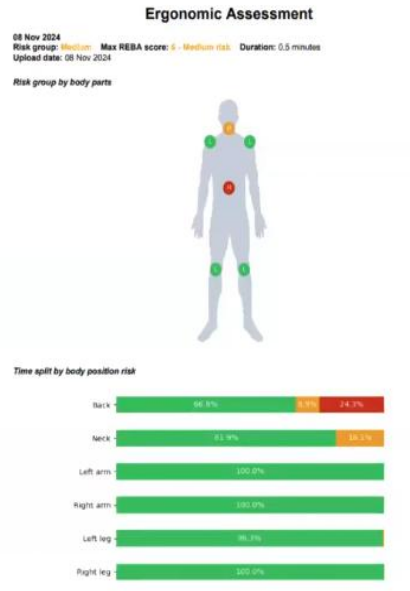
Ergonomic Assessment through Computer Vision



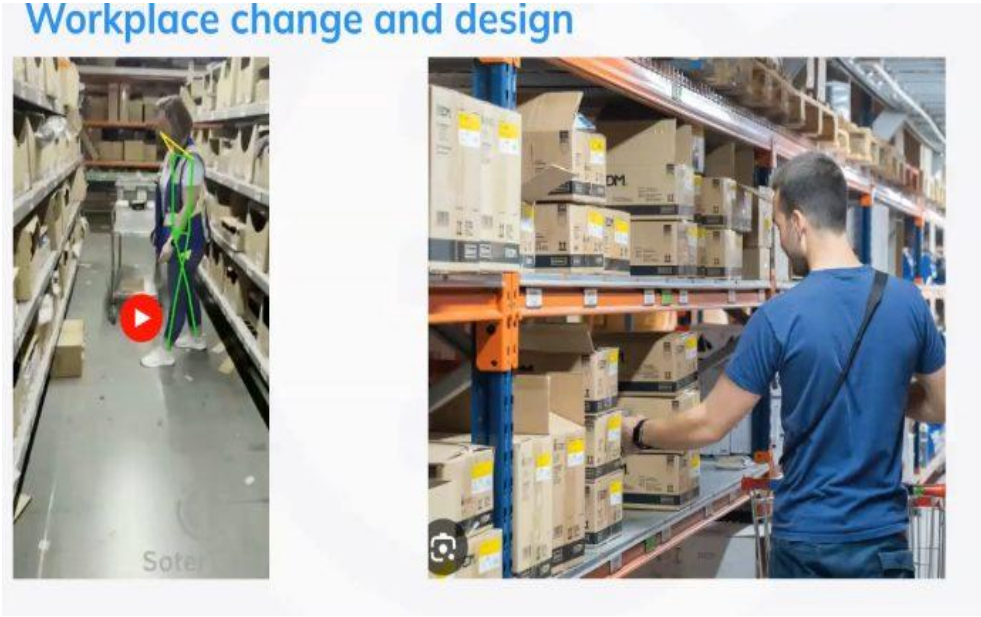












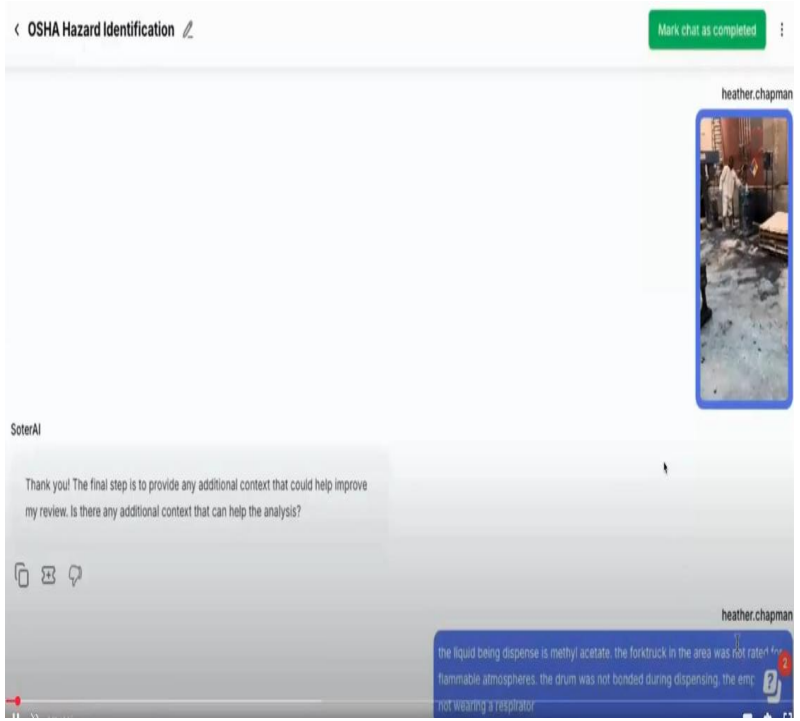
AUGMENTED WORKER – OCCUPATIONAL HEALTH AND SAFETY - ERGONOMICS

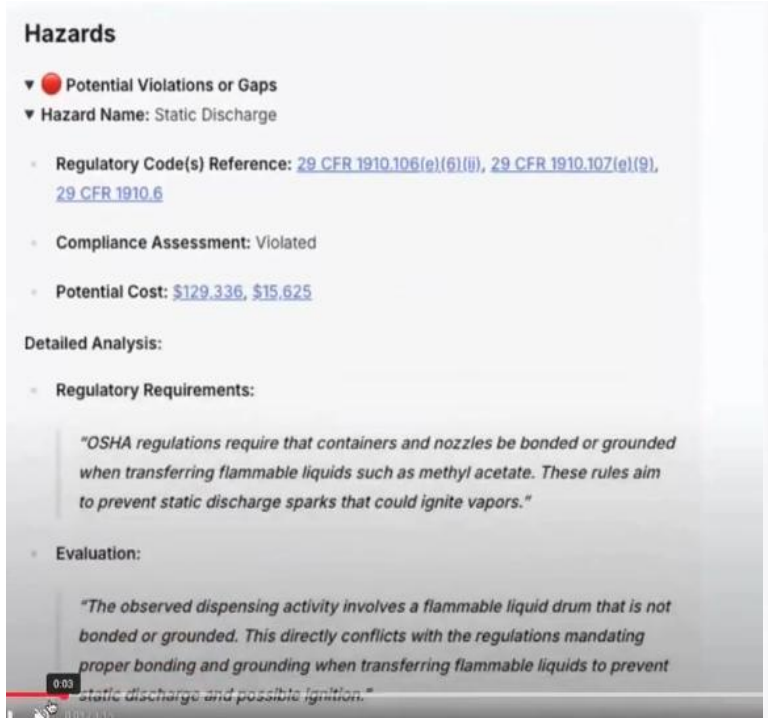


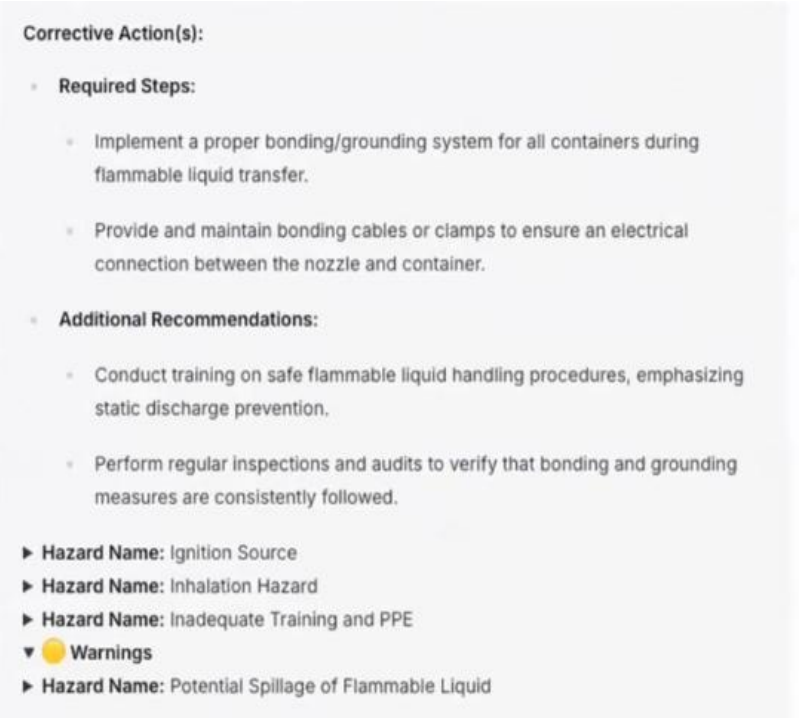


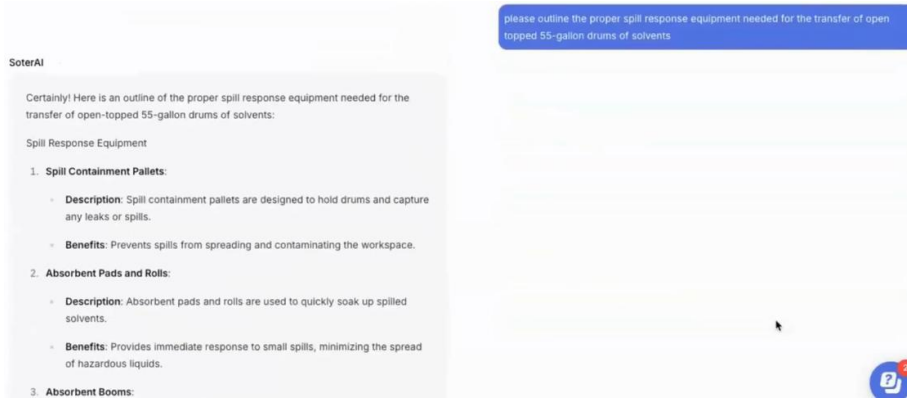
AUGMENTED WORKER – OCCUPATIONAL HEALTH AND SAFETY - REGULATIONS

Identification of work area non-compliance risks from video images: Corrective Actions and Recommendations on demand









AUGMENTED WORKER – OCCUPATIONAL HEALTH AND SAFETY - REGULATIONS

AI Assessment and Recommendations on the Plant Health and Safety Manual

please do a policy gap analysis and identify any compliance gaps and suggest recommended content improvements

Overview of Key Findings

Overall, Company C's Safety & Health Manual covers many foundational requirements, but several elements required by MIOSHA's Hazard Communication (HCS / "Right To Know") provisions 29 CFR 1910.1200 and parallel Michigan rules e.g., MCL 408.1014a et seq. are either missing or need clarification. Below are major gaps and concise improvement suggestions.

Identified Compliance Gaps

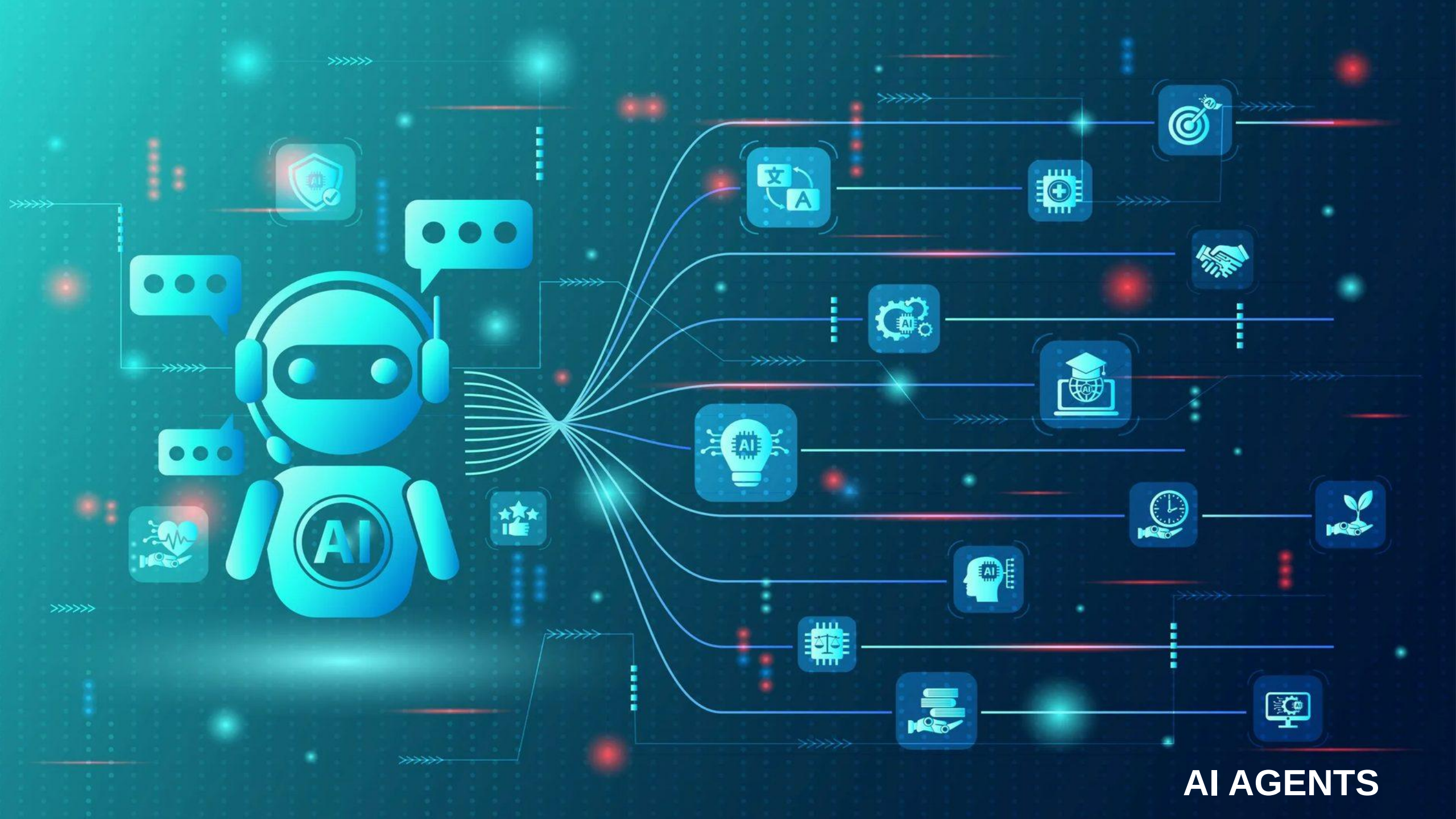
1. Hazard Communication Labels and Piping
   • Current policy does not clearly outline a GHS-aligned container labeling system or address unlabeled pipes (required by 29 CFR 1910.1200(f)).
2. Employee Right-to-Know and Non-Routine Tasks
   • There is no explicit process for informing employees of chemical hazards for non-routine tasks, nor are there clear statements on their right to obtain SDSs via the state if the employer cannot provide them (MCL 408.1014b).
3. Posting Notices of New or Revised SDS
   • While SDS management is mentioned, the policy lacks a requirement to post notice within five working days of new/revised SDS availability (MCL 408.1014k).
4. Anti-Retaliation Provisions
   • No explicit statement assures employees they are protected from discrimination for reporting safety concerns or seeking hazard information (MCL 408.1054).

- **Policy Reviews & Updates**
  - Scheduled
  - Compliance
  - Cross-referencing
- **Training & Document Generation**
  - New Programs
  - Outline
- **Current Uses**
  - Environmental Program
  - ISO 14001
  - Annual Policy Reviews
  - Corporate EHS Handbook

Recommended Short-Form Improvements

1. Implement GHS Labeling System
   • Detail how primary and secondary containers will be labeled (pictograms, signal words, hazard statements) and clarify pipe labeling procedures.
2. Add Non-Routine Task Communication Steps
   • Insert a short section explaining how employees get hazard information before starting infrequent or special tasks (e.g., cleaning reactors).
3. Post SDS Updates Promptly
   • Require posting that a new or updated SDS is available within five working days and keep the notice posted for at least ten days.
4. Include Employee Rights and Anti-Retaliation Language
   • Explicitly state that employees can request SDSs (including from the state if needed) and are protected from retaliation when reporting hazards.

By integrating the above elements into the existing Safety & Health Manual, Company C will address key MIOSHA HCS requirements, reinforce a strong safety culture, and reduce compliance risk.


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AI AGENTS

AI AGENTS - GENERAL

DEFINITION

An AI agent is a software system that uses artificial intelligence to autonomously perceive its environment, reason, plan and act to achieve predefined goals without constant human intervention.

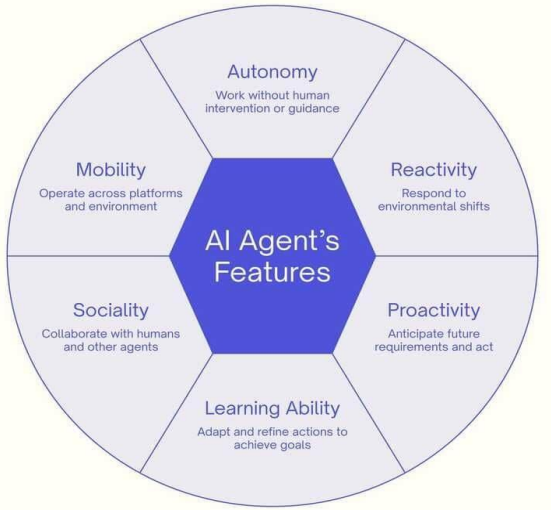
Using techniques such as machine learning, natural language processing and large language models (LLMs), agents can process multimodal inputs, learn over time and coordinate with other agents to execute complex, multi-step tasks and workflows.

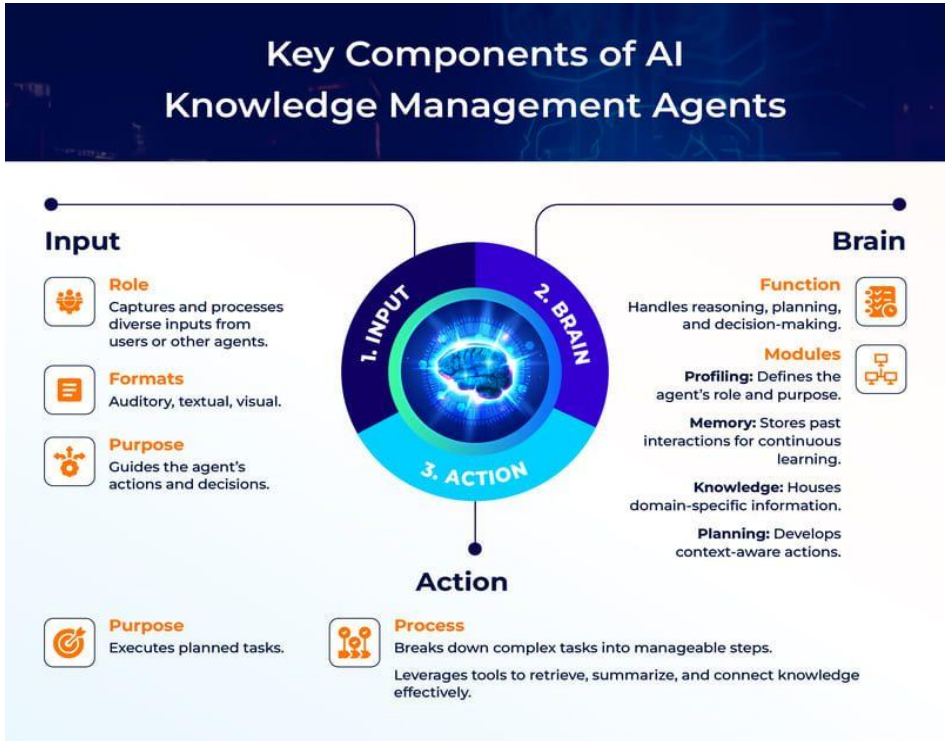
While a standard generative AI model (LLM) can process a request and generate a response, an AI agent goes further by orchestrating the use of multiple tools, planning complex actions and making decisions to fulfil a higher-level goal.

KEY FEATURES

- **Perception:** AI agents collect and process data from their environment through various interfaces, such as text, voice or images.
- **Reasoning and planning:** they use AI models to analyze information, make decisions and design workflows to achieve their goals.
- **Autonomy:** once assigned a mission, an AI agent can act independently, without the need for continuous human guidance.
- **Action:** agents execute tasks by interacting with their environment, using tools or calling APIs to complete complex workflows.
- **Learning:** many AI agents can adapt and improve their performance over time by learning from new data and feedback.
- **Context and memory:** advanced agents have memory to maintain context and integrate information from various sources, enabling more sophisticated interactions and decision-making.

https://xmpro.com/agentic-ai/

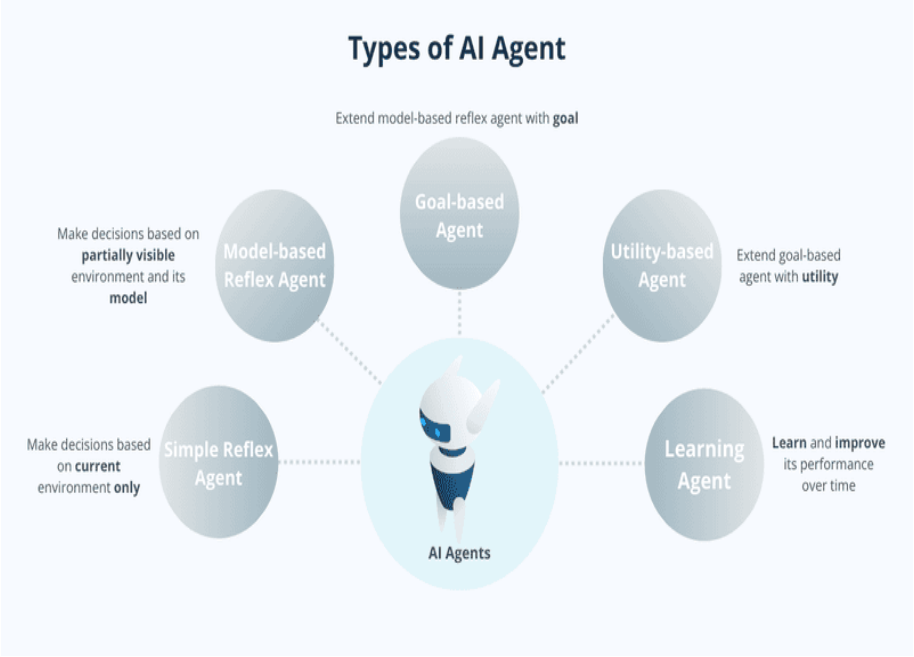


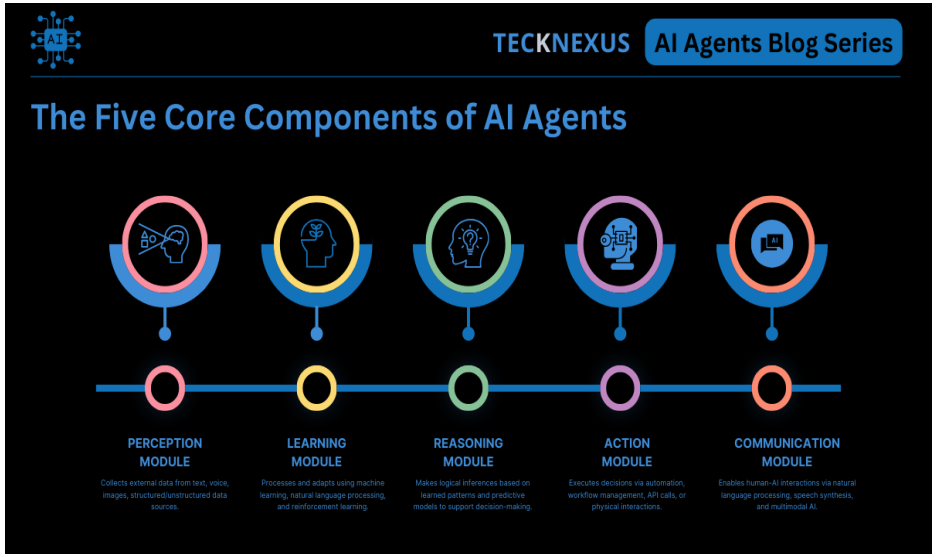


AI AGENTS - TYPES

TYPES

- **Simple Reflex Agents**: operate on the "condition-action" principle and respond to their environment according to simple pre-programmed rules, like a thermostat that turns on the heating at exactly 8 p.m. every night. The agent retains no memory, does not interact with other agents without information and cannot react appropriately to unexpected situations.
- **Model-Based Reflex Agents:** use their cognitive capabilities and memory to create an internal model of the world around them. By storing information in memory, these agents can operate effectively in changing environments, but they are still limited by pre-programmed rules. For example, a robot vacuum cleaner can detect obstacles while cleaning a room and adjust its path to avoid collisions. It also remembers the areas it has cleaned to avoid unnecessary repetition.
- **Goal-Based AI Agents:** are governed by one or more specific goals. They look for the appropriate actions to achieve the goal and plan ahead before executing them. For example, when a navigation system suggests the fastest route to your destination, it analyzes different routes to find the optimal one. If the system detects a faster route, it updates and suggests an alternative route.
- **Utility-Based Agents:** evaluate the outcomes of decisions in situations with multiple viable paths. They use utility functions to measure the usefulness each action could bring. This evaluation system helps identify the ideal option: Is the best option the cheapest? The fastest? The most efficient? For example, a navigation system considers factors such as fuel savings, reduced travel time and toll costs to select and recommend the most favorable route for the user.
- **Learning Agents:** learn through concepts and sensors, while using feedback from the environment or users to improve their performance over time. New experiences are automatically added to the Learning Agent's initial knowledge base, helping it operate effectively in unfamiliar environments. For example, e-commerce websites use Learning Agents to track user activity and preferences and then recommend suitable products and services.





AI AGENTS - ARCHITECTURE

HOW IT WORKS

Upon receiving an order (goal) from a user (Prompt), AI Agents immediately start the goal analysis process, passing the prompt to the core AI model (LLM) and starting to plan actions. The Agent will break complex goals down into specific tasks and subtasks, with clear priorities and dependencies. For simple tasks, the Agent may skip the planning phase and directly improve responses through an iterative process.

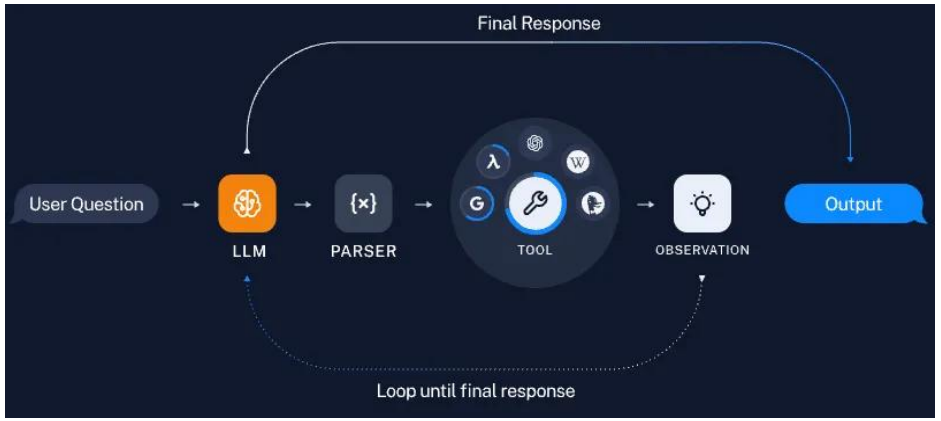
Perception: thanks to sensors, AI agents gather information (transaction data, customer interaction history) from various sources (including external datasets, web searches, APIs and even other agents). During this gathering process, the AI Agent continuously updates its knowledge base, self-adjusts and corrects errors if necessary.

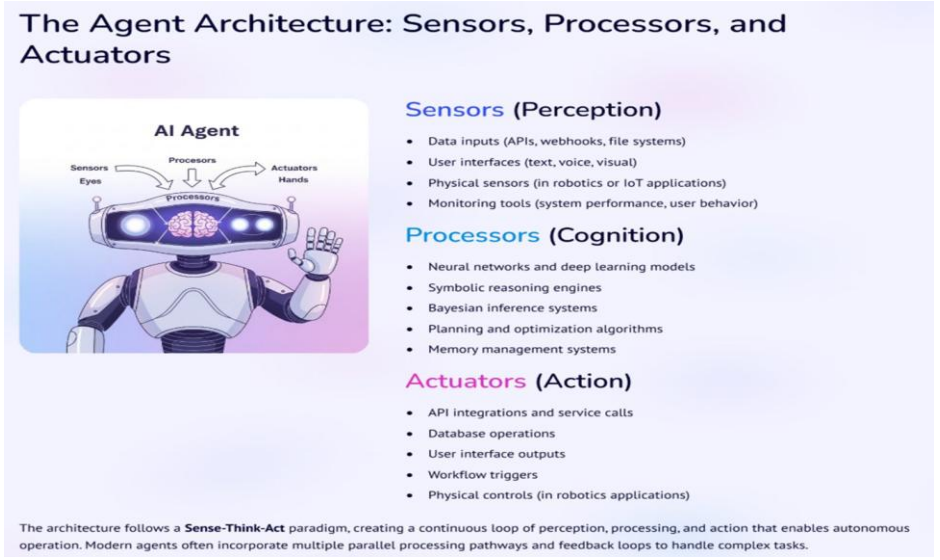
Processing: AI agents use algorithms, deep neural networks, machine learning models and artificial intelligence to analyze information and calculate the necessary actions.

In addition, during this process, the agent's memory continuously stores information (such as the history of decisions made or learned rules), and agents also use user feedback, feedback from other agents and human intervention (HITL) to adjust and improve performance over time, avoiding repeating the same mistakes.

Action: finally, through actuators, AI agents carry out actions based on their decisions. For software agents, this may consist of sending information or executing commands in systems.



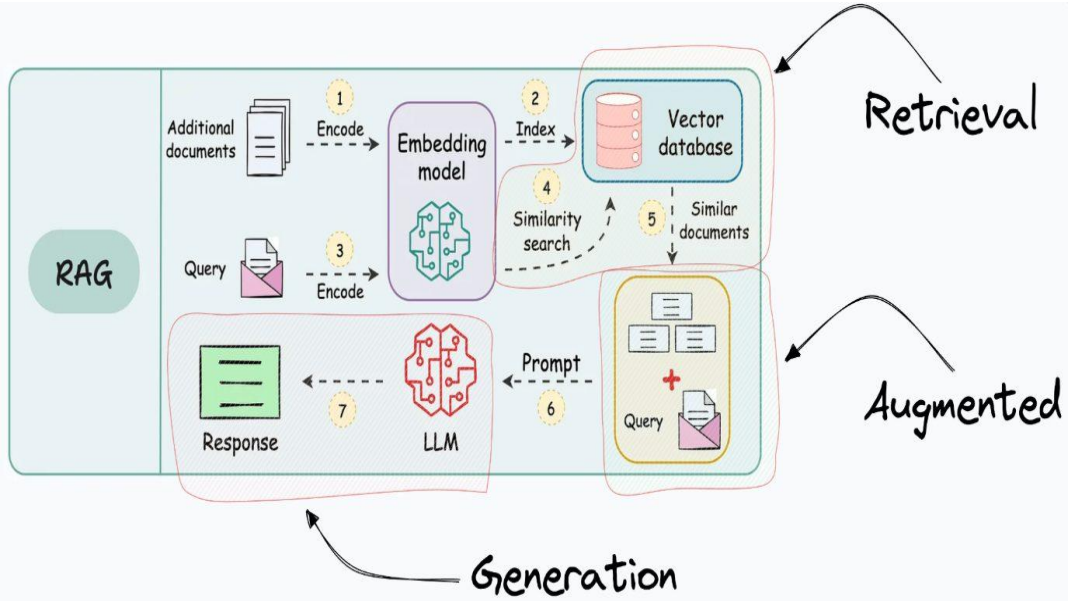




AI AGENTS - RAG

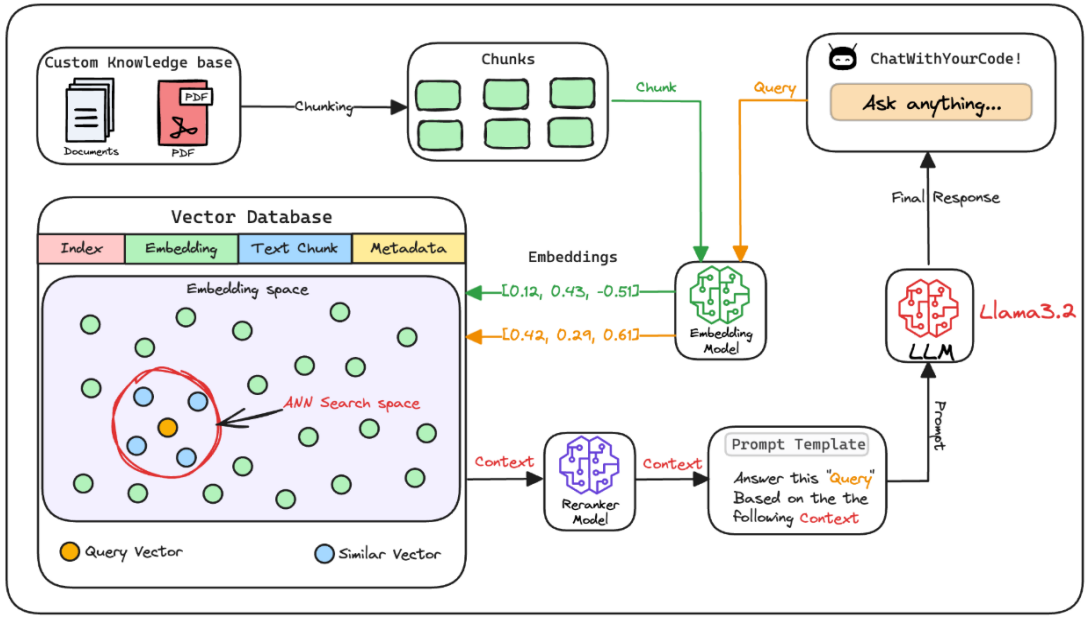
RAG DEFINITION

- **RAG:** Retrieval-augmented generation (RAG) is an effective technique that combines large language models (LLMs) with real-time data retrieval to generate more accurate, up-to-date and contextually relevant answers. A RAG agent does the following:
- **Retrieval:** the user's request is used to query an external knowledge base, such as a vector store, a keyword search or a SQL database. The goal is to obtain supporting data for the LLM's response.
- **Augmentation:** this supporting data is combined with the user's request, often using a template with formatting and additional instructions for the LLM, to create a prompt.
- **Generation:** the prompt is passed to the LLM to generate a response to the user's request.



RAG AGENT ADVANTAGES

- **Proprietary knowledge:** RAG can include proprietary information not initially used to train the LLM, such as demos, emails and documents, to answer domain-specific questions.
- **Up-to-date information:** A RAG application can provide the LLM with information from an up-to-date knowledge base.
- **Citing sources:** RAG allows LLMs to cite specific sources, enabling users to check the factual accuracy of answers.
- **Data security and access control lists (ACL):** the retrieval step can be designed to selectively retrieve personal or proprietary information based on the user's credentials.
- **The RAG architecture:** can work with unstructured or structured supporting data. The data you use with RAG depends on your use case.

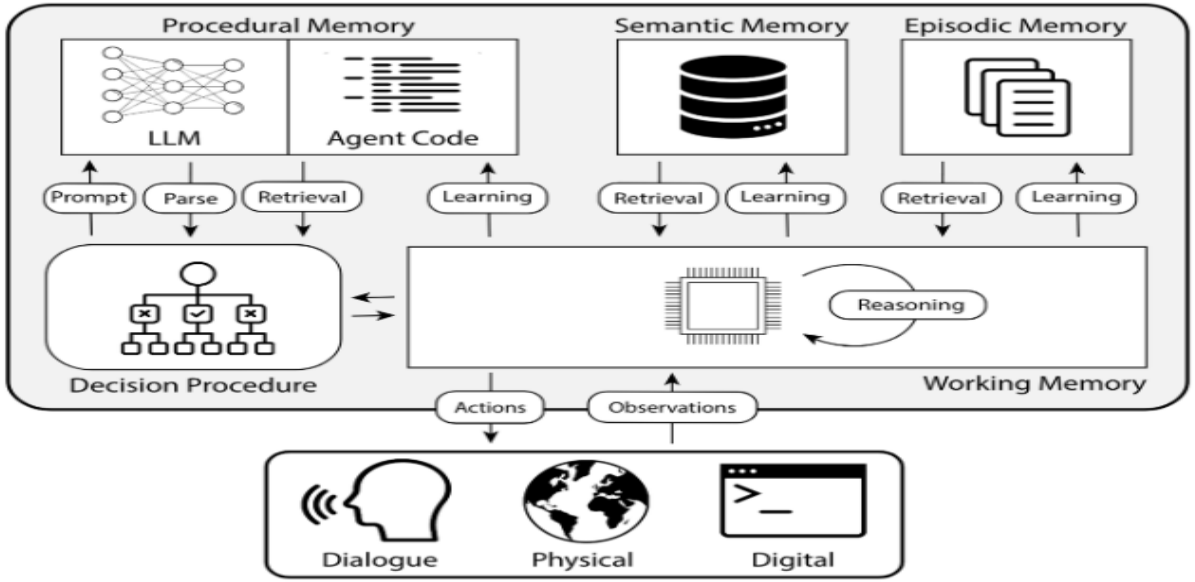


AI AGENTS - MEMORY

SHORT-TERM OR WORKING MEMORY (STM)

Temporary storage of information for a short period, usually seconds to minutes. Working memory acts as a workspace to manipulate and process this information and complete a specific task.

- **Context Window:** they have a limited context window size that acts as working memory and contains the current request and the conversation history.
- **Conversation Buffers:** keep a buffer of recent interactions, either a fixed number of interactions (ConversationBufferWindowMemory) or a fixed token length (ConversationTokenBufferMemory).
- **Conversation Summaries:** periodic summaries of the conversation history using the LLM itself (ConversationSummaryMemory or ConversationSummaryBufferMemory)



LONG-TERM MEMORY (LTM)

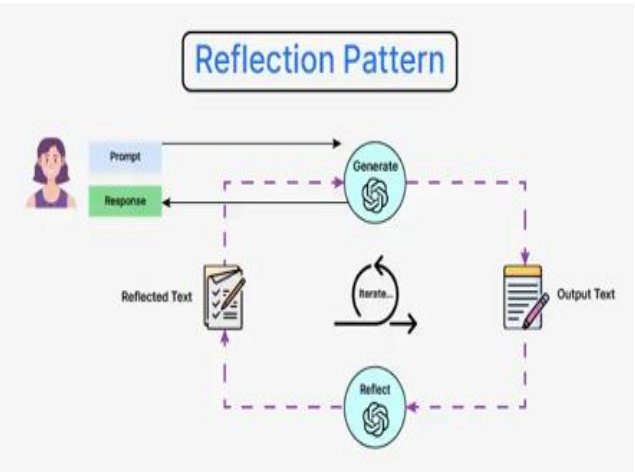
A broader category covering several types of memory relevant to AI agents:

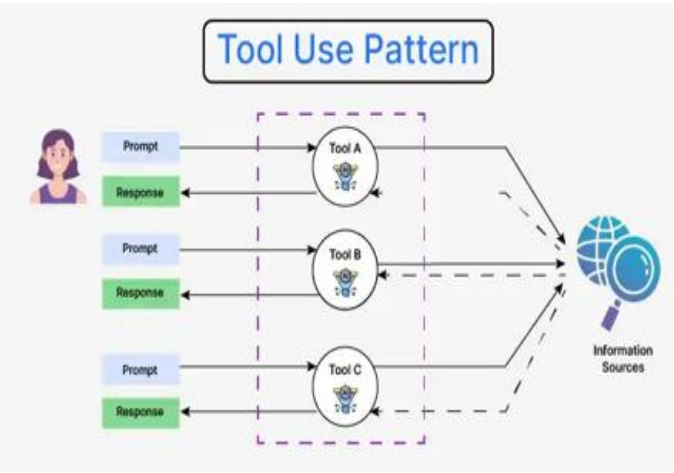
- **Semantic memory:** stores general knowledge and concepts, independent of specific events. This is factual information about the world, the relationships between concepts and the meaning of words. Semantic memory is crucial for understanding context and responding effectively to user queries.
- **Episodic memory:** stores memories of specific events and experiences. It could contain past interactions with users, allowing the agent to learn from successes and failures in similar situations. It can be implemented using relational databases, file storage or vector databases.
- **Procedural memory:** represents the agent's procedures for thinking, acting, making decisions, etc. This type of long-term memory stores practical knowledge, such as motor skills or step-by-step instructions. This could involve learning optimal strategies within a game or environment through reinforcement learning.

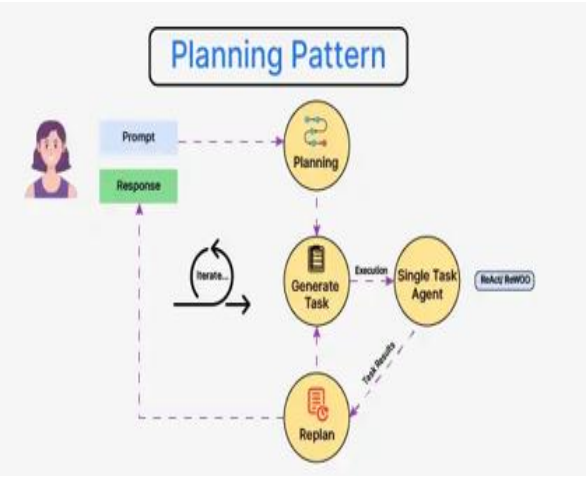
AI AGENTS – DESIGN PATTERNS

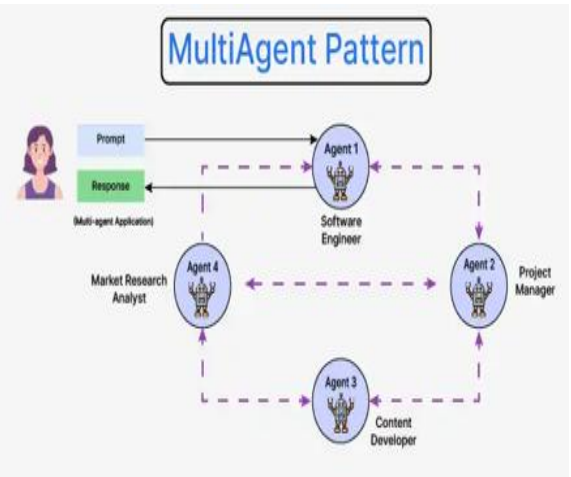
PATTERNS

- **Reflection and Self-Reflective RAG:** focuses on improving the AI's ability to evaluate and refine its own outputs. Imagine an LLM reviewing the generated content or code as if it were a human reviewer, identifying errors, gaps or areas needing improvement and offering suggestions to improve**.** The AI can repeat the reflection process as many times as necessary to achieve a refined, polished result.
- **Tool Use:** allows interaction with external tools and resources to improve problem-solving capability. Instead of relying solely on calculations or internal knowledge, it can access databases, search the web or even execute complex functions with Python. By incorporating tool use, LLMs evolve from static knowledge banks into dynamic agents capable of interacting with external systems to achieve goals.
- **Planning:** allows an LLM to break down large, complex tasks into smaller, manageable components. Planning gives the agent the ability to react to requests and strategically structure the steps needed to achieve a goal. There are two types: ReAct allows the LLM to dynamically alternate between reasoning (analyzing the problem) and acting (performing specific tasks), enabling more adaptive and flexible planning. ReWOO, on the other hand, improves the planning pattern by using an open-world ontology to guide reasoning. This means the LLM can incorporate broader contextual information and knowledge from various domains, resulting in more informed decision-making.
- **Multi-agent:** based on the concept of delegation, similar to project management in human teams. This pattern involves assigning different agents (instances of an LLM with specific roles or functions) to handle various subtasks. These agents can work independently on their tasks while communicating and collaborating to achieve a unified result. They can be Collaborative, Supervised or Hierarchical.









AI AGENTS - MULTI-AGENT

NEED

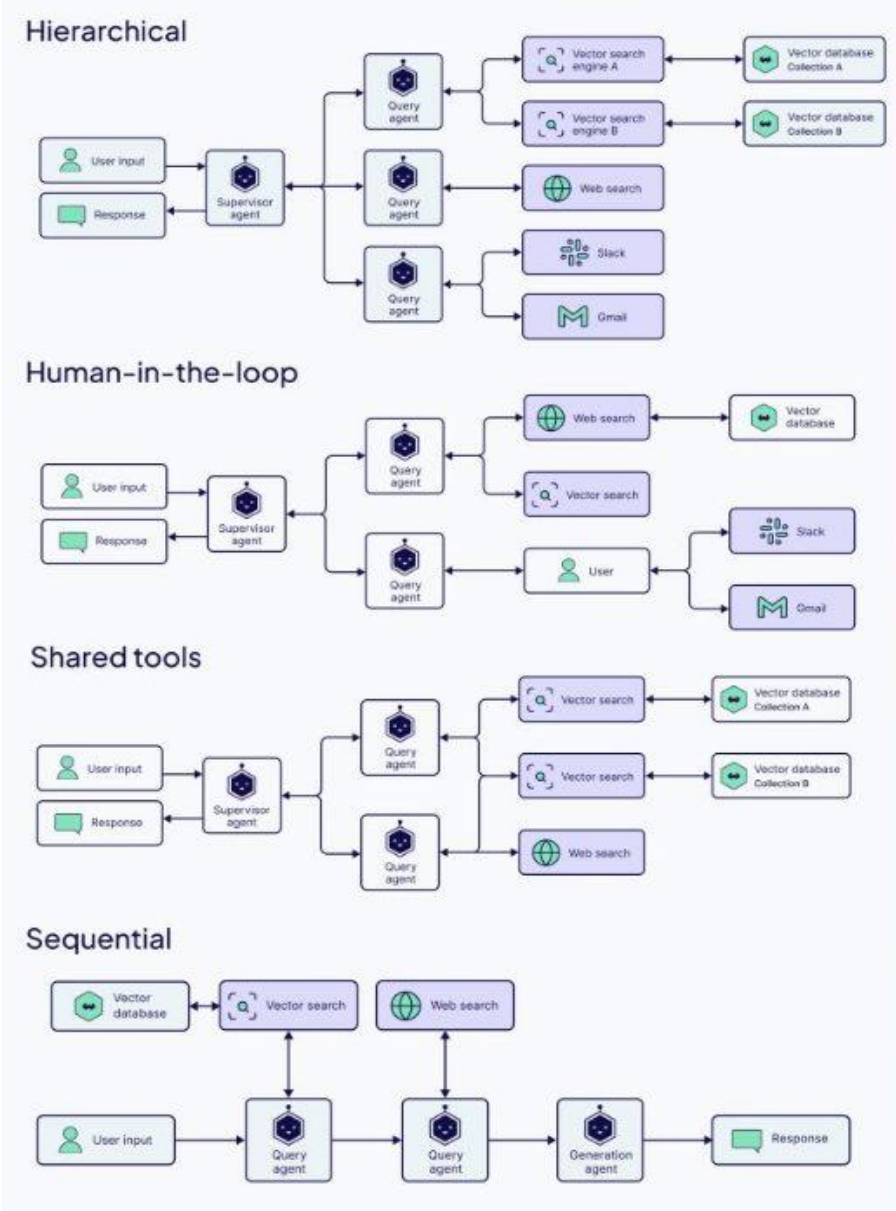
Sometimes a single agent is not enough because the problem demands scalability, specialization or parallel thinking. Examples:

- **Parallelization:** agents work on parts of a task simultaneously.
- **Specialization:** one agent is an expert in legal jargon, another in writing emails.
- **Tool independence:** Each agent can have its own tools and memory.

TYPES

Depending on the hierarchical model:

- **Hierarchical patterns (more controllable):** an orchestrator agent delegates subtasks to others, sees the big picture and controls the flow. This model is useful when tasks can be clearly decomposed, when strict control is desired, or when the agents' roles are known (e.g. summarizer, generator, verifier). Think of business workflows, toolsets, parallel processes.
- **Flat patterns (more dynamic):** agents communicate with each other as peers, without bosses. Useful when tasks require creativity or debate, or when you want agents to evaluate each other, or when there is no single "correct" answer path. Think of: brainstorming, ranking options, multi-perspective reasoning.



AI AGENTS - 2025 STATUS AND FUTURE

- By 2030, more than 60% of enterprise applications will include AI agents as standard features.
- AI copilots integrated into every workflow, from finance and HR to logistics and customer service. These copilots will not only automate tasks but also provide real-time insights and recommendations, enabling employees to work smarter and faster.
- Fully autonomous companies, where multi-agent systems will handle everything from procurement and regulatory compliance to customer interaction, with minimal human intervention.
- Instead of isolated bots, companies will orchestrate networks of collaborating agents that negotiate, share knowledge and solve problems collectively, much like digital departments operating 24/7.

Problem Type	Suggested Agent Type	Enterprise Example
Structured, rules-based	Rule-Based System	Automatically approve reimbursements under $100 or trigger invoice processing based on fixed thresholds
Human-led process, needs speed	Workflow Agent	Suggesting first-draft responses to support tickets in Zendesk, or summarizing meeting transcripts for Slack follow-up
Multi-step, repeatable, bounded	Semi-Autonomous Agent	Lead enrichment and follow-up: fetch CRM info → draft outreach → send email → log interaction
Complex, async, cross-system	Autonomous Agent	A competitive intelligence agent that monitors public news, pulls data from multiple APIs, clusters insights, and generates weekly market reports

Tools

Level 8 – Alignment → Guardrails AI, Arize, Langfuse, Helicone

Governance and oversight — enforce rules, monitor behavior, trace decisions.

Level 7 – Memory → Zep, Mem0, Cognee, Letta

Persistent memory — store context so agents remember past interactions.

Level 6 – Data Extraction → Firecrawl, Scrapy, Docling, Llamaparse

Pull in and clean data from the web or documents for AI use.

Level 5 – Embedding → Nomic, Ollama, Voyage AI, OpenAI

Convert data into vectors for semantic search and retrieval.

Level 4 – Vector DB → Pinecone, Chroma, Milvus, Weaviate

Store embeddings for fast, scalable similarity search.

Level 3 – Framework → LangChain, LlamaIndex, Haystack, DSPy

Orchestrate LLMs, tools, and data sources into workflows.

Level 2 – LLMs → Llama 4, Gemini 2.5 Pro, Claude 4, GPT-4o

The "brains" — interpret, reason, and generate responses.

Level 1 – Evaluation → LangSmith, Phoenix, DeepEval, Ragas

Test and validate AI outputs for quality and accuracy.

Level 0 – Deployment → Groq, AWS, Google Cloud, Together AI

Infrastructure — run and scale models on specialized hardware and cloud platforms.

AI Agent Frameworks (Langgraph +---)

Framework	Best For	Learning Curve	Community	Open Source
Microsoft AutoGen	Collaborative problem-solving, human-in-the-loop workflows, and dynamic group chats.	Moderate	Active (Microsoft and community-driven)	Yes
ZenML	Enterprise teams and projects require a full MLOps pipeline, governance, and compliance.	Moderate to Steep	Active (Company and user community)	Yes
CrewAI	Teams building collaborative AI systems, content generation, and research automation.	Gentle	Active and Growing	Yes
OpenAI Swarm	Prototyping, non-complex workflows, and teams already invested in the OpenAI ecosystem.	Gentle	Growing (as it's in active development)	No
Akka	High-throughput, mission-critical applications and large-scale enterprise deployments.	Steep	Very Active and Mature	Yes
TigerGraph	AI applications in finance, supply chain, and telecom that require real-time analysis of relationships.	Moderate to Steep	Active (Commercial support)	No
SuperAGI	Experimental AI projects and research applications focused on building truly autonomous systems.	Moderate	Active and Engaged	Yes
Haystack	Document-heavy applications, enterprise search, and complex Q&A systems.	Moderate	Active and Mature	Yes
GraphSAGE	Applications with constantly changing data structures, like social networks and recommendation systems.	Moderate to Steep	Active (Research and academic community)	Yes
DSPy	Developers building complex LLM applications who want to simplify prompt engineering and improve output reliability.	Moderate	Active (Stanford-led and community-driven)	Yes

AI AGENTS – MULTI-AGENT EXAMPLE – FAL OPERATOR TRAINING ACADEMY

ACADEMY

The assembly line training department, also called the "Academy", ensures that factory staff have the knowledge and skills needed to carry out the work assigned to them.

In the case of assembly operators, depending on the type of activity (electrical assembly, mechanical assembly, testing, inspection,..) they will have different theoretical and practical training.

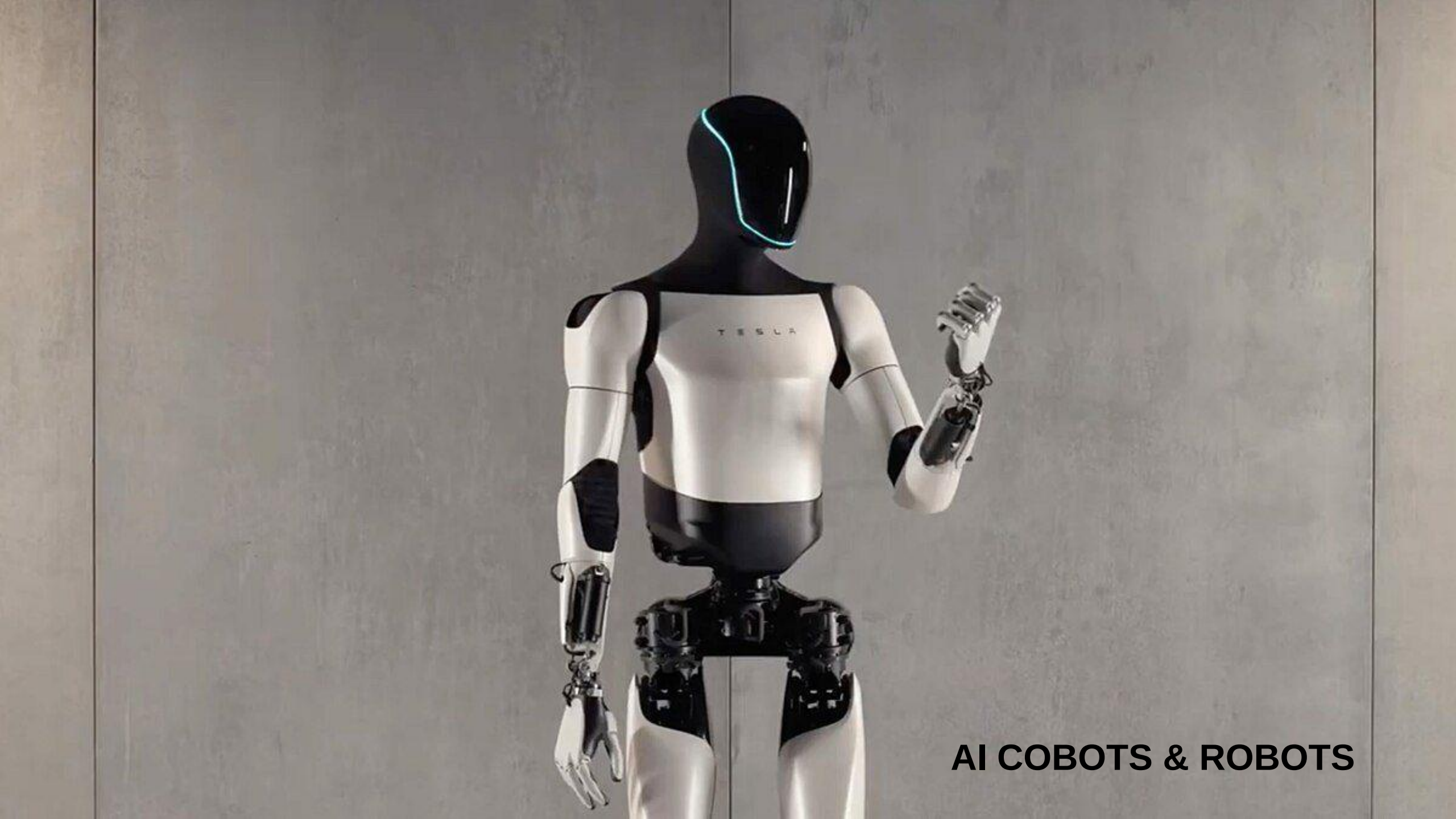
Practical training is usually OTJ ("On the Job"), carried out while doing the work. Theoretical training requires an infrastructure similar to a school or university where, given the specificity of the aeronautical business, it is necessary to prepare tailored courses (specific aircraft), specific documentation and appropriate assessment tests.

This requires a considerable specialist staff including organizers, documentation writers and teachers, which represents a cost that adds no value to the product but is essential.

Training costs can be significantly reduced if AI agents are used for the bulk of the work, leaving a reduced human staff to supervise processes and content..

AGENTS

- **Organizer Agent:** detects training needs and designs the "learning paths" for each operator, requests the creation of the course documentation from the technical author and organizes the training (operators who need it, requirements, teachers, estimated duration,..).**Output:** individual operator plan document.
- **Author Agent:** based on general aeronautical assembly documentation, techniques, standards and assembly-line-specific documentation… prepares the information in an organized, progressive way that helps operators master and remember technical knowledge of aeronautical assembly**.** It also prepares the exam for each course. **Output:** course text documents and teacher's booklet with exam questions.
- **Teacher Agent:** based on the course document generated by the technical author, and also on general and specific aeronautical documentation, helps operators with their questions and carries out knowledge assessments at the end of the course. **Output:** online technical support and online exam.



AI COBOTS & ROBOTS

COLLABORATIVE ROBOTS AND AI HUMANOID ROBOTS

Definition

A cobot (short for collaborative robot) is a robot designed to work safely alongside a human in a shared workspace.

The difference from industrial robots is:

- **Human-robot collaboration:** They work side by side with humans in the same space.
- **Safety:** They incorporate sensors and other safety features to detect human presence and react safely, such as stopping or slowing down.
- **Flexibility:** They are lightweight, adaptable and can be easily integrated into different applications.
- **Ease of use:** They are easier to program and can even learn from operators.
- **Task automation:** They handle repetitive, monotonous, physically demanding or precise tasks, such as packaging, assembly or quality control.











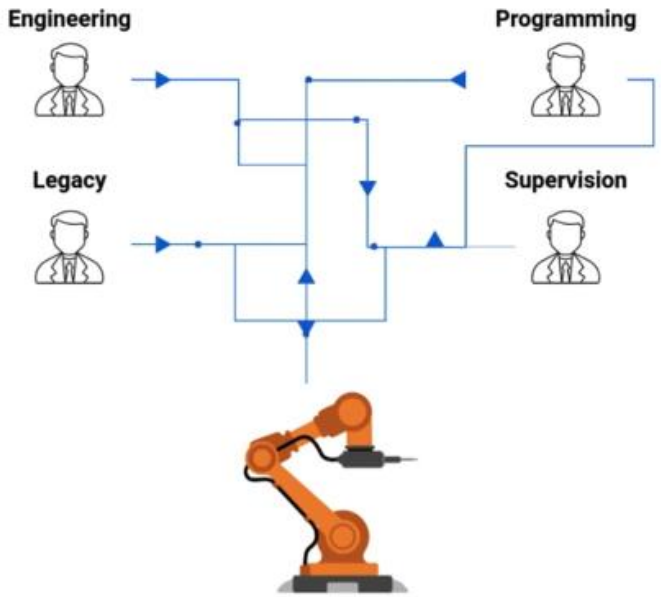


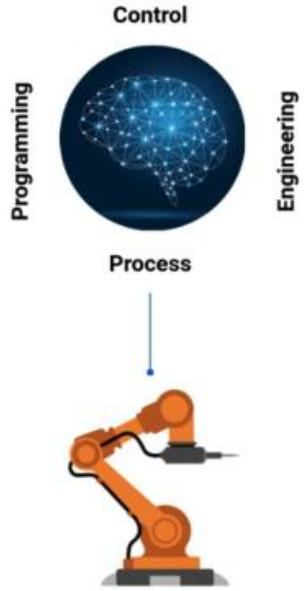
Definition

A humanoid robot is a machine designed to imitate the form and functioning of a human being, with elements such as a head, torso, arms and legs, although not all models have all of them.

- **These robots seek to** interact with humans and their environment naturally, using artificial intelligence, computer vision and locomotion systems to walk, recognize faces and communicate.
- **Types:** they are mainly classified by purpose and level of sophistication: androids are completely artificial and seek a realistic human resemblance, while others focus on specific tasks such as research, assistance or security, and may be more generic or specialized in certain human functions.

AI COLLABORATIVE ROBOT - ROBOTICS 4.0 vs 5.0





Robotic Automation 4.0

- Mechatronic Adaptation (Robotics Eng.)
- Complex Programming (Computer Eng.)
- Constant Supervision (Production Eng.)
- Rigid, non-adaptive logic (Not flexible)
- No data from previous executions (No history)

Robotic Automation 5.0

- Mechatronic Adaptation (AI-proposed)
- Autonomous Programming (Programmed by AI)
- Self-supervision (AI-based sensing)
- Real-time adaptive (AI control)
- History and feedback (AI-linked databases)

AI COLLABORATIVE ROBOT - SENSORY SKILLS

Vision

3D Object Detection

- Advanced vision sensors identify and track objects in real time, ensuring millimeter precision in complex environments.

Gesture Control

- Intuitive hand-movement recognition enables programming without coding knowledge.

Touch

Tactile Force Sensor

- Detects pressure changes for precise hand guidance and force feedback, enabling delicate handling of objects.

Dynamic Payload Adjustment

- Automatically detects and adapts to variable loads, optimizing performance without manual reconfiguration.

Hearing

3D voice recognition

- 360° microphones process commands from any direction, enabling natural communication in noisy industrial environments.

Multilingual understanding

- Processes instructions in several languages, removing language barriers in global environments.

Cognitive Intelligence

Multimodal Integration

- Combines visual, voice and tactile information through on-device neural processing to achieve real-time situational awareness without relying on the cloud.

Self-optimization

- Continuously refines performance through reinforcement learning and adapts autonomously to changing demands.



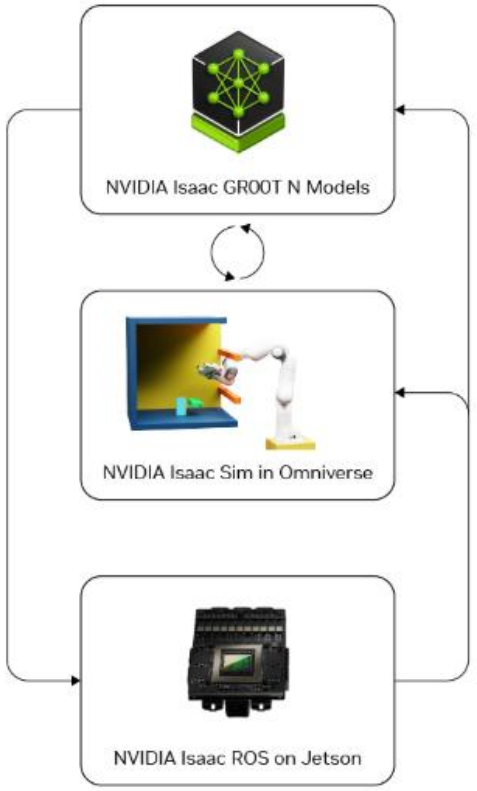


Sensory Skill Kit for Making Cobots Intelligent

AI COLLABORATIVE ROBOT – LLM GEN AI









AI Accelerator Kit for Classic Cobots

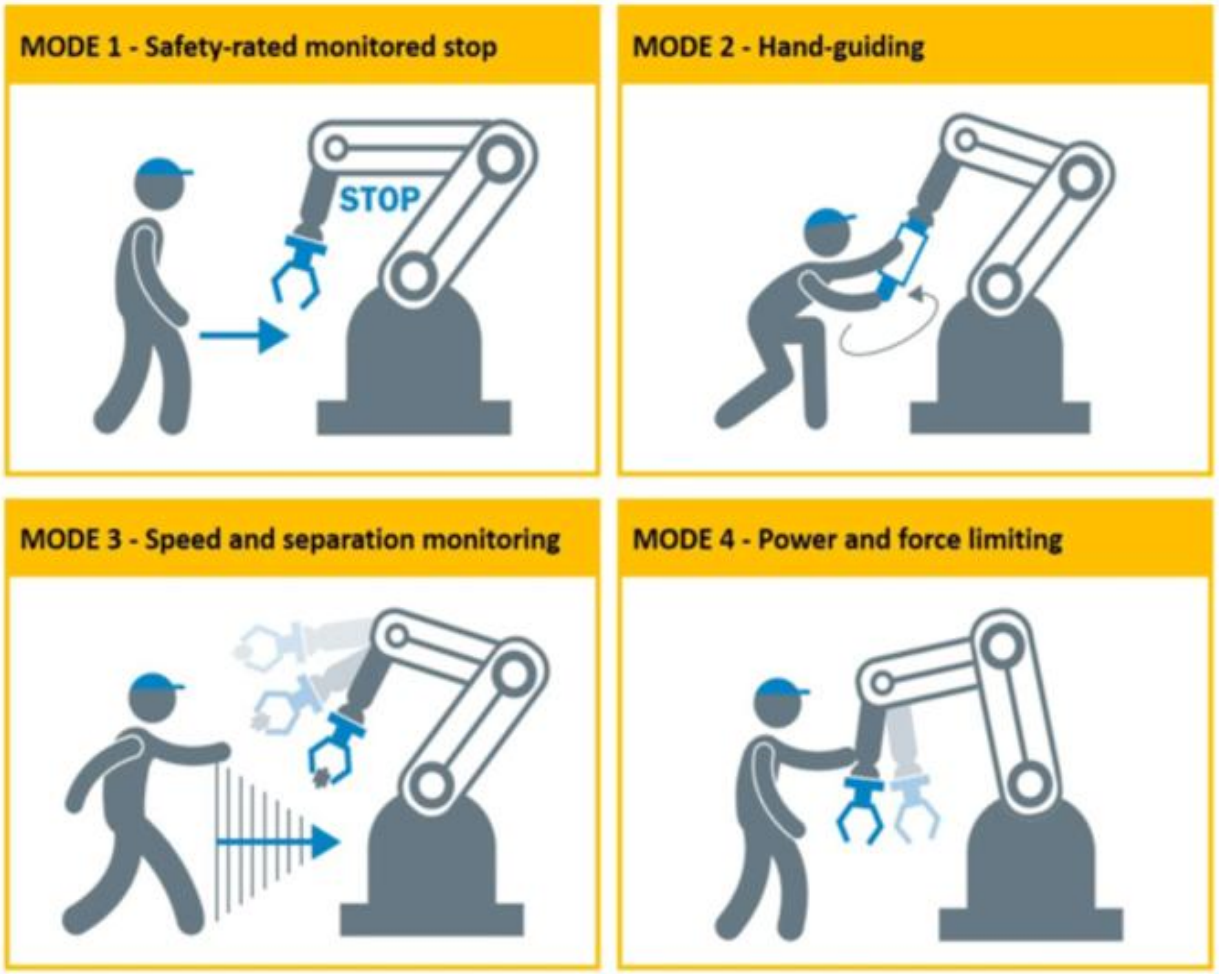
- The accelerator (NVIDIA Jetson AGX Orin + Gemini 335Lg 3D camera) provides built-in demo programs enabling pose estimation, tracking, object detection, path planning, image classification, quality inspection, state detection and more.

AI COLLABORATIVE ROBOT - SAFETY

Safe Contactless Human Detection

- Detection of humans and body parts within a given radius.
- Distinguishing humans from other inanimate objects.
- Speed reduction or even stop





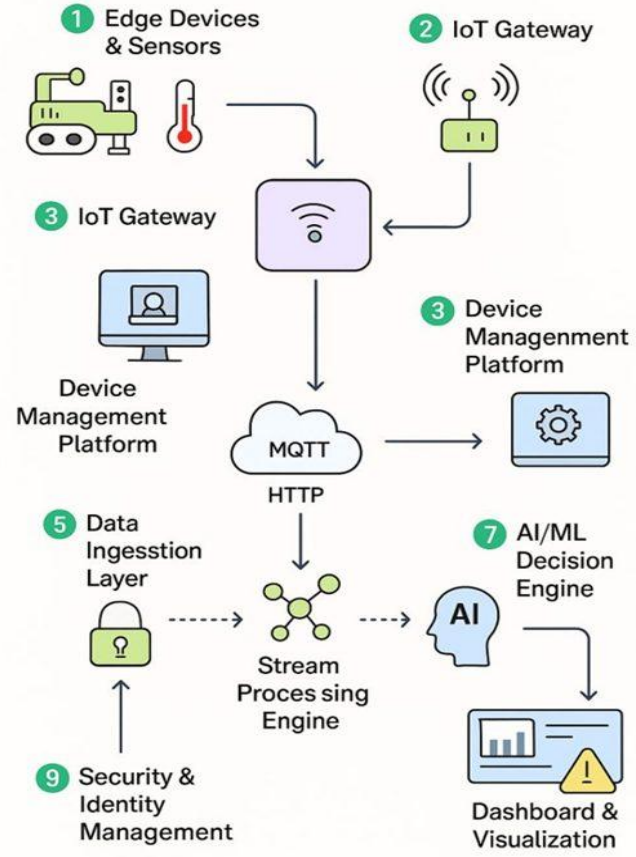
Robot Safety Standards


COGNITIVE FACTORY

COGNITIVE FACTORY

Definition

- A cognitive factory is a manufacturing plant that uses advanced technologies such as IoT, Robotics and AI to create a self-optimizing, adaptive, human-centric production environment.
- It is characterized by its ability to perceive, learn and interact with humans and machines in a fluid and adaptive way, ultimately resulting in greater efficiency, flexibility and sustainability.
- In the aeronautical industry, where there is demand for customized products with short lead times and reasonable costs, the cognitive factory combines the advantages of automated systems with the flexibility, adaptability and responsiveness of human workshops.



Embodied Intelligence

- The key to developing the cognitive factory lies in transferring cognitive skills, attributed to human workers, to technical systems such as machines, robots and industrial facilities.
- Such technical systems must be able to perceive and react appropriately to their environment, and even cooperate with people in a natural way (NLP). Therefore, cognitive technical systems (CTS) can be built as information capture and processing systems (IoT) with cognitive capabilities (AI).
- In addition, systems with embodied intelligence refine their actions based on feedback from their environment, improving performance over time.
- The development of the cognitive factory complements the advance of increasingly intelligent products that integrate and collaborate in their own production process.

Intelligent Systems (Cognitive Factory)

- **Intelligent Environment:** Plant industrial facilities (Assembly Jigs/Stands, Storage Systems, Access Systems…)
- **Cognitive Machines:** machines for Assembly, Testing, Inspection, Handling..)
- **Cognitive Systems:** systems and processes that continuously learn, reason, predict and adapt to the operator and the product in real time
- **Intelligent Product:** endowed with intelligence or connectivity, the product interacts with the assembly line



INTELLIGENT ENVIRONMENT

INTELLIGENT ENVIRONMENT

Ambient Intelligence (Artificial Intelligence of Things AIoT)

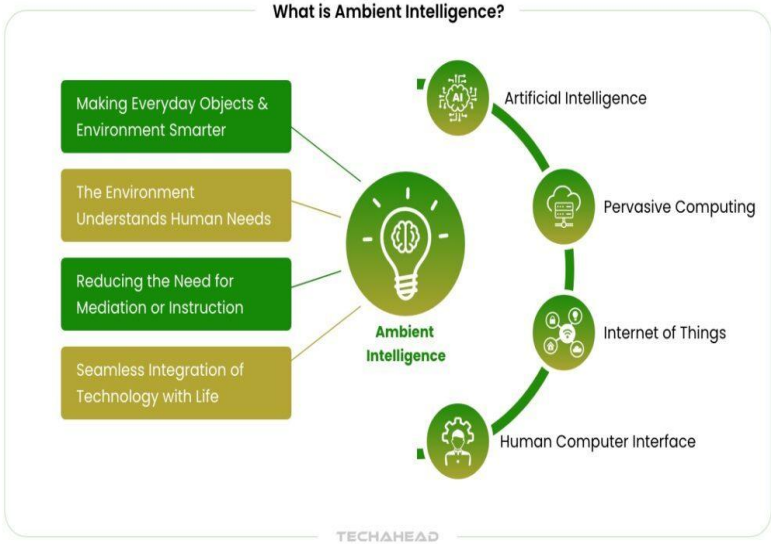
- Industrial AIoT combines the Industrial Internet of Things (IIoT) with Physical Artificial Intelligence (PAI) in an I5.0 industrial environment to create autonomous, intelligent systems that provide an optimal and safe working environment for industrial activities under the Industry 5.0 (human-centric) paradigm.

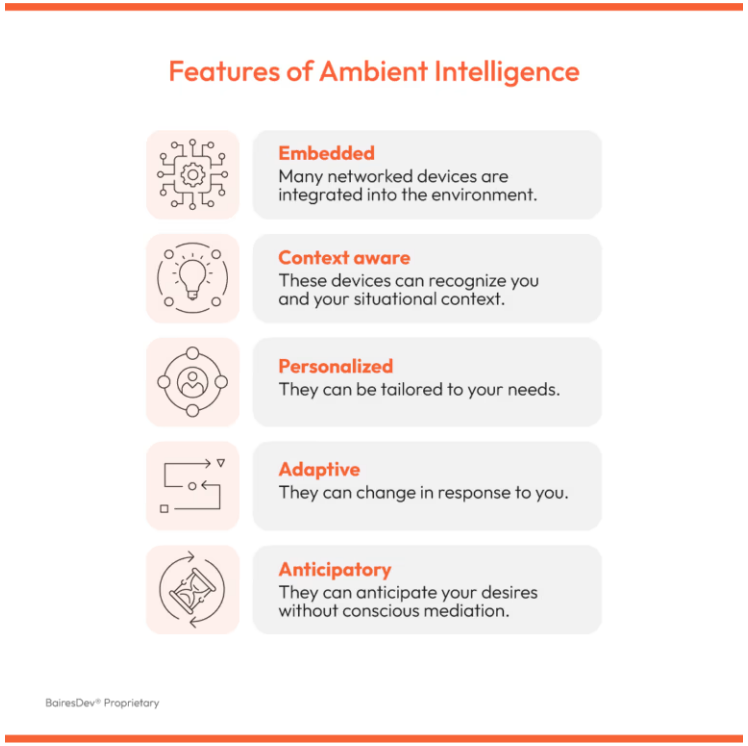
Key Components

- **Sensors and Devices:** The eyes and ears of the system, collecting data from physical equipment.
- **Edge Computing:** Data processing closer to the source to reduce latency and cloud dependency, enabling faster and more efficient operation.
- **Cloud Computing:** Storage and processing of large datasets for deeper analysis, model training and remote updates.
- **AI Algorithms:** Machine learning and deep learning models that interpret data and drive intelligent actions.

Benefits

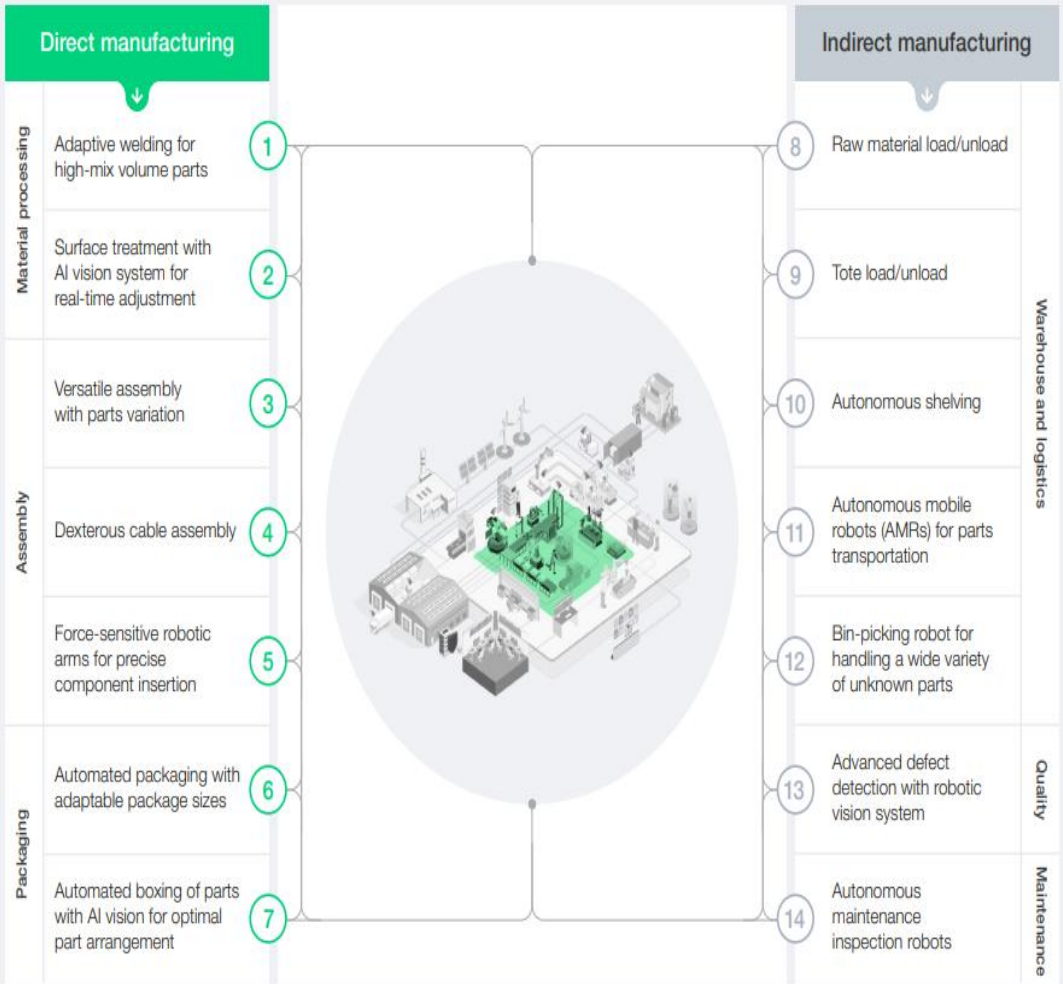
- **Operational Efficiency:** optimizes processes and resource allocation, resulting in higher throughput.
- **Predictive Maintenance:** identifies potential equipment failures before they occur, minimizing downtime and repair costs.
- **Reduced Costs:** lowers operating expenses by automating tasks, optimizing energy consumption and preventing costly breakdowns.
- **Improved Decision-Making**: provides real-time information and automated recommendations, improving strategic planning and responsiveness.
- **Greater safety:** automates hazardous tasks and improves human-machine interactions, creating a safer work environment.





INTELLIGENT ENVIRONMENT – PHYSICAL AI

How physical AI is reshaping factory and warehouse operations



Physical AI automation frees up humans for higher-level tasks

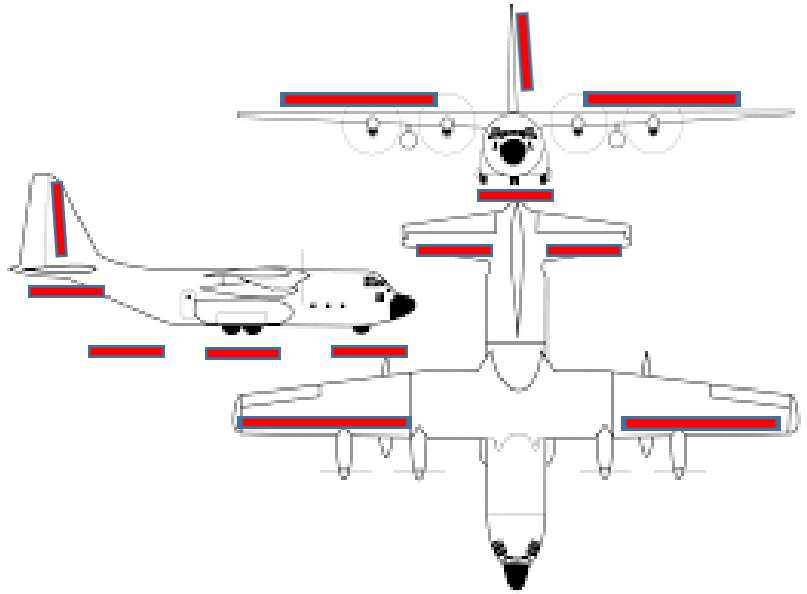
	Direct manufacturing	Indirect manufacturing			
	Material processing, assembly and packaging	**Manufacturing engineering**	**Logistics**	**Quality control**	**Maintenance**
Today	Operators manually process and assemble parts. Tasks require precision, manual machine operation and full attention. Changeovers and packaging are fully manual and labour-intensive.	Manufacturing engineers design, implement and optimize production systems, largely experience-driven. Updating inflexible automation requires high manual effort.	Workers manually move materials using forklifts or carts. Human coordination is needed, leading to inefficiencies, accidents and misplacements.	Products are visually inspected by workers. This is slow, subjective and prone to human fatigue. Random sampling may miss defects.	Reactive or preventive maintenance schedules are followed (independent of machine conditions). Engineers rely on visual checks and logs, making issues hard to foresee.
Future	**Scenario**				
	AI-enabled robots perform welding, assembly and packaging using real-time system feedback. Systems adapt to product and task variations automatically.	Production systems are software-driven, intelligent and adaptive. Engineers plan factories, integrate intelligent robots and visual-based quality control systems.	AI-driven autonomous mobile robots handle materials end-to-end, ensuring the right parts are delivered on time.	Vision-guided robotic arms scan every unit in real time and perform automated inspection tests.	AI-powered systems are becoming self-sufficient, with robots and drones handling some maintenance and inspection tasks while continuously generating data.
	Tasks				
	– Supervise robotic workflows – Train robots for new tasks – Resolve exception cases	– Optimize adaptive system – Oversee digital process design – Cross-disciplinary integration	– Monitor robotic fleets – Reconfigure robot routes – Manage system alerts	– Strategic oversight – Audit flagged defects – Optimize quality control	– Perform complex repairs – Interpret data – Improve system
	Skills				
	– Systems thinking – Digital literacy – Robot programming/ teaching	– Advanced digital and AI skills – Systems thinking – Creativity	– Systems thinking – Digital literacy – Workflow design and safety	– Analytical thinking – AI and big data – Creativity	– Technical skills – Analytical thinking – AI and big data

INTELLIGENT ENVIRONMENT – EXAMPLE: ASSEMBLY STATION SAFETY SYSTEM

Typical Risks

- **Landing Gear:** during extension and retraction tests of both the main and nose landing gear, nobody must inadvertently enter the area due to the risk of fatal entrapment. There is also a risk of unintended retraction with the aircraft dropping to the ground, so safety pin control is required.
- **Flight Controls:** during movement of flight controls such as ailerons, flaps, rudders,. there is a risk of hand entrapment, so the area must be cordoned off.
- **Ramp:** opening and closing creates blind spots that can cause fatal accidents.
- **Electrical:** the presence of current inside the aircraft can cause electrocution (28 VDC or 115 VAC)
- **Hydraulic:** the presence of hydraulic power in the aircraft during assembly can cause eye injuries,..
- **Pressurization:** during leak-tightness tests nobody should be nearby to avoid being hit by swarf or loose rivets..
- **Aircraft Jacking:** lifting the aircraft on jacks must be done with the aircraft level to avoid it shifting and falling off the jacks
- **Stairs:** falls when going up and down stairs
- **Hoses**: in walkways they cause falls.





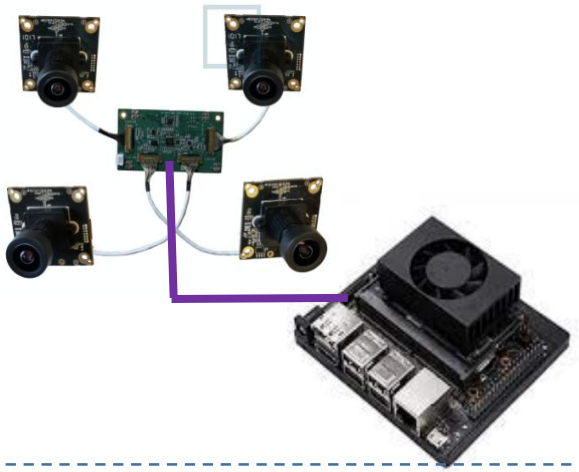
C-130 HERCULES

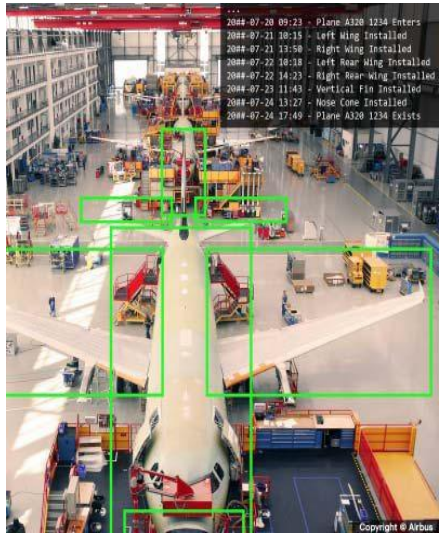
INTELLIGENT ENVIRONMENT – EXAMPLE: ASSEMBLY STATION SAFETY SYSTEM

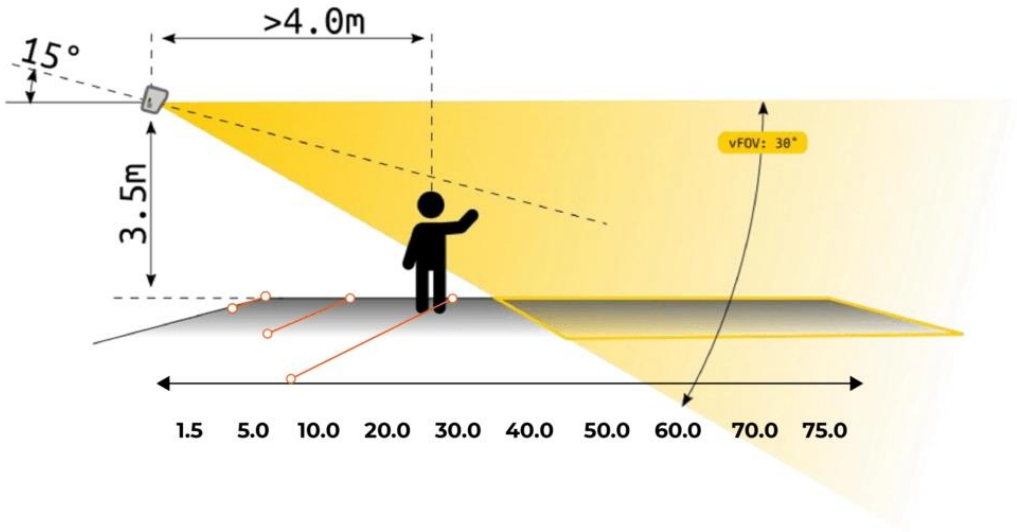
Architecture

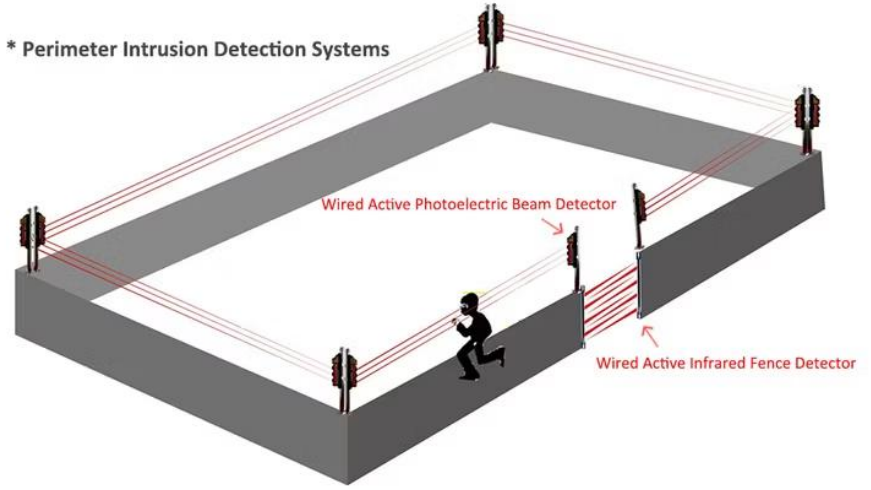
- Central Computer with AI Agent
- Sensing of access to danger areas
- Noise sensing (activation of hydraulic pumps, landing gear movements, flaps, ramp,..)
- Cameras identifying movement of people, objects and aircraft parts.
- Global alert systems, customized by area or activity.

LOCAL EDGE COMPUTING -- NVIDIA JETSON XAVIER

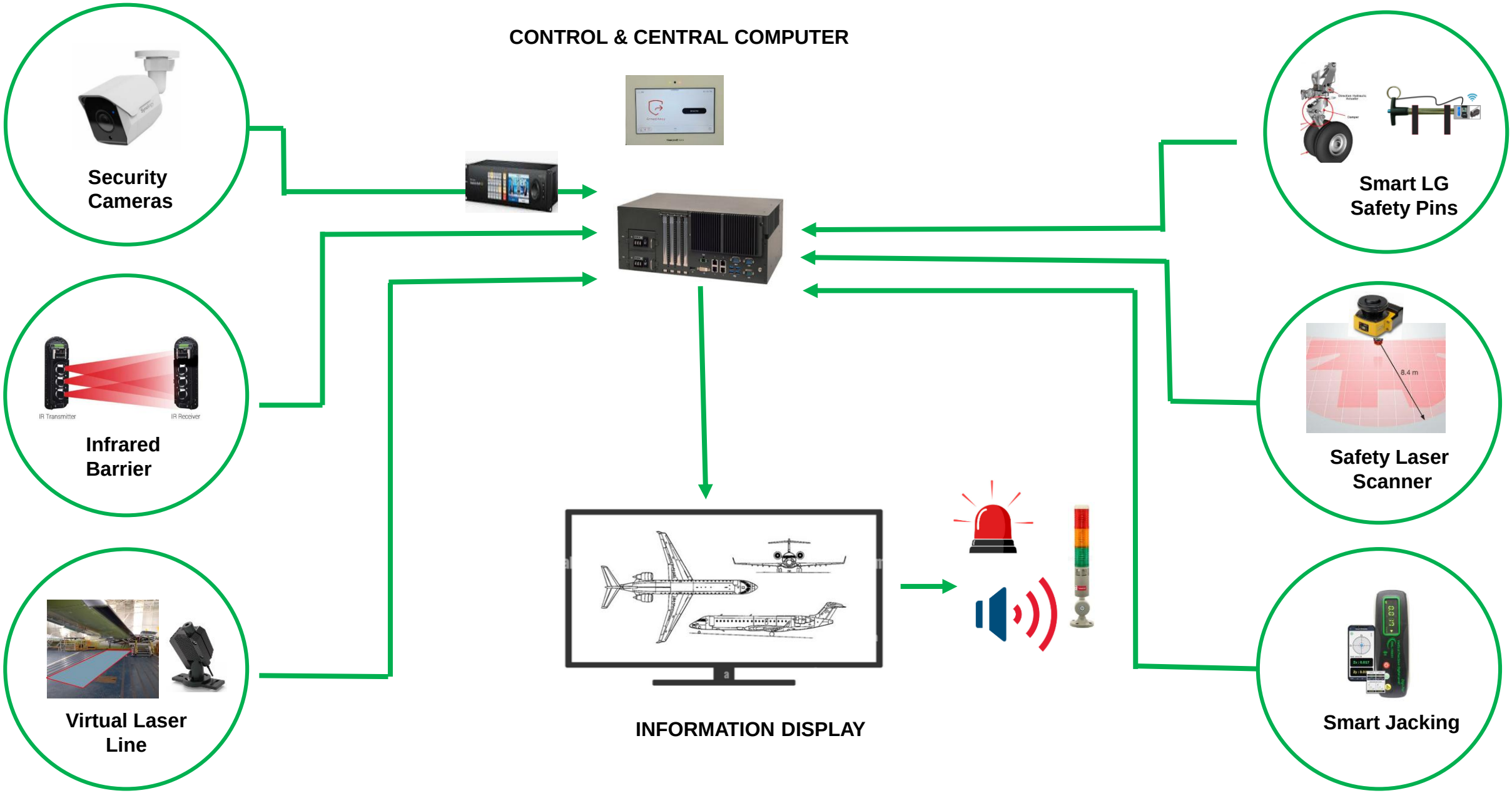




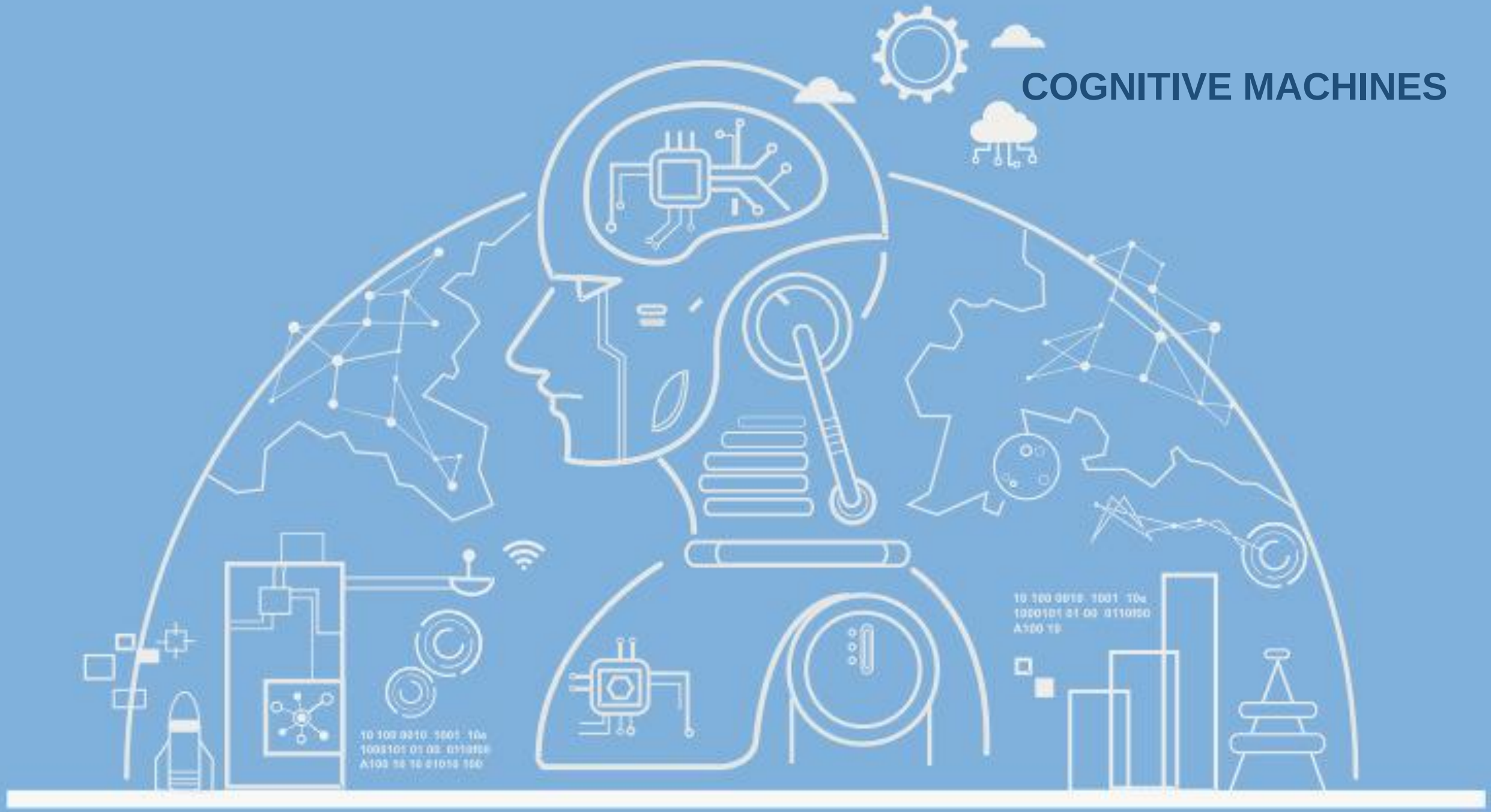




INTELLIGENT ENVIRONMENT – EXAMPLE: ASSEMBLY STATION SAFETY SYSTEM



COGNITIVE MACHINES



COGNITIVE MACHINES

Definition

- While an intelligent machine is a machine that uses artificial intelligence (AI) to learn from experience and perform tasks more accurately and effectively, often imitating human behavior, a cognitive machine, more specifically, is a type of intelligent machine designed to simulate human thought and reasoning processes.
- While AI provides the basic capabilities of intelligent machines, cognitive machines go further by focusing on understanding complex data, learning from interactions and optimizing human decision-making, rather than merely making automated decisions.
- In Industry 5.0 the intelligent machine has been originally designed or transformed (AIoTization) to be sensitive to values such as safety, sustainability and resilience.
- In aircraft assembly lines, machines are usually related to supplying electricity, fuel or hydraulic oil to the aircraft or to testing aircraft systems. Such machines are not intelligent and have only been IoT-enabled according to the I4.0 model

Transformation - AIoTization (Original + 4.0 IoTization + 5.0 AIoTization)

- **Cognitive machine built** with socio-technical values from the factory.
- **IoTized Machine** under Industry 4.0. Modification: internal and external sensing, introduction of microprocessor-based systems and communication systems (Ethernet, Wifi, Lora…)
- **AIoTized Machine:** a second modification includes the introduction of AI locally or in the cloud, as well as databases, memory, prompt-based humanization,… .

Benefits

- **Efficiency:** reduction in the number of operators needed to perform tasks, as the asset acts as a collaborator in the operation. Also reduction in the operator's required handling skills.
- **Autonomy:** Intelligent reactive/corrective/autonomous actions
- **Memory:** previous actions, user knowledge, environment, primary skills…
- **Sustainability:** reduced consumption (smart auto shut-off..), noise adjustment (rpm..)…
- **Safety:** increased safety through natural-language alerts, risk sensing, reduced operator-machine-environment stress….



Commercial Aircraft Ground Support Equipment



Non-AIoTized Ground Power Unit

COGNITIVE MACHINES
IOTIZATION + AIOTIZATION

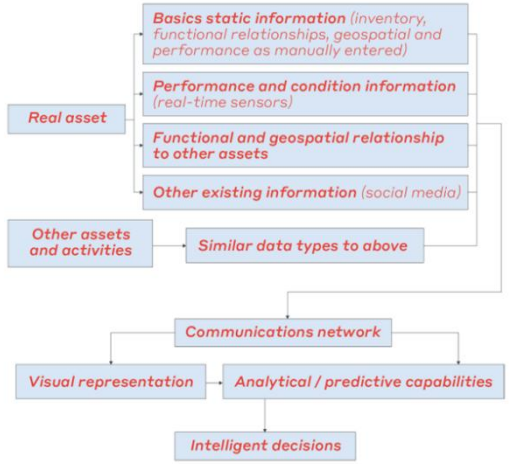
IoTization 4.0

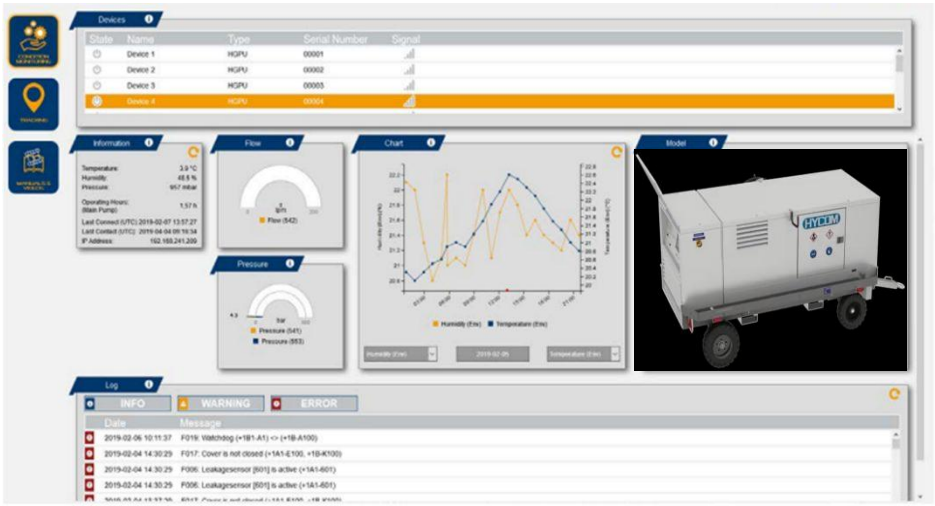
- Sensing system: monitoring of key variables for the mission and maintenance of the industrial asset
- GPS sensing
- Ethernet, wifi, lora, bluetooth connectivity..
- Edge computing (Micro)
- Digital twin
- Predictive and prescriptive capabilities

AIoTization 5.0

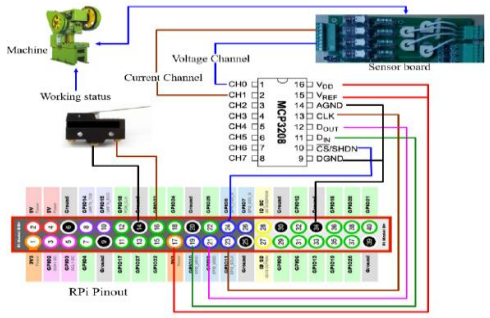
- AI hardware deployment (e.g. Jetson + speaker/microphone, camera,..) and GPIO connection with IoT hw. Not needed in the case of cloud AI.)
- Deployment of local LLM models (Ollama, LM, GPT4All,..) or cloud model (Gemini, ChatGPT,..)
- Memory: Databases
- Personalization, explainable AI…

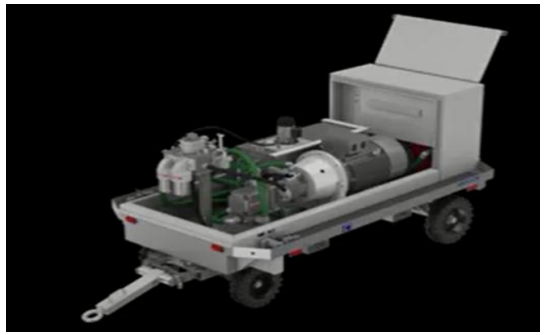
IoTization 4.0 + Digital Twin





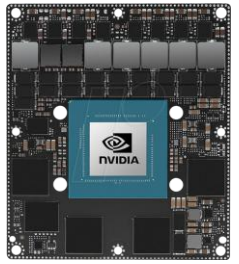






AIoTization 5.0











The Cognitive Factory – AI, ESG & The Future of MES

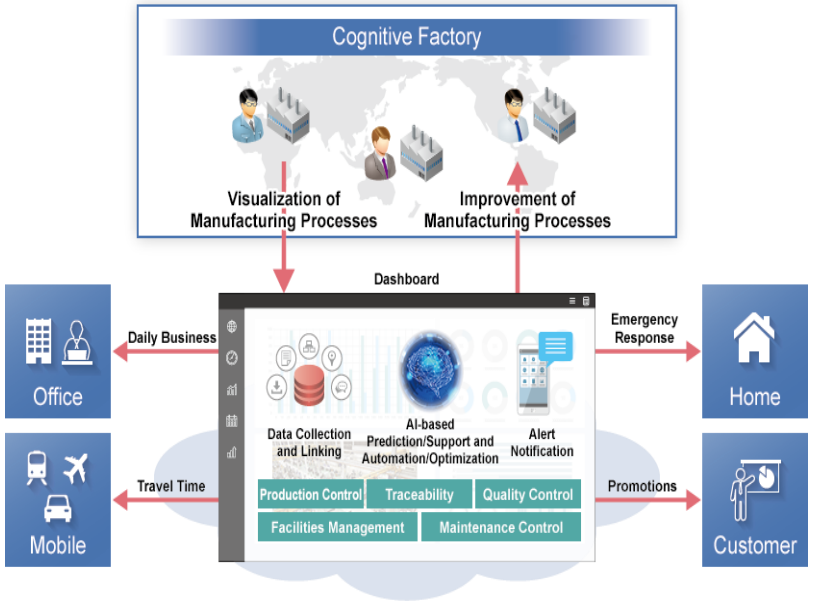


From Iron to Cloud: The Blueprint for MES Integration

COGNITIVE SYSTEMS

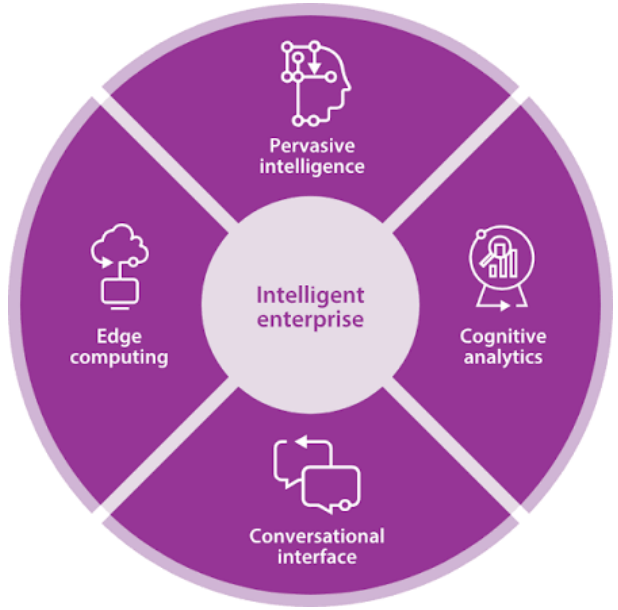
Definition

- Cognitive systems are intelligent, integrated software systems with flexible architectures that are combined for application with a wide range of systems engineering and safety methods and processes.
- They give the line the ability to perceive, reason, learn and, above all, collaborate seamlessly with the aeronautical technician, placing the human at the center of the process with the support of physical and agentic intelligence.
- They understand, learn and make decisions. These systems are oriented towards human skills and capabilities. They can perceive and understand things, draw conclusions and learn. They can also react reliably to unexpected events.
- Cognitive systems often depend on large amounts of unstructured information, such as sensor data, which is frequently incomplete or imprecise and therefore unreliable. Cognitive systems use this data as a basis for learning and making decisions.
- The system also has an overview of its own context, such as environmental influences, interacts with its environment, draws the corresponding conclusions and optimizes its actions.



Cognitive System Networks

- Cognitive systems are not only used in standalone environments. Many of them have to work together within a higher-level system, known as a "system of systems" or connected intelligent systems. Individual systems can react to each other optimally and reliably, even within a network.



Benefits

- **Maximum efficiency and error-free quality:** Artificial intelligence instantly adapts the line to any product, detecting and correcting faults in real time to eliminate waste and delays.
- **Operator well-being and safety:** The system reduces mental stress with visual guides and delegates physical effort to robots, improving ergonomics and accelerating learning.


ASSISTED
MANUFACTURING

AI-ASSISTED OPERATIONS

GENERAL

- AI actively contributes to manufacturing operations by driving efficiency, optimizing production and enabling innovation through generative design, better supply chain management, assisted assembly and intelligent quality control.
- Key trends include greater autonomy, integration into workflows, the growth of agent-based AI and the use of natural language for workforce training and the development of intuitive interfaces.
- Leading manufacturers focus on strategic AI adoption to achieve measurable business impact, address challenges such as security and complexity, and integrate AI to meet sustainability goals and support a skilled workforce.





INTERNAL LOGISTICS

AI-ASSISTED INTERNAL LOGISTICS

GENERAL

- From material handling to inventory planning, artificial intelligence is transforming the way factories operate.
- Traditional logistics often suffers from bottlenecks: delays in material flow, inefficient warehouse layout, labor shortages and high operating costs.
- With AI-based logistics solutions, factories can turn these challenges into opportunities for efficiency, accuracy and scalability.



USE CASES

- **Smart Warehouses:** the whole process of making warehouses smart through IoT (RFID, BLE, LORA/GPS,..) enables integrated location tracking and, through AI, makes it easier to locate materials and optimize their storage location within the assembly line.
- **Intelligent Route Planning**: AI algorithms, combined with Automated Guided Vehicles (AGV) and Autonomous Mobile Robots (AMR), make material transport within the factory smarter. AI continuously calculates the best route within the factory, allowing materials to reach workstations faster and reducing downtime on production lines.
- **Warehouse Optimization with AI:** warehouse management is often a bottleneck in production logistics. Traditional systems rely on error-prone manual scheduling. AI can perform intelligent space allocation, automated inventory counts, demand forecasting.. The result is greater storage efficiency and smoother material flow, perfectly aligned with production schedules.
- **Predictive Maintenance of Logistics Equipment**: reduces downtime of conveying machines through data collection, anomaly detection, maintenance scheduling…. This approach reduces unexpected breakdowns and extends equipment life, directly improving return on investment (ROI).
- **Human-robot collaboration for safer workflows:** AI robots create a safer environment by sharing operators' workload, deciding which transport tasks require human intervention, avoiding obstacles more easily... In this environment workers can focus on supervision and decision-making instead of lifting heavy objects.
- **Multi-robot coordination and fleet management:** when several AGVs or transport robots are managed, AI fleet management systems facilitate joint operations by assigning and distributing tasks in real time. They also optimize charging schedules to avoid downtime.

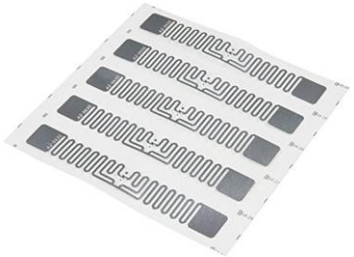
INTERNAL LOGISTICS – AGENT L EXAMPLE

Agent L (logistics)

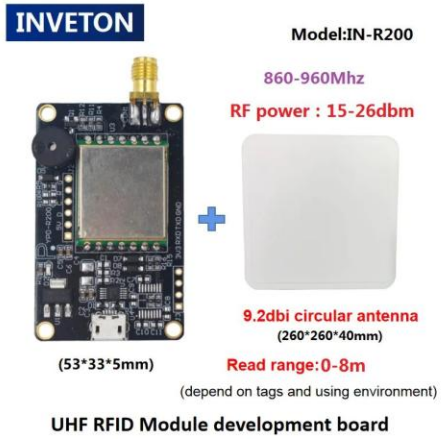
Possible functionalities would be:

- **Material Location Query:** allows users to find out in natural language where material is, based on the operation code or material description. Material is stored in RFID-tagged boxes and is detected via checkpoints (RFID gates), smart shelves….
- **Material Request to FAL Stores:** allows material to be requested in natural language either by P/N or by description. The intermediate store is controlled by a human, and the AGV will go to the store, ask the logistics operator for the material and take it to the position where it was requested.
- **Lost Material Search:** allows searching the different areas for lost material marked with RFID tags. To do so, it must carry an RFID sensor that will accept information in natural language from the operator and then travel along marked paths until it finds it.
- **TBD**












Goal Achieved
Analyze Data
Brailx Data
Execute Task
Execute Task
Optimize Route
Spbiming
REAL TIME
AI
CORE
ASSISTED ASSEMBLY
ai

AI-ASSISTED ASSEMBLY

General

Aircraft assembly operations require a high level of compliance in time, quality and cost, all under the umbrella of strict industrial and safety regulations.

This means operators must be highly qualified, which is generally achieved through extensive training and many years of experience.

This requirement makes it difficult to find staff prepared for this work, and generational renewal is also complicated without incurring high knowledge-transfer costs.

Using generative artificial intelligence, through the creation of appropriate software agents, makes it possible to have staff who, without being highly qualified, can carry out operations with a high degree of reliability.

It also provides high flexibility, allowing new products or versions to be introduced without a major training effort.

M Agents

It is autonomous software that assists the operator in assembly tasks. It has an extensive database of all global and specific assembly regulations, as well as of the operations in progress and the characteristics of both the product and the tools to be used for the job.

It is a virtual co-worker that not only answers questions but also looks after the operator's overall safety and the global status of the station if connected to the intelligent environment.





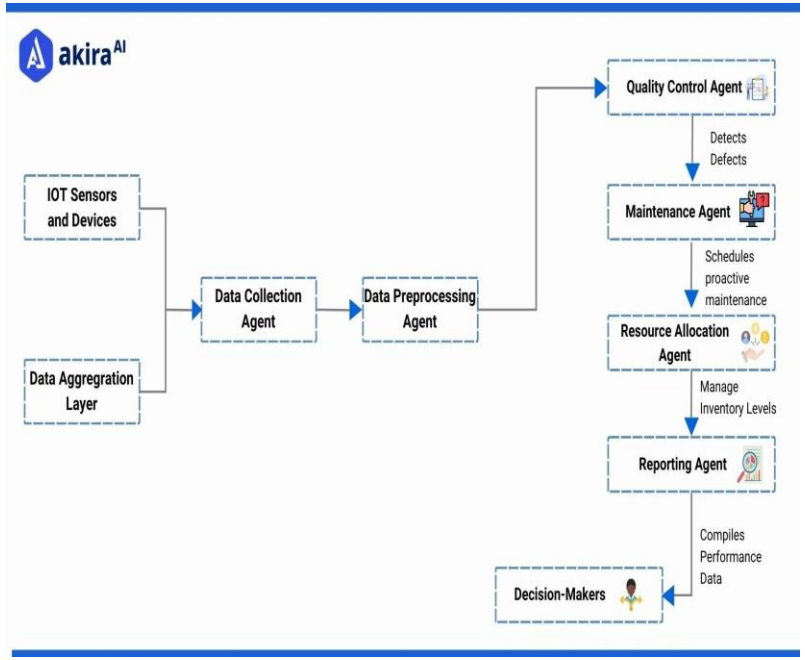
AI-ASSISTED ASSEMBLY – AGENT M EXAMPLE



AGENT M

It supports the operator during assembly operations as follows:

- **Environment:** assists the operator by indicating the production status of the line (simultaneous operations in progress, incompatibility of work zones on the aircraft, specific environmental safety situations,....)
- **Materials:** helps the operator by indicating the materials and tools needed to carry out the work order
- **Safety:** assists the operator on safety matters associated with the work to be done in that operation.
- **Technical:** provides technical assistance to the operator based on the internal documentation associated with the operation, history of other executions of the same operation and general data on the type of operation, technology,..
- **Improvement:** at the operator's request, records information on experience from current or previous executions that may need to be added to the operation. Records anomalies during execution if the operator deems it appropriate. These anomalies, suggestions,.. can reach, in real time, the production engineer responsible for the activity or the departments related to the problem (logistics, industrial services,..).
- **Performance:** if the operator wishes and provides the required data, once the operation is completed it will show the associated performance compared to previous executions and, based on the observed data, will propose whether additional or specific training is needed or recommend actions regarding the operator and operation (e.g. depending on age, certain operations may not be suitable).
- **Miscellaneous:** If connected to MES/ERP it can assist in filling in corporate forms associated with the operation (start, end,..). It can also help with translation if needed.




FUNCTIONAL TESTS
ASSISTED BY AI

AI-ASSISTED FUNCTIONAL TESTS - GENERAL

Functional Tests

- Aircraft functional tests verify the correct operation of all aircraft systems, both on the ground and in flight.
- These tests range from basic system checks (electrical, hydraulic,..) to complex functional check flights in extreme conditions (in-flight engine shutdown, stall,..), ensuring regulatory compliance and the safety and operational integrity of the aircraft.

Test Systems

- It comprises the hardware and software systems that allow the correct operation of the aircraft systems to be checked under conditions as close as possible to their operating environment.
- It includes measurement and simulation systems, generation of navigation signals, communications,..
- It also includes computer-assisted execution systems and results recording systems. In this model the central computer connects to the aircraft systems, simulating and sensing the different behaviors of the aircraft system. This connection is made through AIM (aircraft-machine interconnection) devices.
- In line with continuous improvement processes, the data obtained is studied by the corresponding engineering departments to understand the problems and to anticipate or avoid them, in some cases leading to modifications in the aircraft design.

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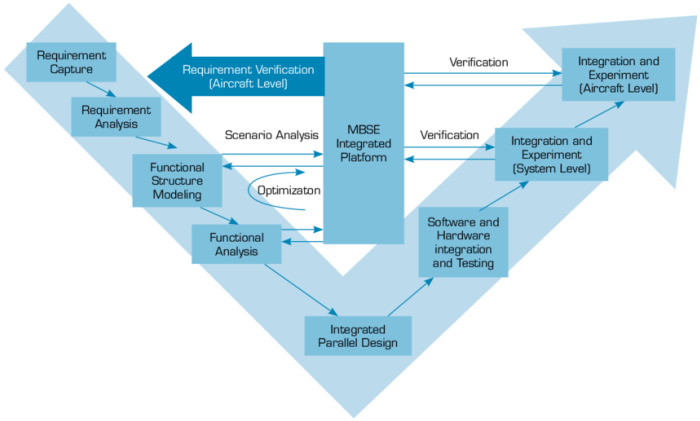


COMPUTERIZED TEST SYSTEM (COMPUTER + AIMS)

AI-ASSISTED FUNCTIONAL TESTS - MBSE

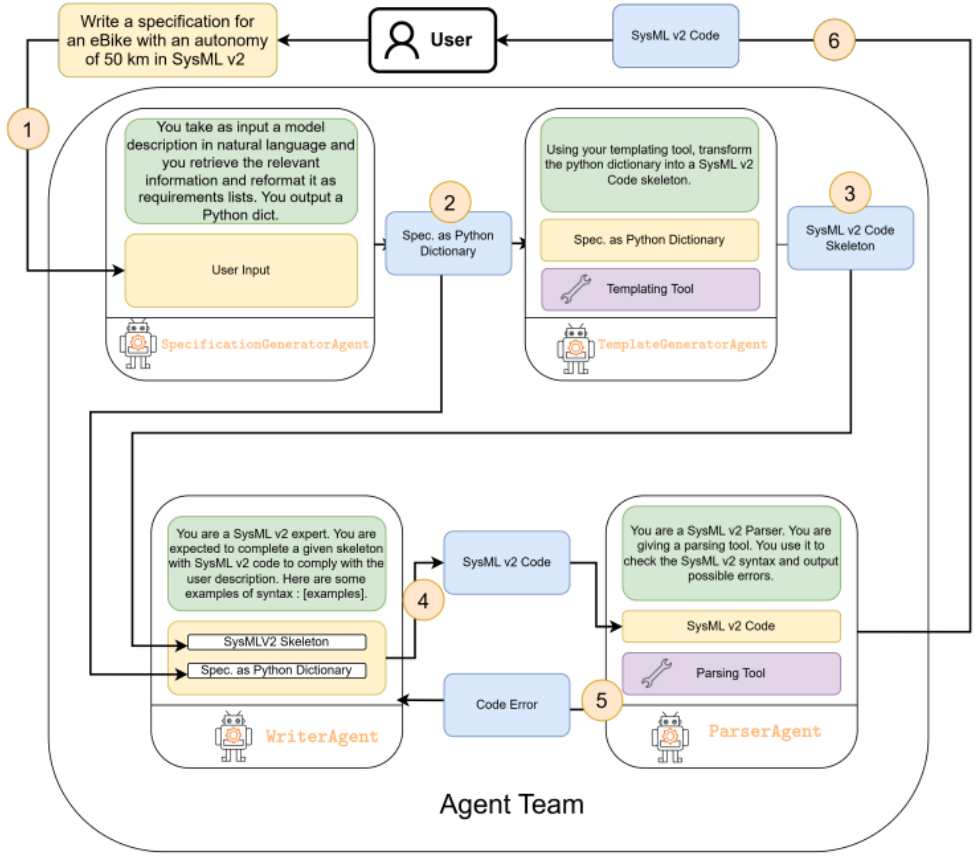
MBSE (Model Based Systems Engineering)

- **Model-Based Systems Engineering (MBSE)** is a methodology focused on creating and exploiting digital models of systems as the primary means of exchanging information, feedback and requirements, rather than relying on documents.
- It involves the entire process of capturing, communicating and ensuring that all the digital models we use to represent a system are coordinated and maintained throughout the system's life cycle.



AI-Assisted MBSE

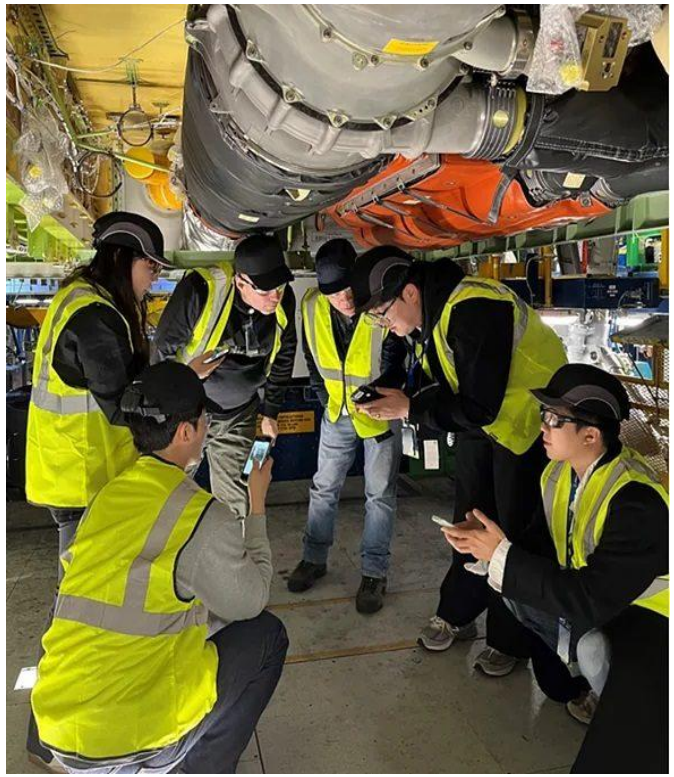
- **AI - Assisted MBSE (SysML):** the problem of automatic model generation (e.g. SysML, SysOn, SysIDE) from natural language specifications represents a major challenge in integrating assistance tools for systems engineering.
- This problem is similar to code generation, as it involves converting a textual description into a formal representation usable by modeling tools.
- For all these reasons, using LLMs to generate the concrete syntax of a model requires fewer resources and avoids problems with long contexts.



AI-ASSISTED FUNCTIONAL TESTS – T Agents

AGENT T

- AI agent assistance for aircraft functional tests not only facilitates their execution by helping the operator carry out the different steps, but can also help in many other aspects:
- **Functional Test Assistance**
  - System Information: the AI must know the mission of the aircraft systems and how they are operated. This information must be provided to the technician as support for any doubts about operation.
  - Test Generation: it must be able to indicate the steps to check a system if there is no documentation.
  - Equipment: it can indicate the test equipment required according to the documentation (work order, maintenance manual...)
  - Performance: it must be able to determine the number of technicians required and the estimated duration of the test if successful.
- **Diagnostic Assistance in Case of Failure**
  - Instruction Generation: based on the AI's knowledge of the system (equipment, schematics...) and the Interface Control Documents between subsystems, it must be able to provide the steps to follow to isolate the system failure.
  - Data Analysis: with the data obtained and historical data (from the aircraft itself or similar ones), a probabilistic analysis of the origin of the failure must be generated.
  - Solution Proposals: it must present one or several options taking into account the costs of opening areas, availability of replacement equipment, …
- **Repair Assistance**
  - Repair Procedure: description of how to carry out the work.
  - Required Tools: determine the tools required, such as scissors, crimpers, soldering iron...
  - Available Materials: determine which material should be used (wire, heat-shrink, connectors, pins…) and where it is located.



AI-ASSISTED FUNCTIONAL TESTS

AGENT T (CONTINUED)

- **Assistance in the Use of Test Equipment**
  - Equipment Selection: select the appropriate equipment for carrying out the test or troubleshooting. It must be within its calibration period and must not be faulty. Equipment that is commonly used is also preferable, as it is familiar to the technicians who will use it.
  - Location and Request: know where the equipment is physically located or even automatically request the equipment from the industrial equipment laboratory.
  - Operating Information: provide information on how the equipment is operated in the technicians' language.
- **Training Assistance**
  - General Training: aeronautics and air navigation systems.
  - Specific Training: on test equipment, repair, safety…
  - Failure and Solution Simulation: for technician training.
- **Human Factors**
  - Risks: determine and communicate the risks posed by the functional test both to the technicians performing it and to the rest of the staff at the station
  - Protective Measures: monitor compliance with personal protective measures (cap, gloves…)
  - Fatigue/Stress: it must be able to determine, from parameters such as tone of voice, language, etc.…, the state of the test crew, helping to ease the situation with appropriate language.
- In short, the goal is to reduce the need for highly qualified staff, reduce downtime, optimize the use of test equipment, increase safety…
- All of this leads to improved efficiency, without requiring additional effort from staff and reducing the technical stress arising from the use of sophisticated tools and the urgency of finding solutions to failures.






QUALITY
ASSISTED
BY AI

AI-ASSISTED QUALITY

FAL Quality Management System

- **EN 9100** is a quality management system standard specific to the aerospace, aviation and defense industry which, based on ISO 9001, adds additional quality and safety requirements for this sector.
- The standard focuses on product safety, prevention of counterfeit products and supply chain management for manufacturers, suppliers and distributors.

Processes regulated by the EN9100 standard

- Quality System
- Contract and Design Review
- Document and Data Control
- Process Control
- Purchasing and Customer-Supplied Products
- Product Identification and Traceability
- Product Handling, Storage and Preservation
- Inspection and Testing/Control of Associated Equipment
- Control of Nonconforming Product
- Quality Records and Internal Audits
- Data Analysis/Statistical Techniques
- Continuous Improvement/Corrective and Preventive Actions
- Training

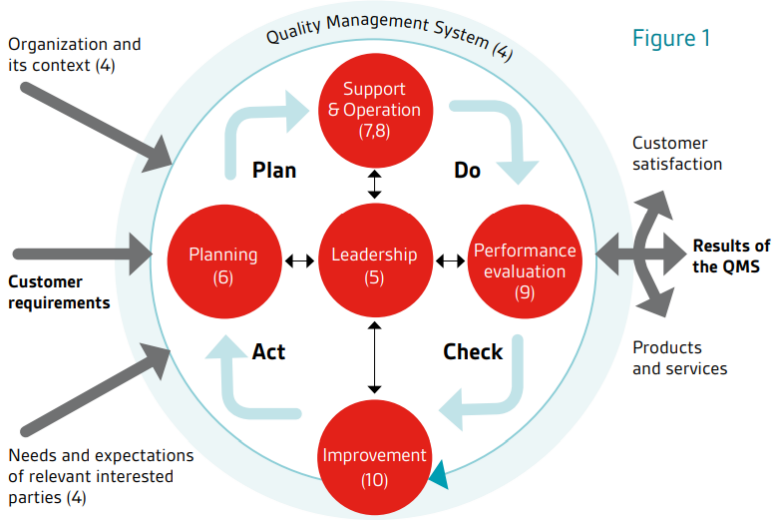
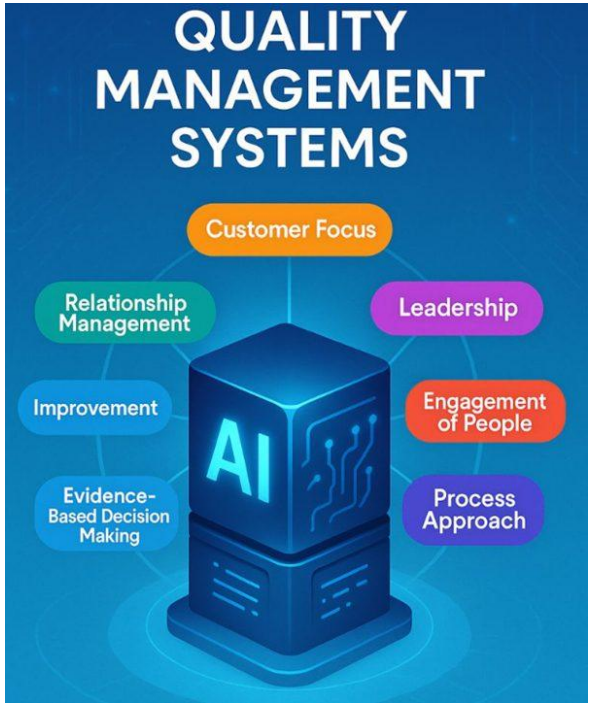


Figure 1

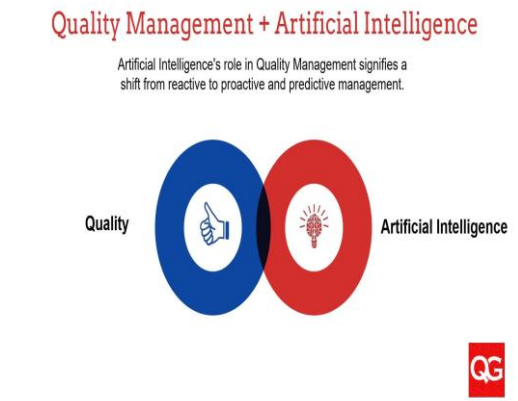


AI-ASSISTED QUALITY – Q Agents

Agent Q

It is an intelligent software system that analyzes production data to monitor, predict and improve product quality in autonomous or assistance mode. Connected to the plant's data sources and systems, it can detect anomalies, forecast possible defects, suggest root causes and automate actions to ensure consistency and higher product standards. Some of the activities it can carry out are:

- **Interpretation of Standards:** AI can provide quick, contextualized explanations of the EN9100 requirements, making it easier for the organization to understand what is needed to establish a quality management system (QMS) in the FAL.
- **Quality Documentation Generation**: AI can generate drafts of quality manuals, procedures and work instructions based on the standard's requirements and the organization's existing processes, speeding up the creation of documents needed for legal compliance.
- **Quality Process Control:** AI can help automate parts of the quality management cycle, providing end-to-end management through a network of AI agents specialized in the different quality tasks.
- **Knowledge Management:** AI agents can act as intelligent assistants, helping employees find relevant information within the QMS documentation or the EN9100 standard itself.
- **Inspection and Testing/Control of Associated Equipment:** Through computer vision and machine learning systems, AI can instantly identify surface defects (dents, corrosion,..). AI also facilitates the control of measuring and test equipment, as well as all metrological control management.
- **Nonconformity Control:** they can dramatically reduce control and management time by automating records, access,… so that we can document non-quality issues in natural language and query them with any filter criteria.
- **Quality Records:** the ease of AI's natural language will speed up and simplify the recording and organization of data, as well as its subsequent access and processing.
- **Data Analysis/Statistical Techniques:** AI's access to statistical tools can provide analyses of the quality situation on the assembly line, which in turn serve as the basis for improvement processes.
- **Continuous Improvement:** with historical data obtained during production processes, AI can identify systemic problems and find root causes. This helps generate corrective and preventive actions in order to mitigate non-quality issues (nonconformities, observations,..)
- **Training**: enables the creation of training materials and interactive tools for staff, ensuring a consistent and effective understanding of the EN9100 principles and requirements.





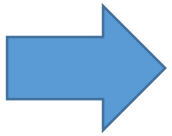


AI-ASSISTED QUALITY EXAMPLE – AGENT Q (NONCONFORMITIES: OBSERVATIONS)

Observations

- **An Observation** is a record of a note or comment derived directly from the inspection of an aircraft during the assembly process.
- **Action:** Each observation will require an action to rework, complete or adjust the part, area or system of the aircraft until it complies with the applicable documentation.
- If, due to the nature of the observation, a specific disposition is required, a Nonconformity will be opened.
- **Recording:** The opening and closing of observations are entered into the corresponding corporate quality systems by the process actors.
- **Procedure:** the quality inspector opens the observation when inspecting an area of the product and detecting a defect, and closes it after verifying that production has carried out the required work.
- **Traceability:** each observation is associated with an aircraft, and both the inspector who opened it and the operator who corrected the defect are recorded.

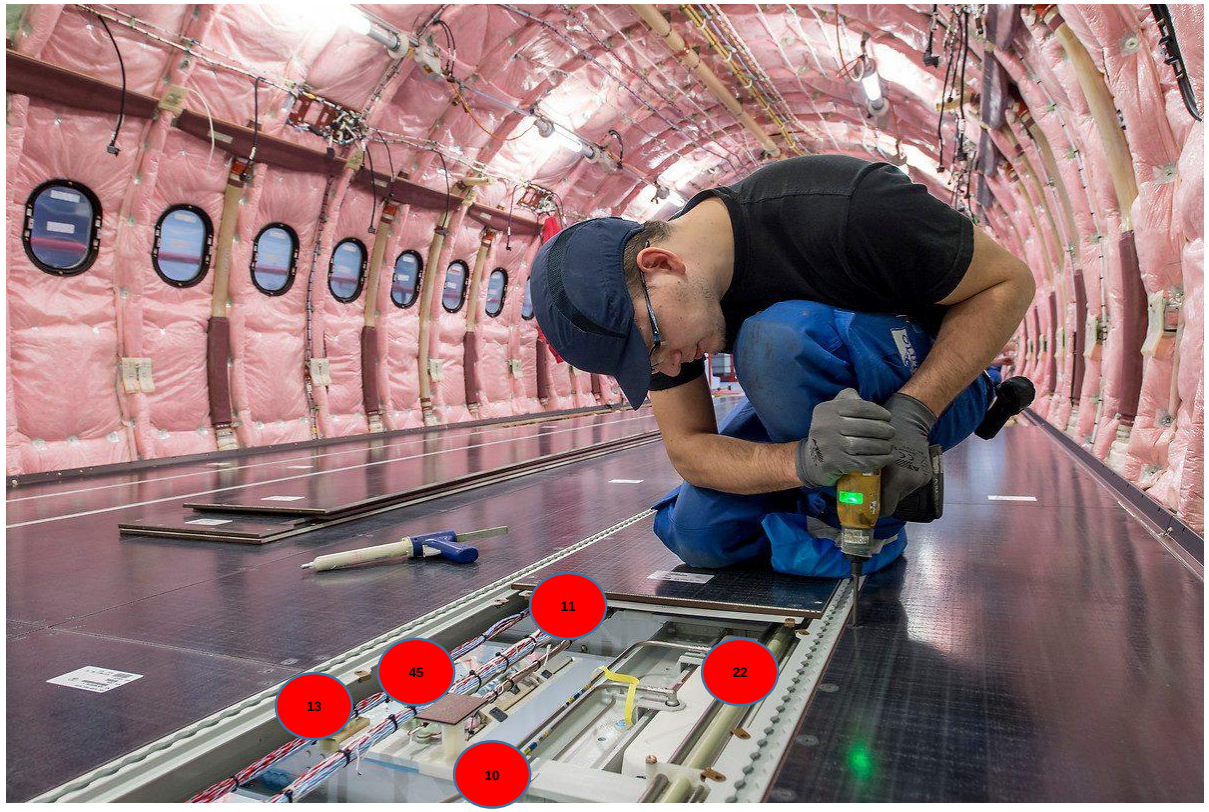
AGENT Q



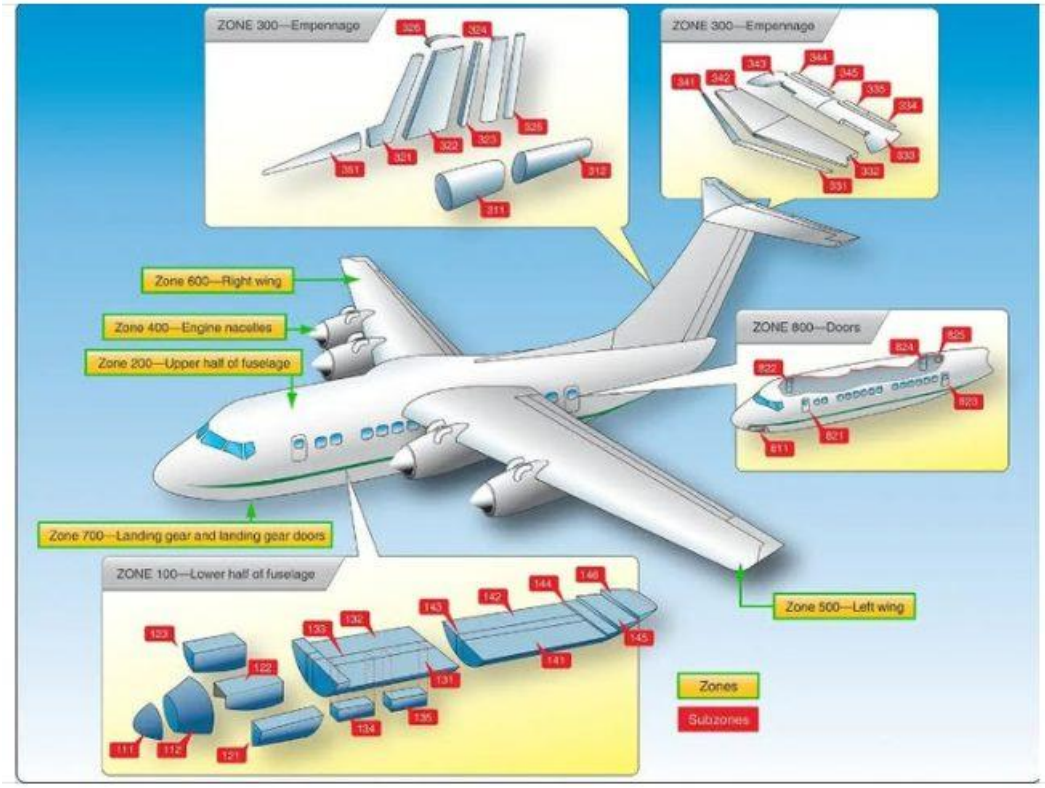
Corrective and Preventive Actions

- In a continuous improvement process, the organization must increase the effectiveness of the quality management system using all the tools available to it (audits, data analysis,..).
- Particularly important are the actions to eliminate the causes of nonconformities and prevent their recurrence.
- **The corrective process** includes:
  - Reviewing nonconformities and determining their causes
  - Determining the necessary actions and implementing them
  - Reviewing the actions taken and their effectiveness
- **The preventive process** complements the corrective one by determining potential nonconformities, causes and preventive actions. It includes:
  - Determining potential nonconformities and their causes
  - Evaluating the need for actions to prevent occurrences
  - Determining and implementing the necessary actions
  - Reviewing the preventive actions taken.

AI-ASSISTED QUALITY EXAMPLE – AGENT Q (NONCONFORMITIES: OBSERVATIONS)



Observations
- The red dots indicate minor defects observed by the quality inspector (stains, missing sealant, dents, clean drains…
- The operator gets ready to correct them.
- A standard aircraft may have thousands of minor defects that must be corrected before delivery to the customer. Such defects do not affect the airworthiness of the aircraft, although they must be corrected as a rule.



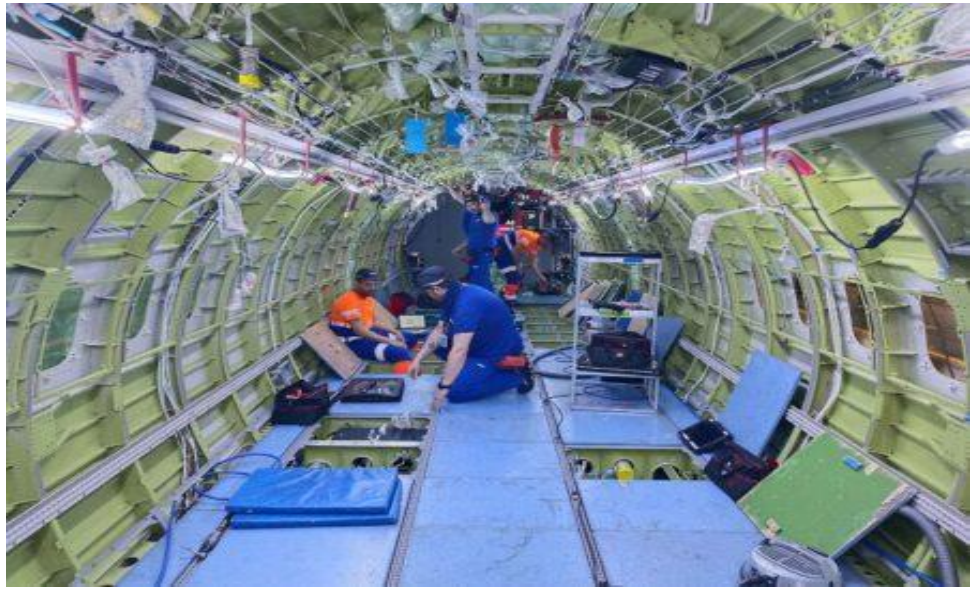
Aircraft Zones and Subzones Map
This method makes it easier to locate components, defects,.. on aircraft. It is divided into major zones, which are in turn subdivided into smaller, sequentially numbered zones and subzones. The numbers assigned to each zone are indexed to indicate the specific location and the type of system related to the component.

AI-ASSISTED QUALITY EXAMPLE – AGENT Q (NONCONFORMITIES: OBSERVATIONS)

Comprobaciones Finales Area de Trenes: Bloc de notas

Archivo Editar Ver

```
Comprobar ausencia de abolladuras, deformaciones, arañazos, incisiones o
Comprobar faltas de remaches
Comprobar sellado correcto y cordones sin resaltes
Ausencia de restos de papel adhesivo sin retirar
Cierre y apertura puertas trenes sin interferencias
Alineación de revestimientos de góndola en todo el contorno
Alignment of Main Landing Gear Fairing skins along the entire contour
Holguras uniformes con el revestimiento de góndola
Estado de la junta de goma en todo el contorno
Correcto ajuste de los pestillos de cierre y ausencia de corrosión
Comprobar ajuste y cierre de tapa de acceso a pin de blocaje de tren
Comprobar estado general de las ruedas y neumáticos (marcas de rodadura)
Protección superficial aplicada en fijaciones de ganchos de tren
Frenado de fijaciones en general
Fijaciones y protecciones de mazos de cables, con especial atención a lo
Tensión y frenado del cable de emergencia de tren y poleas
Engrase de todas las articulaciones
Indicios de pérdidas de líquido hidráulico acumulado en el revestimiento
Síntomas de pérdidas de líquido hidráulico
Limpieza de amortiguadores
Comprobar estado general de las ruedas y neumáticos (marcas de rodadura)
Frenado de eje de rueda auxiliar
Frenado de fijaciones en general
Fijaciones, protecciones, roces y proximidad de mazos de cables en la in
Tensión y frenado del cable de emergencia de tren
Centrado de micro con las pistas de actuación e identificación adecuada
Faros de taxi por condición y estado
Pintura de acabado en todo el compartimento del tren auxiliar
Engrase de todas las articulaciones
Síntomas de pérdidas de líquido hidráulico
Limpieza de amortiguador
```



Observaciones: Bloc de notas

Archivo Editar Ver

AVION	NUMERO	F.APERTURA	INSP. APERTURA	LOCALIZACION	DESCRIPCION	RESOLUCION	OPERARIO	INSP.CIERRE	ESTADO
987	4800210345	24.02.2025	I123	Cabina Carga	Falta por instalar tuercas remachables en ambas cajas de contactores	Montarlas	P312	I123	Cerrada
987	4800210346	24.02.2025	I123	Plano medio	Falta cama sellante	Echarla	P303	I123	Cerrada
987	4800210347	24.02.2025	I123	Empenaje	Bollo en timon de altura	Reparar	P401	I123	Cerrada
988	4800210348	21.02.2025	I126	Plano medio	Mancha de Pintura en el asiento copiloto	Cambio	P312	I123	Cerrada
988	4800210349	24.02.2025	I223	Plano medio	Pin roto en conector J32	Grapar	P312	I223	Cerrada
988	4800210360	25.02.2025	I123	Plano medio	Falta por instalar mascarilla	Instalar	P312	I123	Abierta
986	4800210361	23.02.2025	I127	Plano medio	Falta por instalar asiento tercer tripulante	Instalar	P314	I193	Cerrada
986	4800210245	24.02.2025	I127	Cabina Carga	Falta por instalar botella oxigeno	Montarlas	P312	I123	Cerrada
986	4800210246	24.02.2025	I123	Plano medio	Falta cama sellante tapa bombas hidraulicas	Echarla	P303	I123	Cerrada
986	4800210247	28.02.2025	I123	Plano medio	Bollo en panza trasera	Reparar	P401	I123	Cerrada
988	4800210248	21.02.2025	I123	Plano medio	Mancha de Pintura volante	Cambio	P312	I123	Cerrada
990	4800210249	24.02.2025	I223	Cabina Carga	Gancho tren delantero no frenado	Frenar	P312	I223	Cerrada
990	4800210260	25.02.2025	I123	Cabina Vuelo	Cristal arañado	Cambiar	P312	I123	Abierta
990	4800210761	23.02.2025	I123	Cabina Carga	Piso 32 falta tornillo	Instalar	P314	I123	Cerrada
990	4800211345	24.02.2025	I123	Cabina Vuelo	Falta por instalar acelerometro	Montarlo	P322	I193	Cerrada
990	4800211346	24.02.2025	I124	Motor Derecho	Falta lazo detector fuego	Montarlo	P313	I123	Cerrada
990	600211347	24.02.2025	I123	Empenaje	Bollo en timon de altura	Reparar	P401	I113	Cerrada
990	600211348	21.02.2025	I123	Cabina Vuelo	Mancha de Pintura en el asiento copiloto	Cambio	P312	I123	Cerrada
986	600211349	24.02.2025	I224	Plano medio	Falta mazo paracaidistas	Instalar	P312	I203	Cerrada
986	600211360	25.02.2025	I123	Plano medio	FOD cinta carrocero	Limpiar	P312	I323	Abierta



OPERATIONAL EXCELLENCE

AI-ASSISTED OPERATIONAL EXCELLENCE

Operational Excellence

- **AI-driven operational excellence** involves using artificial intelligence to optimize business processes, increase efficiency and improve decision-making by automating tasks and predicting outcomes.

Means

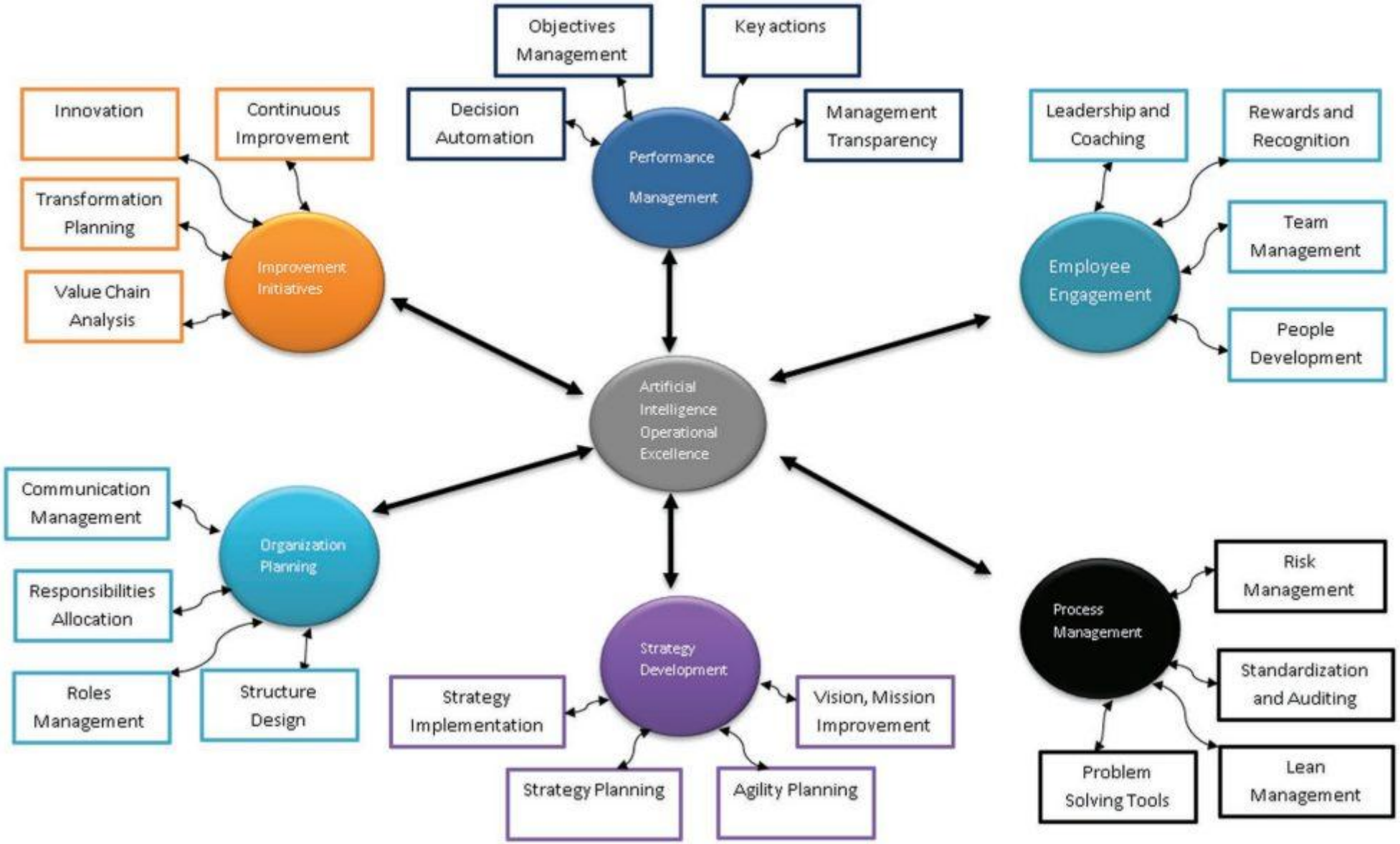
- **Process automation:** AI automates repetitive, rule-based tasks, allowing employees to concentrate on higher-value strategic tasks.
- **Predictive analytics:** AI analyzes data to forecast possible equipment failures, identify market trends and predict operational disruptions before they occur**.**
- **Real-time decision-making:** By analyzing data in real time, AI provides immediate insights, enabling more informed, data-driven decisions to adapt to changing market conditions.
- **Optimized resource allocation**: AI uses intelligent forecasting to optimize resource allocation, ensuring efficient use of assets and staff.
- **Improved quality control:** AI algorithms can detect defects and quality deviations with high accuracy, improving product quality and reducing waste.
- **Improved supply chain management:** AI helps optimize supply chain logistics, making it more efficient and resilient.
- **Agile operations:** AI enables continuous learning from data, helping organizations adapt their processes, stay agile and respond effectively to external changes.



Benefits

- **Higher productivity and efficiency:** optimizing workflows and automating tasks results in higher output.
- **Cost savings and error reduction:** automation and error detection minimize operating costs and waste.
- **Improved customer experience:** faster support, higher-quality products and more reliable services improve customer satisfaction.
- **Strategic advantage:** AI provides a competitive advantage by allowing organizations to operate more efficiently and innovate faster.

AI-ASSISTED OPERATIONAL EXCELLENCE

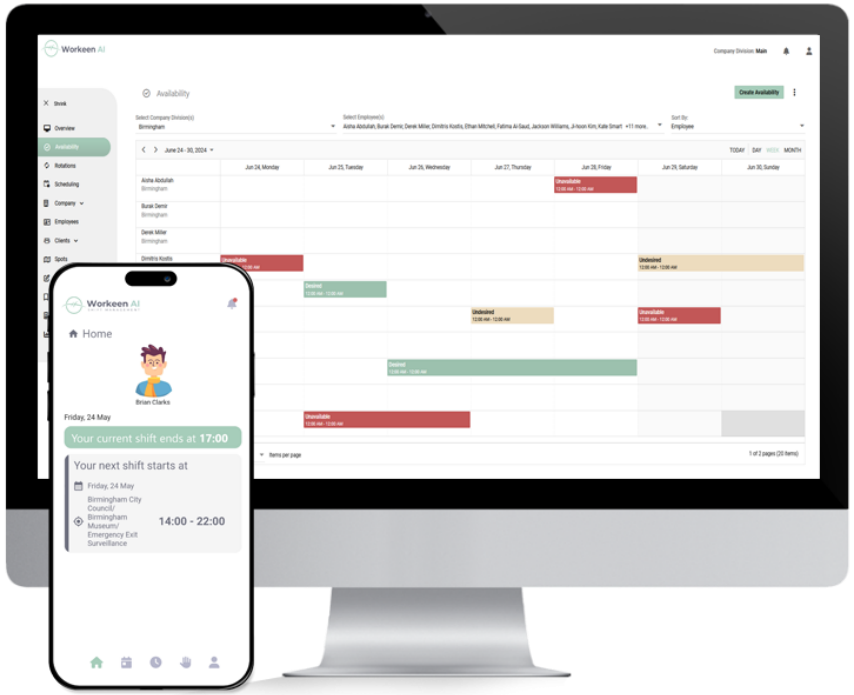



WEEK 1
WEEK 2
WEEK 3
WEEK 4
WEEK 5
S M T W Th F Sa
AI-ASSISTED PLANNING

AI-ASSISTED PLANNING

Production Planning with AI

- **Transforms** production planning by leveraging advanced algorithms and data analysis to optimize demand forecasting and decision-making, improve efficiency and adapt to dynamic production.
- **Improves** production planning by analyzing historical data and market trends more accurately**.**
- **Optimizes** the Master Production Schedule (MPS) by considering various factors such as demand forecasts, resource availability and production constraints, generating flexible plans that adapt to changes.
- **In addition, AI-based Material Requirements Planning (MRP) systems** optimize inventory levels by analyzing real-time data and predicting future material needs more accurately, thus minimizing stockouts and storage costs.
- **Likewise, capacity planning** to reduce costs benefits from AI simulations, which identify constraints and offer resource allocation recommendations for a feasible plan.
- **Optimizes** task sequencing to minimize downtime according to production requirements, takt and lead time.
- **They monitor** production data in real time and make the necessary adjustments to manage disruptions arising from shortages of staff or material, tools,..





AI-ASSISTED PLANNING - ARCHITECTURE

Operational Flow

- **Query:** the process starts when the user submits a query.
- **APP:** this is the main application, allowing natural language input and output of information in both text and voice.
- **Output:** the LLM generates an output based on the query and the data received. This output can take various forms, such as production schedules, inventory forecasts or supply chain performance reports. User feedback on the LLM's output improves the accuracy and relevance of the result
- **LLM Processing:** once received, the application passes the query to the orchestration layer.
- **Orchestration layer**: This layer combines the query with the relevant data from the vector database and the LLM cache and sends them to the LLM. Its role is to orchestrate the flow of data and tasks, ensuring smooth coordination between all components of the architecture (Langchain, Langgraph, n8n,..)
- **LLM:** they are essential for executing production planning tasks (ChatGPT, Gemini, Claude..).
- **Agent:** tackles complex problems, interacts with the external environment and improves learning through post-deployment experiences.

Data Storage

- **LLM Cache:** caches frequently accessed information, speeding up response time (SQLite, GPTCache..)
- **Logging/LLMOps:** LLM operations (LLMOps) tools such as Weights & Biases, MLflow, Helicone and Prompt Layer help log actions and monitor performance through feedback loops.
- **Validation:** A validation layer is used to validate the accuracy and reliability of the LLM (Guardrails, Rebuff, Guidance..)

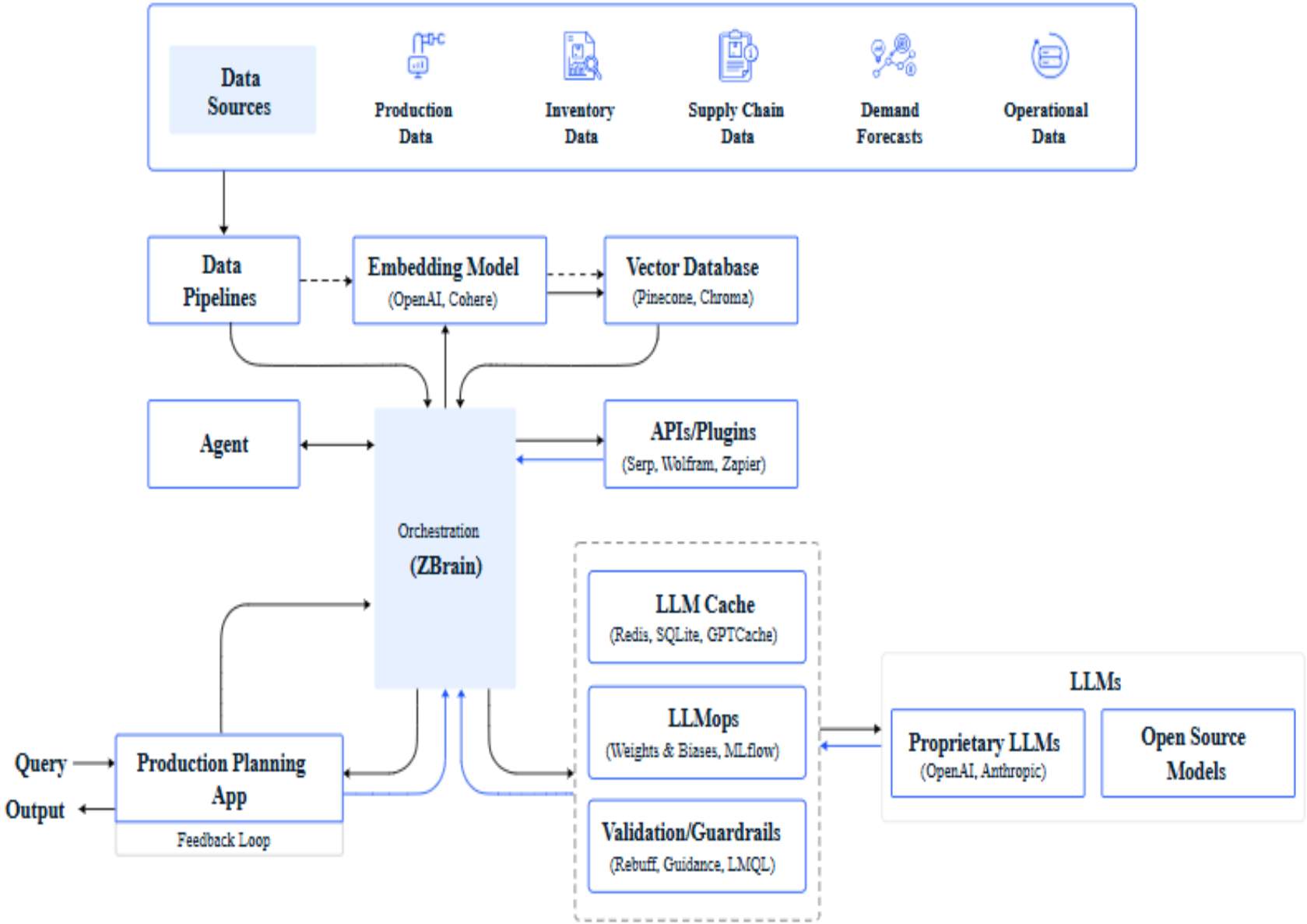
Data Sources

- **Demand forecasts:** projections of future demand based on historical sales data and market trends.
- **Production data:** information on times, costs and quality of the FAL's assembly, testing and delivery processes.
- **Inventory data:** material inventory levels (shortages) from inventory management systems.
- **Supply chain data:** delivery lead times of parts (MCA) through the supply chain.

Data Flow

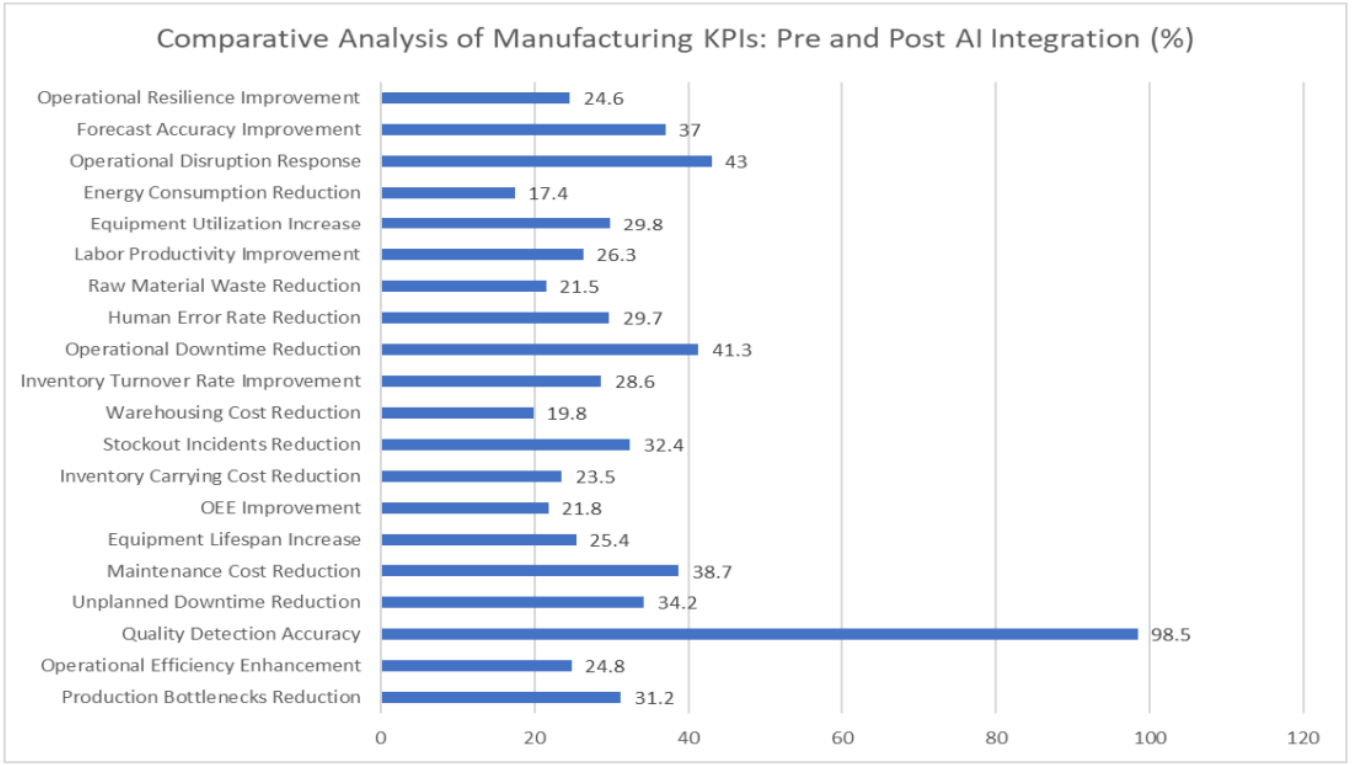
- **Pipelines:** manage the ingestion, cleaning and structuring of data, preparing it for subsequent analysis.
- **Embedding:** the prepared data is processed by an embedding model. This model converts textual and numerical data into representations called vectors, which AI models can understand (OpenAI Embed., Gemini Embed.,...)
- **Vector Database:** the generated vectors are stored in a vector database, enabling efficient querying and retrieval (Pinecone, Chroma, Qdrant).
- **API&Plugins:** connects different components and enables additional functionality, such as access to additional data, connection to LLMs or simple execution of specific tasks.

AI-ASSISTED PLANNING - ARCHITECTURE

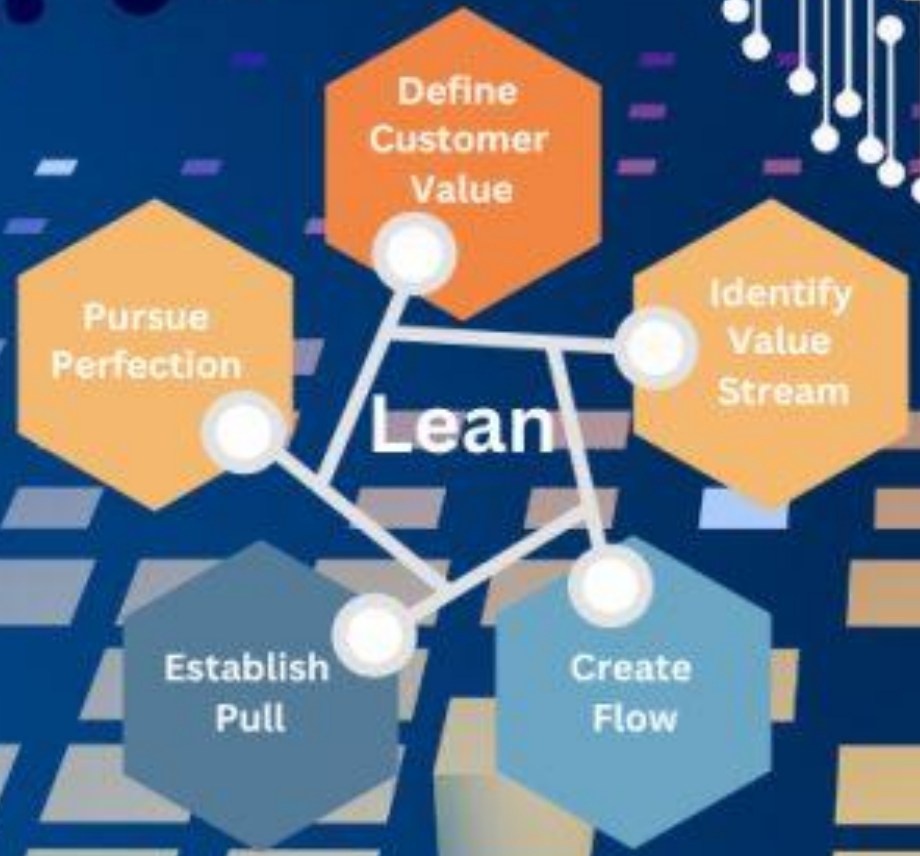


PRODUCTION PLANNING





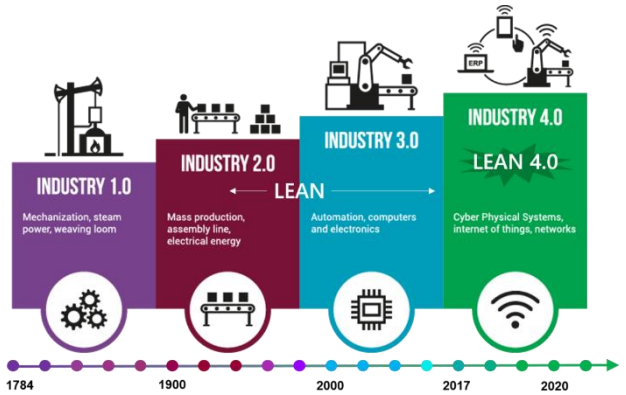
GEN AI LEAN


Define Customer Value
Identify Value Stream
Create Flow
Establish Pull
Pursue Perfection
Lean

LEAN MANUFACTURING - EVOLUTION

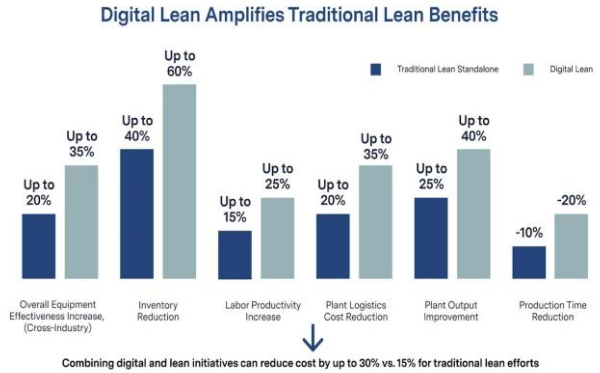
Traditional Lean (Industry 2.0 & 3.0)

- Lean principles are the foundation of modern manufacturing practices. They drive productivity by eliminating waste, and improve quality, safety and employee engagement.
- The main inefficiencies of traditional lean compared with digital lean are the difficulty adapting to dynamic environments, the lack of flexibility and the difficulty processing large volumes of data.
- In addition, new generations of workers are used to real-time digital interfaces. The pen-and-paper approach of many lean practices is less attractive, and the lack of engagement jeopardizes the sustainability and quality of the method.



Digital Lean (Industry 4.0)

- Digital Lean can be defined as extending traditional lean activities with I4.0 technologies such as IoT, Computer Vision, AI/ML, Data Analytics,. to increase factory visibility, productivity and agility.
- While traditional lean alone can typically produce a 15% reduction in the costs of key operations, it is the combined effect of lean and digital tools that doubles the savings, to a 30% cost reduction.



AI Lean (Industry 5.0)

- Although both Digital Lean and Gen AI Lean leverage technology to enhance lean manufacturing principles, they differ in approach and scope, the latter focusing on the application of generative AI.
- By harnessing the power of AI, companies can automate routine tasks, identify patterns and trends, and gain detailed insight into factory behavior.
- Generative AI can also help companies optimize their operations, reduce costs and improve overall efficiency.

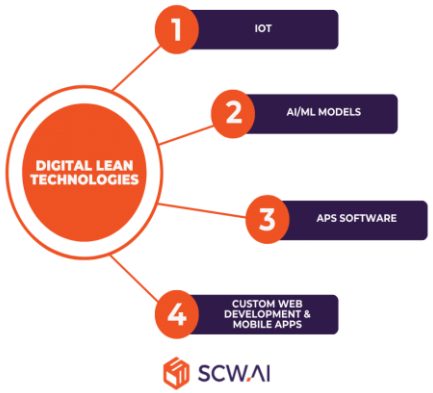


DIGITAL LEAN – INDUSTRY 4.0

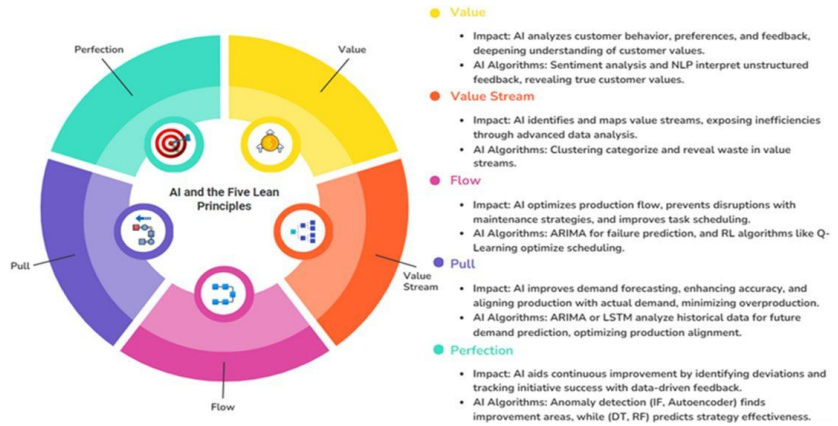
Traditional Lean vs Digital Lean

#	Lean Tool	Industry 4.0 Tool	Objective
1	Gemba Walk	Digital Twin	Enable real-time monitoring and analysis to identify inefficiencies and optimize processes.
2	5S	IoT Sensors	Maintain workplace organization while monitoring environmental conditions to enhance safety and efficiency.
3	Standardized Work	Workflow Automation Software	Standardize work processes to reduce variability and improve consistency in production.
4	8 Lean Waste Analysis	Big Data Analytics	Leverage data analytics to identify and eliminate waste, improving efficiency.
5	Kaizen	Collaborative Platforms	Facilitate continuous improvement through team collaboration and innovation.
6	Value Stream Mapping (VSM)	Process Mapping Software	Visualize and optimize material and information flows across production systems.
7	Just-In-Time (JIT)	Automated Inventory Systems	Align production with real-time demand, minimizing inventory and reducing lead times.
8	Kanban	Digital Kanban Boards	Improve scheduling and task visibility using digital tools.
9	Poka-Yoke (Error Proofing)	Sensor-Based Error Detection	Automate error detection and correction to prevent defects in the production process.
10	Jidoka (Autonomation)	AI-Powered Monitoring Systems	Empower machines to autonomously detect and respond to production anomalies.
11	Root Cause Analysis (RCA)	Machine Learning Algorithms	Use advanced analytics to identify root causes and predict potential failures.
12	Bottleneck Analysis	Simulation and Modeling Tools	Identify and eliminate bottlenecks to improve production flow and overall efficiency.
13	Total Productive Maintenance (TPM)	Predictive Maintenance Tools	Predict and prevent equipment failures to increase reliability and minimize downtime.
14	Takt Time	Production Planning Tools	Sync production rates with customer demand to ensure timely delivery and avoid overproduction.
15	Andon	Real-Time Alert Systems	Provide immediate notifications for rapid response to operational issues.
16	Quality Assurance / Quality Control (QA/QC)	Automated Inspection Systems	Ensure consistent product quality through real-time automated inspections.
17	Cellular Manufacturing	Smart Manufacturing Cells	Enhance production flexibility and efficiency with intelligent, modular setups.
18	Continuous Flow	Smart Conveyor Systems	Enable uninterrupted, seamless material flow using connected systems.
19	Visual Management	Augmented Reality (AR) Displays	Provide real-time visual feedback and operational guidance to workers.
20	SMED (Single-Minute Exchange of Dies)	IoT-Enabled Tool Tracking	Optimize tool availability and reduce changeover times through real-time monitoring.
21	Hoshin Kanri (Policy Deployment)	Decision Support Systems	Align organizational strategies with operational goals using advanced decision-making tools.
22	Heijunka (Production Leveling)	ERP Systems	Balance production schedules to manage demand fluctuations and reduce waste.
23	Total Maintenance System (TMS)	Cloud-Based Maintenance Platforms	Centralize and streamline maintenance management using cloud-based platforms for data-driven decisions.









TRADITIONAL vs DIGITAL vs AI - LEAN MANUFACTURING

Traditional Lean Tools	Digital Lean Tools	Gen AI Lean Tools	Objective
Value Stream Mapping (VSM)	Process Mapping Software	AI VSM	Dynamic, proactive real-time optimization
5S	5S IoT	AI 5S	Smart 5S leveraging IoT data, computer vision and LLM optimization
Standardized Work	Workflow Automation Sw.	AI Dynamic Standard Work	Digital/dynamic work instructions that change in real time
Kaizen	Collaborative Platforms	AI Kaizen	Identification of history, solution proposals and follow-up
Andon	Real Time Alert Systems	AI Andon	Predictive Andon
Root Cause Analysis (RCA)	Machine Learning Algorithm	Real Time AI RCA	Real-time cause analysis and immediate action proposals
8 Lean Wastes	Big Data Analytics	AI Wastes	Identification and elimination with AI
Just In Time (JIT)	Automated Inventory System	Intelligent AI Inventory	Dynamic, automatic stock adjustment based on history, ETA..
SQCDP Panel	SQCDP Panel sw	Intelligent SQCDP	Automatic completion based on data, technical selection
Visual Management	Mixed Reality Devices	NLP Support+ Visual	Visual guidance of operations and feedback panels
…..			

AI VALUE STREAM MAPPING

AI VSM (Value Stream Mapping)

VSM is a visual tool that analyzes and represents the flow of materials and information in a production process, from start to delivery to the customer.

Its main objective is to identify non-value-adding activities (waste) in order to eliminate or minimize them, thereby improving efficiency and productivity.

The role of generative AI is:

- **Automated data collection and analysis**: generative AI can automate this process by collecting real-time data from various sources, such as IoT devices, MES/ERP systems and other digital tools. Once collected, AI can analyze it to identify trends and anomalies.
- **Identification of hidden patterns and bottlenecks**: generative AI, with its advanced pattern recognition capabilities, can uncover these hidden problems. By analyzing large datasets, AI can identify inefficiencies that may not be evident at first glance, allowing organizations to address them more effectively.
- **Predictive Analytics:** Generative AI can also enhance VSM analysis through predictive analytics. By examining historical data, AI can forecast future trends and possible bottlenecks, as well as suggest preventive measures.
- **Enhanced Collaboration and Visualization:** AI can create and display dynamic VSMs that update in real time, reflecting the latest data and analysis to the different production teams. This fosters a culture of continuous improvement, as teams can quickly identify and address problems as they arise.



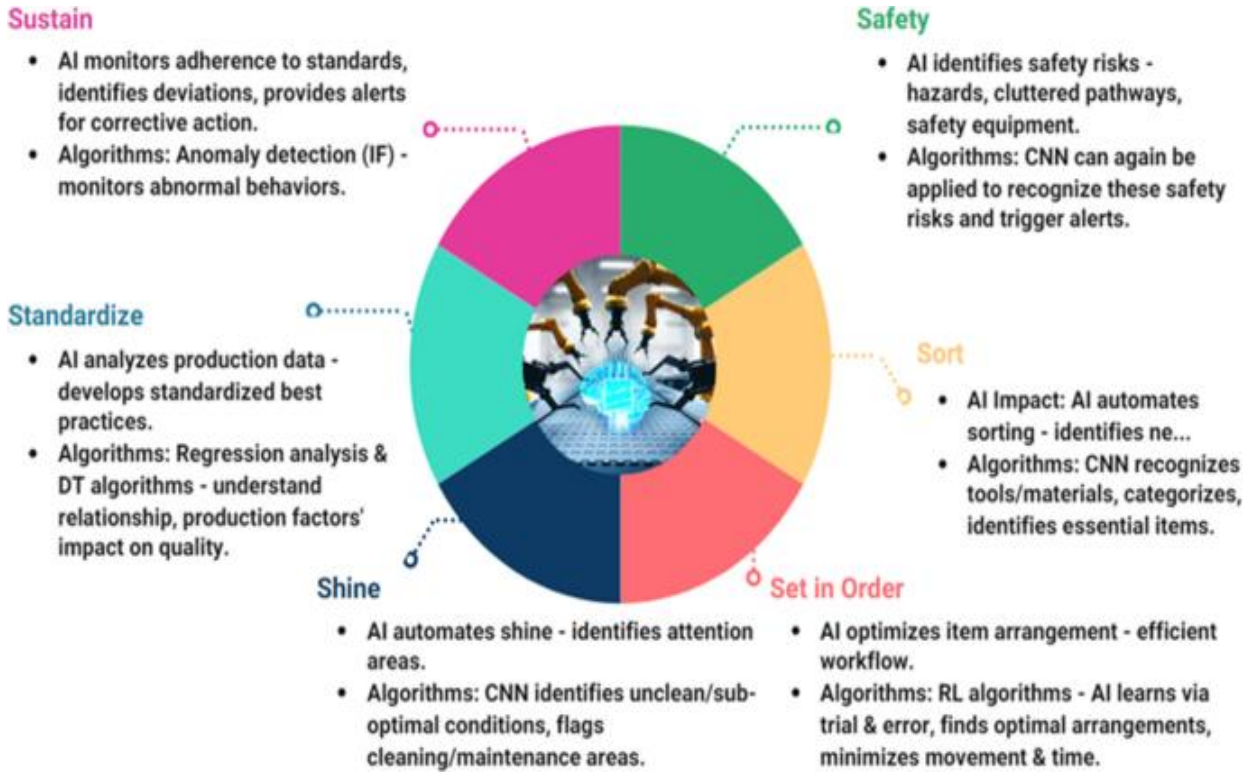


AI 5S

AI 5S (Organization, Cleanliness and Standardization)

5S with sensors (IoT) and AI makes it possible to organize the work environment (station) for maximum efficiency and safety.

- **Seiri (Sort):** 4.0 sensing of assets makes it possible to know the geoposition of production elements and therefore, through AI, to know how they are used and to classify them.
- **Seiton (Set in Order):** both the use and the tracking of assets will allow AI to obtain an optimized map of asset positions as well as real-time status control.
- **Seiso (Shine):** real-time geo-control of assets together with visual sensing (area cameras) makes it possible, through computer vision, to know the cleanliness status of the station, the presence of FOD, ... AI can suggest or command cleaning and tidying according to the established standard.
- **Seiketsu (Standardize):** AI can optimize the standard procedures for maintaining cleanliness and order, or even carry out some of the procedures itself.
- **Shitsuke (Sustain):** AI can foster discipline through training processes, reminders,.. in situations suited to these suggestions (new hires,..).

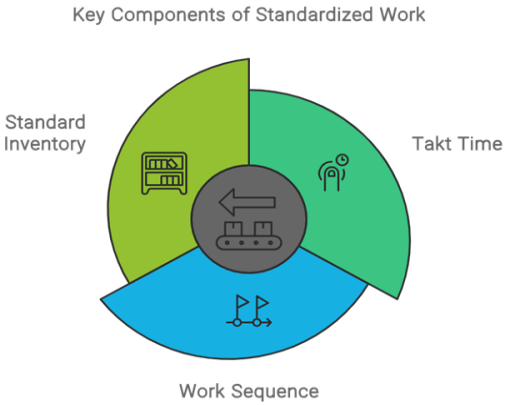


STANDARDIZED WORK - LEAN GENERATIVE AI

Standardized Work

Standardized work is a tool that defines the best and safest method for performing a task, ensuring consistency and efficiency in production processes.

- AI-driven standardized work, known as Digital Standard Work (DSW), transforms static paper instructions into dynamic, interactive digital guides that adapt in real time.
- AI converts, translates and updates procedures, while enabling instant access via QR codes and providing data-driven insights to optimize processes and close skill gaps. This approach ensures consistent quality, improves safety, accelerates training and creates "living" documentation that evolves with the process.



AI Improvement

- **Digitizes and centralizes instructions:** AI converts traditional manuals into structured digital work instructions, eliminating the downtime associated with searching for paper documentation.
- **Creates interactive guides:** Instead of static instructions, DSW uses AI-based tools and augmented reality to provide interactive, real-time help to workers, often via tablets or wearables.
- **Enables real-time data integration:** AI connects to shop-floor sensors to capture process data, which is then incorporated into digital work instructions to provide immediate feedback and enforce standards.
- **Automates updates and translation:** AI can automatically update procedures when processes or regulations change and translate instructions into several languages for a global workforce.
- **Optimizes training and compliance:** AI can trigger automatic requalification when standard operating procedures are updated, ensuring the team complies with current regulations.

Benefits

- **Greater efficiency:** task automation and real-time guidance reduce errors and allow workers to focus on value-adding activities.
- **Better quality and consistency:** by ensuring all workers follow the most optimized and up-to-date processes, AI reduces process variation and ensures consistent results.
- **Greater safety:** AI-based guidance and real-time data capture help eliminate unsafe practices and ensure compliance with safety regulations.
- **Faster onboarding and training**: interactive, personalized digital work instructions accelerate worker training and reduce the time new employees need to get up to speed**.**
- **Agility and adaptability:** dynamic documentation allows organizations to quickly adapt their standard operating procedures to changes in processes, technologies and regulations.

AI KAIZEN

Kaizen AI

The Kaizen AI methodology focuses on the continuous improvement of the assembly line through small changes enhanced by the computing power of AI.

- **Identify the problem or opportunity:** AI can identify, through historical data, the operations with the most deviations, quality problems, etc.., which will be the target of improvement.
- **Analyze the current process:** once the problem is identified, AI, using data, can help thoroughly understand the current process.
- **Develop or identify solutions:** AI, through its knowledge, can help find solutions.
- **Implement the solution:** once the solution is identified, the next step is to implement it. This may involve making changes to processes, procedures or equipment, also aided by AI with its generative power.
- **Analyze the results:** after implementing the solution, it is essential to analyze the results and make the necessary adjustments. AI should collect data, monitor the process and implement further improvements based on the statistical findings.
- **Standardize the solution:** the final step is to standardize the successful solution. This involves documenting the new process, training employees in the new procedures and ensuring the improvements are sustained over time. This can also be done by leveraging generative power to develop procedures and courses.

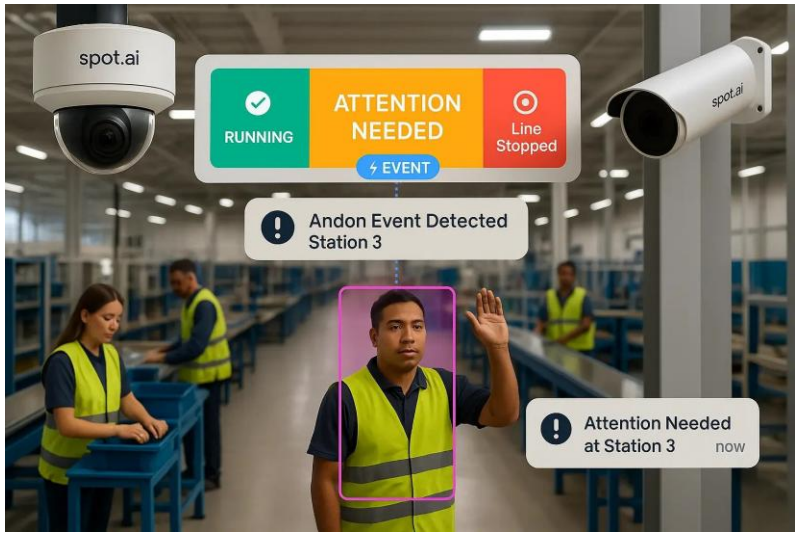


AI ANDON

Andon AI

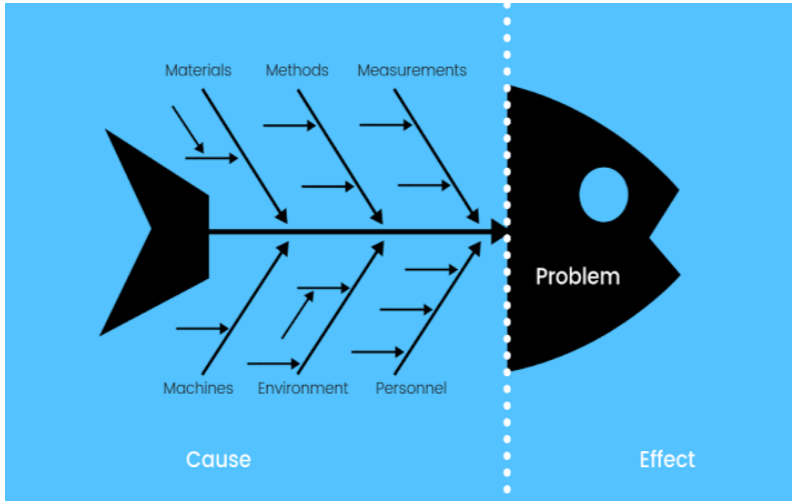
Upgrades the traditional Andon system by integrating artificial intelligence to optimize monitoring and management of the assembly line. It enables:

- **Real-time monitoring and alerts:** AI Andon systems provide continuous monitoring of the production line, ensuring any problem is detected and resolved promptly. This reduces downtime and minimizes disruptions.
- **Predictive Andon**: by analyzing historical data, AI Andon systems can predict when equipment is likely to fail. This allows maintenance to be scheduled, reducing unexpected breakdowns and extending machinery life.
- **Improved decision-making:** AI algorithms analyze large amounts of data to provide insights that help managers make informed decisions. This leads to optimized production processes and better resource allocation.
- **Improved quality control:** AI Andon systems can detect deviations in product quality in real time, ensuring defects are identified and corrected immediately. This improves overall product quality and reduces waste.
- **Greater efficiency:** By automating routine tasks and providing real-time information, AI Andon systems streamline operations and improve overall efficiency. This translates into higher productivity and lower operating costs.
- **Continuous improvement:** AI Andon systems support continuous improvement initiatives by providing data-driven insights into production processes. This helps identify areas for improvement and track the effectiveness of implemented changes.





AI ROOT CAUSE ANALYSIS



Traditional Root Cause Analysis (RCA)

Root Cause Analysis is a structured approach to identifying the underlying causes of problems. Rather than simply addressing symptoms, RCA seeks to address the fundamental issues that cause failures, defects or inefficiencies. The problems posed by the traditional RCA method are:

- **Data overload:** as industries digitalize, the volume of data generated can overwhelm traditional RCA methods.
- **Bias and subjectivity:** human analysts can unconsciously introduce bias into their analyses.
- **Reactive nature:** traditional RCA usually focuses on past incidents, limiting its ability to prevent future problems.

AI is addressing these limitations, improving RCA thanks to its ability to process large datasets, identify patterns and predict possible problems.

Root Cause Analysis (RCA) by AI

- **Enhanced data analysis:** AI systems excel at processing large volumes of structured and unstructured data.
- **Speed and automation**: AI drastically reduces the time needed to identify root causes. By automating tasks such as data collection, categorization and preliminary analysis, organizations can respond to problems more quickly.
- **Predictive analytics:** one of AI's most significant contributions to RCA is its predictive capability. Machine learning models can analyze historical data to forecast possible problems before they occur.
- **Multidimensional analysis:** traditional root cause analysis (RCA) methods usually focus on individual factors or categories. AI can simultaneously analyze multiple dimensions, such as time, location and operating parameters, providing a more complete understanding of the problem.

8 WASTES - LEAN GENERATIVE AI

8 Lean Wastes

AI facilitates the identification and elimination of lean wastes.

- **Overproduction**: excessive production of products as well as unrequired activities or services (aircraft, parts, reports, meetings..).
- **Transportation:** moving materials or products unnecessarily. AI-based analytics can track movement in real time, identifying bottlenecks and inefficiencies.
- **Motion**: movement of people… AI together with computer vision technology can monitor employee movements and optimize the plant layout to reduce travel times.
- **Waiting**: waiting in production processes. By using AI to analyze workflow processes, waiting times can be identified and solutions implemented.
- **Overprocessing:** doing more work than necessary or adding complexity without adding value. AI can analyze operations data to accurately balance production sequences.
- **Defects:** errors, rework,… due to lack of information or lack of standardization. AI technologies, such as machine learning, can predict possible defects by analyzing quality control data.
- **Underused talent:** failing to leverage employees' skills, creativity and problem-solving potential. AI can identify skill gaps and suggest personalized training programs.
- **Inventory:** accumulation of materials, documents, etc.. AI analytics can detect historical and seasonal problems.



SQCDP AI

SQCDP AI

Basically, it is a key visual management tool in the Lean philosophy that collects and displays the status of Safety, Quality, Cost, Delivery and People (S-Q-C-D-P) factors, providing visibility of metrics, rapid identification of problems and continuous improvement of production processes.

In an AI-assisted model, current data, predictions and suggested solutions are presented:

- **Safety:** Real-time presentation of data from incident records, H&S reports, new regulations, suggestions and risk prediction based on environmental sensing (computer vision monitoring of personal protective equipment or of entry into a hazardous work area).., etc. Automatic presentation of possible actions to be validated by the human team. It connects with the personal AI assistants of the production actors.
- **Quality:** Real-time presentation of data not only on incidents and non-quality cases but also on deviations and predictions against the established quality standards and levels.
- **Cost:** Shows daily cost data obtained from the MES, in real time, indicating deviations or predictions. Detects root causes (operations with the greatest deviation and critical events) based on historical data and conditions (poor training, poor industrialization of the operation,..). Proposes mitigation actions.
- **Delivery:** evaluates whether work orders are delivered on time without possible penalties. Various KPIs are used to assess delivery performance, such as schedule adherence, which shows the alignment between actual and planned schedules. Other metrics, such as OEE (overall equipment effectiveness), cycle time and takt time, make it possible to assess efficiency losses and production speed, providing insight into late deliveries.
- **People:** helps maximize labor efficiency by assessing the general physical condition of the teams (absences, sick leave) and tracking worker performance. It manages capabilities based on training data and operation execution data, determining each worker's skill level and training needs.



SIX SIGMA Gen AI

AI-supported Six Sigma

Six Sigma focuses on eliminating defects and optimizing processes. Combining this methodology with Generative AI produces a powerful improvement tool:

- **Define Phase (Problem Identification):** in the Define phase, AI enables more precise identification of problem areas and project objectives through the use of advanced data mining and processing techniques. AI algorithms can process large, complex datasets with a speed and accuracy that exceed traditional statistical methods. This makes it possible to identify subtle patterns in real time, uncover hidden correlations and detect anomalies that might otherwise go unnoticed.
- **Measure Phase (Problem Quantification):** Gen AI-based solutions offer real-time data (structured or not), acquired from sources such as IoT devices, ERP systems and manual inputs, enabling, once processed, accurate monitoring of assembly line operations. Rapid identification of variations allows, with the help of AI, immediate corrective action to be taken, minimizing disruptions.
- **Analyze Phase (Cause Identification**): AI can filter large volumes of data to uncover the important factors causing errors or inefficiencies. These models can detect trends that traditional statistical methods (Fishbone Diagrams, Pareto,..) might miss, speeding up root cause discovery. Beyond root cause analysis, AI-based analytical models can predict possible problems, allowing the organization to address and resolve potential problems before they develop.
- **Improve Phase (Solution Implementation and Verification):** AI enables virtual testing of possible changes before implementing them in real operations. After entering the constraints of the activity (cost, time, materials,..) it can create a simulation, making it easier to evaluate process changes without disruption. AI further automates process optimization through reinforcement learning and evolutionary algorithms, which explore a wide range of possible solutions and automatically select the most effective one.
- **Control Phase (Continuous Monitoring and Dynamic Control):** unlike traditional monitoring methods (control charts), AI improves the sustainability of improvements by providing continuous monitoring through real-time analysis and machine-learning-based anomaly detection systems. These AI systems automatically detect deviations from optimal process performance and generate alerts, enabling rapid corrective action. In addition, AI creates self-regulating feedback loops that modify process parameters in response to changing situations.

